Coefficient Sweeps for Anchored Derivative-Closed Pfaffian Families

July 29, 2026

1 Theoretical Setup

1.1 Problem Setting

Fix integers $N \geq 1$, $q \geq 0$, and $m \geq 0$, a number $T > 0$, and a nondegenerate compact interval $\Theta \subseteq [-T, T]$. Let $U \supseteq \Theta$ be an open interval. We use the one-dimensional Balcan–Nguyen–Sharma common-chain convention (Balcan et al., 2024). Let $\eta = (\eta_1, \dots, \eta_q)$ be a common triangular chain on U satisfying

$$\eta_j'(\theta) = P_j(\theta, \eta_1(\theta), \dots, \eta_j(\theta)), \qquad 1 \leq j \leq q, \tag{1.1}$$

and let $Q_i(\theta, y_1, \dots, y_q)$, $0 \leq i \leq N$, be output polynomials. With total degree,

$$M := \begin{cases} \max_{1 \leq j \leq q} \deg P_j, & q \geq 1, \\ 0, & q = 0, \end{cases} \qquad \Delta := \max_{0 \leq i \leq N} \deg Q_i, \tag{1.2}$$

define

$$F_i(\theta) := Q_i(\theta, \eta_1(\theta), \dots, \eta_q(\theta)), \qquad \widetilde{F} := (F_0, F_1, \dots, F_N), \qquad F := (F_1, \dots, F_N). \tag{1.3}$$

The coordinate F_0 is deterministic and only the coefficients multiplying $F_1, \dots, F_N$ are random.

Let $B : U \to \mathbb{R}^{(N+1)\times(N+1)}$ be a supplied polynomial matrix,

$$B_{rs}(\theta) = \sum_{\ell=0}^{m} b_{rs,\ell}\theta^\ell, \qquad 0 \leq r, s \leq N, \tag{1.4}$$

and set $T_* := \max\{1, T\}$. Its static coefficient-height certificate is

$$\widehat{\Lambda}_{B,T} := \left(\sum_{r=0}^{N} \sum_{s=0}^{N} \left(\sum_{\ell=0}^{m} |b_{rs,\ell}| T_*^\ell \right)^2 \right)^{1/2}. \tag{1.5}$$

For $R > 0$ and $0 < \kappa < \infty$, let $\mathcal{D}_{N,R,\kappa}$ be the class of Borel probability laws on $\mathbb{R}^N$ having one Lebesgue density f_μ supported on $[-R, R]^N$ with $0 \leq f_\mu \leq \kappa$ almost everywhere. Coordinates may be arbitrarily correlated. Assume this class is nonempty, write $\mathcal{D} := \mathcal{D}_{N,R,\kappa}$, and set

$$A := (2R)^N \kappa. \tag{1.6}$$

The affine event and its section are

$$\phi_\alpha(\theta) := F_0(\theta) + \langle \alpha, F(\theta) \rangle, \qquad H_\theta := \{a \in \mathbb{R}^N : F_0(\theta) + \langle a, F(\theta) \rangle = 0\}. \tag{1.7}$$

Whenever $F_j(\theta) \neq 0$, write the nonpivot coordinates as $\beta = (\beta_i)_{i\neq j} \in [-R,R]^{N-1}$ and define

$$T_j(\theta,\beta) := -\frac{F_0(\theta)}{F_j(\theta)} - \sum_{i\neq j} \beta_i \frac{F_i(\theta)}{F_j(\theta)}. \tag{1.8}$$

Let $\Psi_j(\theta,\beta)$ insert T_j in coordinate j and β_i in every other coordinate. The associated graph Jacobian is $|\det D\Psi_j| = |\partial_\theta T_j|$.

The anchor and closure are theorem-critical conditions. Under them, $F \neq 0$ on Θ, so define

$$\gamma_F(\theta) := \frac{F(\theta)}{\|F(\theta)\|_2}, \qquad \Gamma_{\mathrm{proj}}(F) := \sup_{\theta\in\Theta} \|\gamma_F'(\theta)\|_2. \tag{1.9}$$

Finally, for positive-length intervals, define

$$C_{\mathcal{D}}^{\mathrm{Pf}}(F;\Theta) := \sup_{\mu\in\mathcal{D}} \sup_{\substack{I\subseteq\Theta \text{ interval} \\ |I|>0}} \frac{\Pr_{\alpha\sim\mu}[\exists\theta\in I : \langle\alpha, F(\theta)\rangle = 0]}{|I|}, \tag{1.10}$$

$$C_{\mathcal{D}}^{\mathrm{aff}}(F_0,F;\Theta) := \sup_{\mu\in\mathcal{D}} \sup_{\substack{I\subseteq\Theta \text{ interval} \\ |I|>0}} \frac{\Pr_{\alpha\sim\mu}[\exists\theta\in I : \phi_\alpha(\theta) = 0]}{|I|}. \tag{1.11}$$

1.2 Assumptions

Assumption 1 (Static one-dimensional parameter regime). *$N \geq 1$, $q \geq 0$, $m \geq 0$, $T > 0$, $R > 0$, and $0 < \kappa < \infty$ are finite fixed instance data; $\Theta \subseteq [-T,T]$ is nondegenerate compact, $U \supseteq \Theta$ is open, and $\mathcal{D}_{N,R,\kappa}$ is nonempty. All presentation degrees, matrix coefficients, and support parameters are fixed before a law or interval is selected.*

Assumption 2 (Balcan common-chain presentation). *The chain and output polynomials use the triangular equations and degree convention in the problem setting, with ambient parameter dimension $p = 1$. Thus q is chain length, M is chain degree, and Δ is output-function degree.*

Assumption 3 (Anchored static derivative closure). *There is a fixed $j_* \in \{1,\dots,N\}$ with $Q_{j_*} \equiv 1$, and the supplied deterministic matrix obeys*

$$\widetilde{F}'(\theta) = B(\theta)\widetilde{F}(\theta) \qquad (\theta \in U). \tag{1.12}$$

Consequently $F_{j_*} \equiv 1$ and $\|F(\theta)\|_2 \geq 1$. No projective-speed, transversality, or swept-area bound is assumed here.

Assumption 4 (Arbitrarily correlated capped joint laws). *The coefficient vector is drawn from an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$. The only distributional inputs are cube support and the one full-joint density cap; no independence, marginal-density, or conditional-density condition is imposed.*

1.3 Formalized Goal

The theorem below proves, without an additional assumption, the exact anchored derivative-closure coefficient-sweep goal. It includes the static certificate and homogeneous normalized-curve identity, the equivalent measurable pivot-chart inequalities, the coordinate-free affine rate and capacity, the sharper homogeneous rate and capacity, the exact affine-monic presentation/certificate and probability recovery, and Counter-example 1 with selected-law probability $\epsilon/(4\delta)$ and positive-length ratio $1/(4\delta)$, all-law upper coefficient $1/\delta$, and raw certificate and projective speed $1/\delta$, with their distinct probabilistic and certificate origins made explicit. All probabilities are ordinary probabilities for a fixed law, all interval suprema precede the outer law supremum, and all stated vector, operator, Frobenius, scalar, Lebesgue, and Hausdorff norms and measures are Euclidean. The conclusion is confined to presentations satisfying the supplied anchor and derivative-closure certificate; it makes no claim that an unrestricted raw Pfaffian presentation admits that certificate or any polynomial presentation-format bound.

2 Preliminaries

All vector norms, inner products, induced operator norms, and Frobenius norms are Euclidean. The symbol λ_k denotes k-dimensional Lebesgue measure in the original coefficient coordinates (and in the nonpivot coordinates when the dimension is $N-1$); $\mathcal{H}^{N-1}$ denotes Euclidean Hausdorff measure on sections. We use $\lambda_0([-R,R]^0)=1$ and regard $[-R,R]^0$ as the singleton containing the empty tuple.

For a fixed deterministic instance, μ and then an interval are selected only after all presentation data $(\Theta, T, q, M, \Delta, N, R, \kappa, A, m, B)$ have been fixed. The capacities in the setup therefore always take the interval supremum for each fixed law before the outer law supremum. Every probability is ordinary probability for that fixed law; no independence is implicit.

For an affine section, H_θ is the actual set defined in the setup. When $F_0 \equiv 0$, it is the central section $F(\theta)^\perp$, and B_F denotes the principal block of B indexed by $1, \ldots, N$. In a monic specialization with degree d, the notation p_α always means the polynomial with the deterministic leading coefficient stated in the theorem, while α contains only the lower coefficients.

The pointwise chart T_j and insertion map Ψ_j are the setting-defined original-coordinate maps. Empty cells are allowed, and no lower bound on a pivot is part of the notation. These conventions are used in the measurable chart statement and in the exact monic recovery.

3 Main Theorem

Theorem 3.1 (Anchored derivative-closure coefficient sweep)**.** *The static and homogeneous clause below holds under Assumptions 1–3. All law-bearing clauses also use Assumption 4. In every case the deterministic presentation is fixed before the law and interval are selected.*

Static and homogeneous certificate. For every $\theta \in \Theta$,

$$\|B(\theta)\|_{\mathrm{op}} \le \|B(\theta)\|_{\mathrm{F}} \le \widehat{\Lambda}_{B,T}, \qquad \sup_{\theta\in\Theta} \|B(\theta)\|_{\mathrm{op}} \le \widehat{\Lambda}_{B,T}. \tag{3.1}$$

If $F_0 \equiv 0$ and B_F is the block indexed by $1, \ldots, N$, then

$$F' = B_F F, \qquad \gamma_F' = (I_N - \gamma_F \gamma_F^{\mathsf{T}}) B_F \gamma_F, \qquad \Gamma_{\mathrm{proj}}(F) \le \widehat{\Lambda}_{B,T}. \tag{3.2}$$

The anchor gives $F_{j_*} = 1$, $\|F\|_2 \ge 1$, and $F \neq 0$.

Equivalent affine chart form. For every $\mu \in \mathcal{D}$ and every interval $I \subseteq \Theta$ with $|I| > 0$, and every Lebesgue-measurable partition $I = \bigsqcup_{j=1}^N E_j$ with $F_j \neq 0$ on E_j,

$$\begin{aligned} \Pr_{\alpha\sim\mu}[\exists \theta \in I : \phi_\alpha(\theta) = 0] &\le \kappa \sum_{j=1}^N \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j(\theta,\beta)| \le R\} |\partial_\theta T_j(\theta,\beta)| \, d\beta \, d\theta \\ &\le \kappa \sum_{j=1}^N \int_{E_j} \int_{[-R,R]^{N-1}} |\partial_\theta T_j(\theta,\beta)| \, d\beta \, d\theta. \end{aligned} \tag{3.3}$$

These are ordinary-probability inequalities in the original N coefficient coordinates; they remain valid with no pivot margin, including tangent, multiple, endpoint, empty-cell, zero-Jacobian, persistent-root, and $N = 1$ cases.

Coordinate-free affine rate. For the same μ and I,

$$\begin{aligned}\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] &\le \kappa \int_I \int_{H_\theta\cap[-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta \\ &\le \kappa\sqrt{2}(2R)^{N-1}(1 + NR^2)\widehat{\Lambda}_{B,T}|I| \\ &= \frac{A(1 + NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}|I|.\end{aligned} \tag{3.4}$$

Consequently,

$$C^{\mathrm{aff}}_{\mathcal{D}}(F_0, F; \Theta) \le \frac{A(1 + NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}. \tag{3.5}$$

If $\widehat{\Lambda}_{B,T} = 0$, the event probability is exactly zero. The section convention is Euclidean $\mathcal{H}^{N-1}$, with $\mathcal{H}^0$ counting measure when $N = 1$.

Sharper homogeneous rate. If $F_0 \equiv 0$, then for $a \in H_\theta$,

$$H_\theta = F(\theta)^\perp = \gamma_F(\theta)^\perp, \qquad \frac{|\langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} = |\langle a, \gamma_F'(\theta)\rangle|, \tag{3.6}$$

and

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle\alpha, F(\theta)\rangle = 0] \le A\sqrt{\frac{N}{2}}\Gamma_{\mathrm{proj}}(F)|I| \le A\sqrt{\frac{N}{2}}\widehat{\Lambda}_{B,T}|I|, \tag{3.7}$$

$$C^{\mathrm{Pf}}_{\mathcal{D}}(F; \Theta) \le A\sqrt{\frac{N}{2}}\widehat{\Lambda}_{B,T}. \tag{3.8}$$

The interval supremum is taken for each fixed law before the law supremum. If $\Gamma_{\mathrm{proj}}(F) = 0$, every nonempty-interval event is one fixed proper central hyperplane and has probability zero under every admissible full density.

Exact affine-monic specialization. For every integer $d \ge 1$, every bounded interval $J \subset \mathbb{R}$, and every possibly correlated law on $[-R, R]^d$ with full density bounded by κ, set

$$F_0(\theta) = \theta^d, \qquad F_{k+1}(\theta) = \theta^k\ (0 \le k \le d-1), \qquad p_\alpha(\theta) = \theta^d + \sum_{k=0}^{d-1} \alpha_k\theta^k. \tag{3.9}$$

Then $q = M = m = 0$, $\Delta = N = d$, $A = (2R)^d\kappa$, the random vector is exactly $\alpha = (\alpha_0, \ldots, \alpha_{d-1})$, and the leading coefficient one is deterministic and outside that vector. The constant $(d+1)$-square shift has only

$$B_{0,d} = d, \qquad B_{k+1,k} = k \quad (1 \le k \le d-1), \qquad \widehat{\Lambda}_{B,T} = \left(\sum_{k=1}^{d} k^2\right)^{1/2}. \tag{3.10}$$

For $d \ge 2$, use $E_1 = J \cap \{|\theta| \le 1\}$ and $E_d = J \cap \{|\theta| > 1\}$ (all intermediate cells are empty), with

$$T_1(\theta, \beta) = -\theta^d - \sum_{k=1}^{d-1} \beta_k\theta^k, \qquad T_d(\theta, \beta) = -\theta - \sum_{k=0}^{d-2} \beta_k\theta^{k-d+1}. \tag{3.11}$$

Their velocities obey

$$|\partial_\theta T_1| \le d + \frac{Rd(d-1)}{2} \quad (|\theta| \le 1), \tag{3.12}$$

$$|\partial_\theta T_d| \le 1 + R\sum_{m=1}^{d-1} \frac{m}{|\theta|^{m+1}} \le 1 + \frac{Rd(d-1)}{2} \le d + \frac{Rd(d-1)}{2} \quad (|\theta| > 1). \tag{3.13}$$

At $d=1$, $[-R,R]^0$ has mass one, $T_1(\theta) = -\theta$, and $|\partial_\theta T_1| = 1$. In every case, including empty and singleton J,

$$\Pr_{\alpha\sim\mu}[\exists\theta \in J : p_\alpha(\theta) = 0] \le \kappa(2R)^{d-1}\left(d + \frac{Rd(d-1)}{2}\right)|J|. \tag{3.14}$$

Counter-example 1. For $0 < \delta \le 1$, take $\Theta = [-1,1]$, $F_0 = 0$, $F = (1, \theta/\delta)$, $R = 1$, $\kappa = 1/4$, and $A = 1$. Then

$$q = M = 0, \quad \Delta = 1, \quad N = 2, \quad m = 0, \qquad \widehat{\Lambda}_{B,T} = \Gamma_{\text{proj}}(F) = \frac{1}{\delta}, \tag{3.15}$$

with the sole nonzero shift entry $B_{2,1} = 1/\delta$. Every possibly correlated admissible law and every positive-length interval $I \subseteq [-1,1]$ obey

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \alpha_1 + \alpha_2\theta/\delta = 0] \le \frac{|I|}{\delta}, \qquad C^{\text{Pf}}_{\mathcal{D}}(F;[-1,1]) \le \frac{1}{\delta}. \tag{3.16}$$

For the selected independent uniform law on $[-1,1]^2$ and $0 < \epsilon \le \delta$,

$$\Pr[\exists\theta \in [0,\epsilon] : \alpha_1 + \alpha_2\theta/\delta = 0] = \frac{\epsilon}{4\delta}, \qquad \frac{\Pr[\exists\theta \in [0,\epsilon] : \alpha_1 + \alpha_2\theta/\delta = 0]}{\epsilon} = \frac{1}{4\delta}. \tag{3.17}$$

All assertions use ordinary probability for each fixed law, arbitrary full-joint coefficient correlation, and literal endpoint conventions. Vector, operator, Frobenius, and scalar norms are Euclidean; the measures are Lebesgue and Euclidean Hausdorff measures. No confidence parameter, hidden constant, chart-count factor, interval-location factor, auxiliary tolerance, random leading coordinate, or second probability conversion is present. Once the supplied certificate is fixed, additional dependence on q, M, Δ is degree zero. The result preserves both baseline-invariance obligations: the exact deterministic-leading-coefficient monic theorem and the distinct Counter-example scales. It makes no claim that an unrestricted raw Pfaffian presentation admits the supplied derivative-closure certificate or a polynomial presentation-format bound.

4 Proof Sketch

The proof first converts the polynomial coefficient list of B into a uniform operator bound and then differentiates $F/\|F\|_2$; this yields the static and homogeneous projective certificate in Lemmas A.2 and Proposition A.1. Persistent affine combinations form a proper affine locus and are null under every full-joint density by Proposition A.3.

For a measurable pivot partition, the coefficient sweep is exhausted by regions on which the pivot is at least $1/n$. The equal-dimensional area formula, with multiplicity, controls every finite chart and monotone convergence removes n; Proposition A.5 gives both chart inequalities. A fixed-section change of variables then cancels the pivot denominator exactly and produces the normal-velocity integral in Proposition A.6.

Brunn–Minkowski centralizes each translated cube section, and Ball's theorem controls its central section by $\sqrt{2}(2R)^{N-1}$ (Ball, 1986). The derivative closure controls the affine normal velocity.

Their product gives the coordinate-free affine rate and ordered capacity supremum in Proposition A.10. In the homogeneous case the radial derivative vanishes on the actual root section, leaving only projective speed; Proposition A.12 and Proposition A.13 give the sharper coefficient.

The monic specialization is handled inside the same affine sweep. The constant and outer monomial pivots give the two location-free velocity estimates, while the leading coefficient remains deterministic. The complete presentation, certificate, and probability recovery appear in Proposition A.18. Finally, the one-entry shear is differentiated directly, and both coefficient wedges are integrated exactly; Proposition A.21 keeps the lower ratio, all-law upper coefficient, and geometric section normalization distinct. The final conjunction is Proposition A.22.

References

Balcan, M.-F., Nguyen, A. T. and Sharma, D. (2024). Algorithm configuration for structured pfaffian settings. 1

Ball, K. (1986). Cube slicing in $\mathbb{R}^n$. *Proceedings of the American Mathematical Society* **97** 465–473. 5, 29

Federer, H. (1969). *Geometric Measure Theory*. Springer, Berlin. 15, 22

Gardner, R. J. (2002). The Brunn–Minkowski inequality. *Bulletin of the American Mathematical Society* **39** 355–405. 28

A Proof Details

A.1 Static Certificate and Projective Closure

Lemma A.1 (Anchor nonvanishing)**.** *Under Assumptions 1, 2, and 3, for every $\theta \in \Theta$,*

$$F_{j_*}(\theta) = 1, \qquad \|F(\theta)\|_2 \geq 1, \qquad F(\theta) \neq 0. \tag{A.1}$$

Consequently $\gamma_F = F/\|F\|_2$ is well-defined and differentiable along Θ.

Proof. By the anchored closure assumption, $Q_{j_*} \equiv 1$. Therefore, using the setting definition of an output feature,

$$F_{j_*}(\theta) = Q_{j_*}(\theta, \eta_1(\theta), \ldots, \eta_q(\theta)) = 1 \tag{A.2}$$

for every $\theta \in U$, and in particular for every $\theta \in \Theta$. Thus

$$\|F(\theta)\|_2^2 = \sum_{i=1}^{N} F_i(\theta)^2 \geq F_{j_*}(\theta)^2 = 1. \tag{A.3}$$

The common-chain presentation makes the outputs differentiable on U; equivalently, Assumption 3 supplies their derivatives there. Since the denominator is bounded below by one on a neighborhood of every point of Θ, normalization is differentiable, including at the endpoints of Θ. This proves the lemma. ☐

Lemma A.2 (Literal coefficient-height matrix certificate)**.** *Under Assumptions 1 and 3, for every $\theta \in \Theta$,*

$$\|B(\theta)\|_{\text{op}} \leq \|B(\theta)\|_{\text{F}} \leq \widehat{\Lambda}_{B,T}. \tag{A.4}$$

Consequently

$$\sup_{\theta \in \Theta} \|B(\theta)\|_{\text{op}} \leq \widehat{\Lambda}_{B,T}. \tag{A.5}$$

Proof. Retain the setting-defined $T_* := \max\{1, T\}$, and introduce only for this proof

$$c_{rs} := \sum_{\ell=0}^{m} |b_{rs,\ell}| T_*^{\ell}, \qquad 0 \leq r, s \leq N. \tag{A.6}$$

For $\theta \in \Theta$, $|\theta| \leq T \leq T_*$. Because ℓ is a nonnegative integer and $T_* \geq 1$, $|\theta|^{\ell} \leq T_*^{\ell}$ also for $\ell = 0$. Hence, entry by entry,

$$\begin{aligned} |B_{rs}(\theta)| &= \left| \sum_{\ell=0}^{m} b_{rs,\ell} \theta^{\ell} \right| \\ &\leq \sum_{\ell=0}^{m} |b_{rs,\ell}| \, |\theta|^{\ell} \\ &\leq \sum_{\ell=0}^{m} |b_{rs,\ell}| T_*^{\ell} = c_{rs}. \end{aligned} \tag{A.7}$$

Summing the squared entrywise bounds over exactly the $(N+1)^2$ entries gives

$$\|B(\theta)\|_{\mathrm{F}}^2 = \sum_{r=0}^{N}\sum_{s=0}^{N}|B_{rs}(\theta)|^2 \le \sum_{r=0}^{N}\sum_{s=0}^{N}c_{rs}^2 = \sum_{r=0}^{N}\sum_{s=0}^{N}\left(\sum_{\ell=0}^{m}|b_{rs,\ell}|T_*^{\ell}\right)^2 = \widehat{\Lambda}_{B,T}^2. \tag{A.8}$$

For completeness, the operator/Frobenius comparison is also direct. For any $x = (x_0, \ldots, x_N) \in \mathbb{R}^{N+1}$, applying the scalar Cauchy–Schwarz inequality separately in each row yields

$$\begin{aligned}\|B(\theta)x\|_2^2 &= \sum_{r=0}^{N}\left|\sum_{s=0}^{N}B_{rs}(\theta)x_s\right|^2 \\ &\le \sum_{r=0}^{N}\left(\sum_{s=0}^{N}|B_{rs}(\theta)|^2\right)\left(\sum_{s=0}^{N}|x_s|^2\right) \\ &= \|B(\theta)\|_{\mathrm{F}}^2\|x\|_2^2.\end{aligned} \tag{A.9}$$

Taking the supremum over unit vectors proves $\|B(\theta)\|_{\mathrm{op}} \le \|B(\theta)\|_{\mathrm{F}}$. Combining the two pointwise inequalities and then taking the supremum over Θ proves the lemma. No coefficient, power, matrix entry, or dimensional factor has been absorbed. □

Lemma A.3 (Homogeneous block extraction)**.** *Under Assumption 3, if $F_0 \equiv 0$ and $B_F(\theta)$ denotes the $N \times N$ block $(B_{rs}(\theta))_{1\le r,s\le N}$, then*

$$F'(\theta) = B_F(\theta)F(\theta) \qquad \textit{for every } \theta \in \Theta. \tag{A.10}$$

Proof. In the homogeneous specialization,

$$\widetilde{F}(\theta) = \begin{pmatrix} 0 \\ F(\theta)\end{pmatrix}, \qquad \widetilde{F}'(\theta) = \begin{pmatrix} 0 \\ F'(\theta)\end{pmatrix}. \tag{A.11}$$

The exact closure identity therefore reads

$$\begin{pmatrix} 0 \\ F'(\theta)\end{pmatrix} = B(\theta)\begin{pmatrix} 0 \\ F(\theta)\end{pmatrix}. \tag{A.12}$$

Taking rows $1, \ldots, N$, the column-zero terms are multiplied by $F_0(\theta) = 0$, while the remaining columns are exactly $B_F(\theta)F(\theta)$. Thus $F' = B_F F$. No vanishing of any off-block entry of B has been assumed. □

Proposition A.1 (Normalized derivative and projective certificate)**.** *Under Assumptions 1, 2, and 3, and Lemmas A.1, A.2, and A.3, if $F_0 \equiv 0$, then for every $\theta \in \Theta$,*

$$\gamma_F'(\theta) = \big(I_N - \gamma_F(\theta)\gamma_F(\theta)^{\mathsf{T}}\big)B_F(\theta)\gamma_F(\theta), \tag{A.13}$$

and

$$\Gamma_{\mathrm{proj}}(F) = \sup_{\theta\in\Theta}\|\gamma_F'(\theta)\|_2 \le \widehat{\Lambda}_{B,T}. \tag{A.14}$$

Proof. Fix $\theta \in \Theta$, and write $r(\theta) := \|F(\theta)\|_2$ within this proof. Lemma A.1 gives $r(\theta) \ge 1$, so all following divisions are valid. Differentiating $r^2 = \langle F, F\rangle$ gives

$$2rr' = 2\langle F, F'\rangle, \qquad r' = \frac{\langle F, F'\rangle}{r}. \tag{A.15}$$

Therefore direct differentiation of $\gamma_F = F/r$ gives

$$
\begin{aligned}
\gamma_F' &= \frac{F'}{r} - \frac{Fr'}{r^2} \\
&= \frac{F'}{r} - \frac{F\langle F, F'\rangle}{r^3} \\
&= \left(I_N - \gamma_F\gamma_F^\mathsf{T}\right)\frac{F'}{r}.
\end{aligned} \tag{A.16}
$$

By Lemma A.3, $F' = B_F F = rB_F\gamma_F$. Substitution proves the exact identity

$$
\gamma_F' = \left(I_N - \gamma_F\gamma_F^\mathsf{T}\right)B_F\gamma_F. \tag{A.17}
$$

It remains to prove the norm bound without hiding a dimensional constant. Since $\|\gamma_F\|_2 = 1$, for every $v \in \mathbb{R}^N$,

$$
\left\|\left(I_N - \gamma_F\gamma_F^\mathsf{T}\right)v\right\|_2^2 = \|v\|_2^2 - |\langle\gamma_F, v\rangle|^2 \le \|v\|_2^2. \tag{A.18}
$$

Thus the projector has Euclidean operator norm at most one. Also, if $x \in \mathbb{R}^N$, embed it as $\bar{x} = (0, x) \in \mathbb{R}^{N+1}$, and let π discard coordinate zero. Then

$$
B_F(\theta)x = \pi B(\theta)\bar{x}, \tag{A.19}
$$

so

$$
\|B_F(\theta)x\|_2 \le \|B(\theta)\bar{x}\|_2 \le \|B(\theta)\|_{\mathrm{op}}\|\bar{x}\|_2 = \|B(\theta)\|_{\mathrm{op}}\|x\|_2. \tag{A.20}
$$

Taking the supremum over unit x proves $\|B_F(\theta)\|_{\mathrm{op}} \le \|B(\theta)\|_{\mathrm{op}}$. Hence the exact identity and Lemma A.2 yield, pointwise,

$$
\begin{aligned}
\|\gamma_F'(\theta)\|_2 &\le \|I_N - \gamma_F(\theta)\gamma_F(\theta)^\mathsf{T}\|_{\mathrm{op}}\|B_F(\theta)\|_{\mathrm{op}}\|\gamma_F(\theta)\|_2 \\
&\le \|B(\theta)\|_{\mathrm{op}} \\
&\le \widehat{\Lambda}_{B,T}.
\end{aligned} \tag{A.21}
$$

The right-hand side is finite deterministic instance data. Taking the supremum over $\theta \in \Theta$ both proves finiteness of $\Gamma_{\mathrm{proj}}(F)$ and gives the claimed bound. □

Proposition A.2 (Boundary and baseline consistency)**.** *Under Assumptions 1, 2, and 3, Lemmas A.1, A.2, and A.3, and Proposition A.1, all conclusions of those results remain valid without modification under the static specializations $q = 0$, $m = 0$, constant B, constant $\widetilde{F}$, $N = 1$, or $\widehat{\Lambda}_{B,T} = 0$, at both endpoints of Θ, and for a stationary γ_F in the homogeneous branch. In particular,*

$$
F_{j_*} = 1, \qquad F \ne 0, \qquad \sup_{\theta\in\Theta}\|B(\theta)\|_{\mathrm{op}} \le \widehat{\Lambda}_{B,T} \tag{A.22}
$$

still hold. Whenever $F_0 \equiv 0$, the same regimes also satisfy

$$
F' = B_F F, \qquad \gamma_F' = (I_N - \gamma_F\gamma_F^\mathsf{T})B_F\gamma_F, \qquad \Gamma_{\mathrm{proj}}(F) \le \widehat{\Lambda}_{B,T}. \tag{A.23}
$$

In addition:

1. If $N = d \ge 1$, $m = 0$, and the only nonzero entries of the constant augmented monomial shift are $B_{0d} = d$ and $B_{k+1,k} = k$ for $1 \le k \le d-1$, then

$$
\widehat{\Lambda}_{B,T} = \left(\sum_{k=1}^{d} k^2\right)^{1/2}. \tag{A.24}
$$

2. If $N=2$, $m=0$, $0<\delta\le 1$, and the only nonzero entry of the constant shear is $B_{2,1}=1/\delta$, then $\widehat{\Lambda}_{B,T}=1/\delta$.

Proof. If $q=0$, the chain is empty and $M=0$; none of the calculations in Lemmas A.1–A.3 or Proposition A.1 uses a chain coordinate. Thus the proof remains unchanged and the additional dependence on q, M, Δ, after B is fixed, is exactly degree zero.

If $m=0$, then $B(\theta)=(b_{rs,0})_{r,s=0}^N$ is constant and the defining certificate becomes exactly

$$\widehat{\Lambda}_{B,T}=\left(\sum_{r=0}^{N}\sum_{s=0}^{N}|b_{rs,0}|^2\right)^{1/2}=\|B\|_{\mathrm{F}}, \tag{A.25}$$

because $T_*^0=1$. This also covers any constant B presented with a larger formal value of m and zero higher coefficients. When $N=1$, the anchor forces the sole feature to be $F_1=1$, so $\gamma_F=1$ and $I_1-\gamma_F\gamma_F^{\mathsf{T}}=0$; the normalized identity and the zero projective speed therefore hold exactly. All derivatives used in these conclusions exist on the open interval $U\supseteq\Theta$, so the same calculations apply at both endpoints of Θ.

If $\widetilde{F}$ is constant, then $\widetilde{F}'=0$; in the homogeneous branch the anchored nonzero vector F and its normalization γ_F are constant, so $\Gamma_{\mathrm{proj}}(F)=0$. More generally, a stationary homogeneous normalized curve, meaning $\gamma_F'\equiv 0$ on Θ, has $\Gamma_{\mathrm{proj}}(F)=0\le\widehat{\Lambda}_{B,T}$ directly from the setting definition. This stationary specialization does not require $B=0$, but the literal anchor rules out nonzero radial motion along the current feature direction. Indeed, write $F=r\gamma_F$, where $r=\|F\|_2\ge 1$. The anchored coordinate identity is

$$1=F_{j_*}=r\gamma_{F,j_*},\qquad \gamma_{F,j_*}=\frac{1}{r}>0. \tag{A.26}$$

Differentiating $F=r\gamma_F$ and its anchored coordinate, and using $\gamma_F'=0$, gives

$$0=F_{j_*}'=r'\gamma_{F,j_*}+r\gamma_{F,j_*}'=r'\gamma_{F,j_*}, \tag{A.27}$$

so $r'=0$. Hence $F'=r'\gamma_F+r\gamma_F'=0$, and Lemma A.3 yields

$$0=F'=B_FF=rB_F\gamma_F,\qquad B_F\gamma_F=0. \tag{A.28}$$

Thus the projector identity permits radial components algebraically for a generally normalized curve, while the anchored stationary branch has zero radial speed. The full matrix B may still be nonzero: annihilating the current augmented feature direction does not force the entire matrix to vanish.

If $\widehat{\Lambda}_{B,T}=0$, the defining sum of squares first gives $\sum_{\ell=0}^m|b_{rs,\ell}|T_*^\ell=0$ for every (r,s). Every term in each of these sums is nonnegative and $T_*>0$, so every coefficient $b_{rs,\ell}$ is zero, and hence $B\equiv 0$. The closure identity gives $\widetilde{F}'=0$ on the interval U, so the tuple is constant; in the homogeneous branch $\gamma_F'=0$ and $\Gamma_{\mathrm{proj}}(F)=0$. Thus the zero right-hand side is valid rather than a degenerate failure of the bound.

For item 1, take the actual augmented monic tuple

$$\widetilde{F}(\theta)=(\theta^d,1,\theta,\ldots,\theta^{d-1}),\qquad d\ge 1. \tag{A.29}$$

This is the $q=0$, $M=0$, $N=d$, $\Delta=d$ specialization. Its derivative is given by the constant shift with $B_{0d}=d$ and $B_{k+1,k}=k$ for $1\le k\le d-1$, because $(\theta^d)'=dF_d$ and $(F_{k+1})'=kF_k$. Thus $m=0$, and the nonzero coefficient magnitudes are exactly $d,1,2,\ldots,d-1$. The literal formula gives

$$\widehat{\Lambda}_{B,T}^2 = d^2 + \sum_{k=1}^{d-1} k^2 = \sum_{k=1}^{d} k^2. \tag{A.30}$$

The sum over $1 \leq k \leq d-1$ is empty when $d = 1$, so that boundary case gives height one. This proves only the certificate specialization allocated to this row; it does not import the later affine-monic sweep.

For item 2, the actual tuple $\widetilde{F} = (0, 1, \theta/\delta)$ has $q = 0$, $M = 0$, $\Delta = 1$, $N = 2$, and $m = 0$, and it has derivative closure under the constant one-entry shear $B_{2,1} = 1/\delta$, with all other entries zero. The same literal formula gives

$$\widehat{\Lambda}_{B,T}^2 = |1/\delta|^2 = 1/\delta^2, \tag{A.31}$$

and $\delta > 0$ gives $\widehat{\Lambda}_{B,T} = 1/\delta$. These are algebraic reductions of the certificate itself: they do not assume the later monomial-instance or counter-example conclusions. $\square$

Synthesis of the static certificate. Lemma A.1 gives $F_{j_*} = 1$, hence $F \neq 0$ and $\|F\|_2 \geq 1$. From the literal coefficient list and $|\theta| \leq T_*$, Lemma A.2 gives

$$\sup_{\theta \in \Theta} \|B(\theta)\|_{\mathrm{op}} \leq \widehat{\Lambda}_{B,T}. \tag{A.32}$$

If $F_0 \equiv 0$, Lemma A.3 gives $F' = B_F F$, and Proposition A.1 therefore gives

$$\gamma_F' = (I_N - \gamma_F \gamma_F^{\mathsf{T}}) B_F \gamma_F, \qquad \Gamma_{\mathrm{proj}}(F) \leq \widehat{\Lambda}_{B,T}. \tag{A.33}$$

Proposition A.2 shows that constant and stationary curves, endpoints of Θ, zero coefficient height, and the two baseline specializations preserve these conclusions with their literal constants. $\square$

A.2 Persistent-Root Nullity

Lemma A.4 (Persistent-root affine translate and anchor exclusion)**.** *Under Assumption 3 and the Anchor nonvanishing Lemma A.1, for every interval $I \subseteq \Theta$ with $|I| > 0$, the set $Z_\infty(I)$ is empty or a proper affine subspace of $\mathbb{R}^N$. More precisely, define the common kernel*

$$K_I := \bigcap_{\theta \in I} \ker(v \longmapsto \langle v, F(\theta) \rangle) = \left\{ v \in \mathbb{R}^N : \langle v, F(\theta) \rangle = 0 \text{ for every } \theta \in I \right\}. \tag{A.34}$$

If $a^0 \in Z_\infty(I)$, then

$$Z_\infty(I) = a^0 + K_I, \qquad e_{j_*} \notin K_I, \qquad a^0 + e_{j_*} \notin Z_\infty(I). \tag{A.35}$$

Proof. For each $\theta \in I$, let

$$H_\theta := \left\{ a \in \mathbb{R}^N : F_0(\theta) + \langle a, F(\theta) \rangle = 0 \right\}. \tag{A.36}$$

The pointwise meaning of identically zero gives the exact identity

$$Z_\infty(I) = \bigcap_{\theta \in I} H_\theta. \tag{A.37}$$

This intersection may be indexed by uncountably many points; no finite-rank reduction is being assumed. If it is empty, the first branch of the target is complete. Suppose it is nonempty and choose $a^0 \in Z_\infty(I)$. For every $a \in \mathbb{R}^N$,

$$
\begin{aligned}
a \in Z_\infty(I) &\iff F_0(\theta) + \langle a, F(\theta)\rangle = 0 \quad \text{for every } \theta \in I \\
&\iff \langle a - a^0, F(\theta)\rangle = 0 \quad \text{for every } \theta \in I \\
&\iff a - a^0 \in K_I.
\end{aligned} \tag{A.38}
$$

Hence $Z_\infty(I) = a^0 + K_I$. Each set in the defining intersection for K_I is the kernel of a linear functional, so their arbitrary intersection contains zero and is closed under addition and scalar multiplication. Thus K_I is a linear subspace.

The anchor is expressed in the original random-coefficient coordinates and gives

$$
\langle e_{j_*}, F(\theta)\rangle = F_{j_*}(\theta) = 1 \qquad \text{for every } \theta \in I. \tag{A.39}
$$

Therefore $e_{j_*} \notin K_I$. In particular $K_I \neq \mathbb{R}^N$, so its translate $a^0 + K_I$ is a proper affine subspace. The required anchor-direction exclusion can also be checked directly:

$$
\begin{aligned}
F_0(\theta) + \langle a^0 + e_{j_*}, F(\theta)\rangle &= F_0(\theta) + \langle a^0, F(\theta)\rangle + F_{j_*}(\theta) \\
&= 0 + 1 = 1 \qquad \text{for every } \theta \in I.
\end{aligned} \tag{A.40}
$$

Thus $a^0 + e_{j_*} \notin Z_\infty(I)$.

The argument does not require the nonanchor features to be distinct, nonconstant, or linearly independent. If $F_0 \equiv 0$ on I, then $0 \in Z_\infty(I)$ and the exact identity becomes $Z_\infty(I) = K_I$, still proper because it excludes e_{j_*}. If the features are constant or dependent, K_I may have any dimension from 0 through $N-1$, and its translate may be unbounded, but properness is unchanged. When $N = 1$, $j_* = 1$ and

$$
K_I = \{v \in \mathbb{R} : vF_1(\theta) = 0 \text{ for every } \theta \in I\} = \{0\}, \tag{A.41}
$$

so a nonempty locus is exactly one point; otherwise it is empty.

Finally, the proof uses the literal point set I. Whether an interval includes neither, one, or both of its relative endpoints changes only which θ's occur in this intersection. No passage to $\overline{I}$, analytic identity theorem, or endpoint continuation is used. This proves the lemma.

□

Lemma A.5 (Borel and Lebesgue nullity of the persistent-root locus)**.** *Under the Anchor nonvanishing Lemma A.1 and Lemma A.4, for every interval $I \subseteq \Theta$ with $|I| > 0$, the set $Z_\infty(I)$ is Borel and*

$$
\lambda_N(Z_\infty(I)) = 0. \tag{A.42}
$$

This conclusion includes empty loci, unbounded nonempty affine loci, and the $N = 1$ empty-or-singleton case.

Proof. For every fixed $\theta \in I$, the map

$$
a \longmapsto F_0(\theta) + \langle a, F(\theta)\rangle \tag{A.43}
$$

is a continuous affine map on $\mathbb{R}^N$, so H_θ is closed. Therefore the exact target set

$$
Z_\infty(I) = \bigcap_{\theta \in I} H_\theta \tag{A.44}
$$

is closed, hence Borel. Arbitrary intersections of closed sets are closed, so this remains valid when I is uncountable and also when the intersection is empty.

Because $|I| > 0$, the interval is nonempty; choose any $\bar{\theta} \in I$. The literal anchor gives $F_{j_*}(\bar{\theta}) = 1$, and hence the fixed-time set has the graph representation

$$H_{\bar{\theta}} = \left\{ a \in \mathbb{R}^N : a_{j_*} = -F_0(\bar{\theta}) - \sum_{i \neq j_*} a_i F_i(\bar{\theta}) \right\}. \tag{A.45}$$

Since a persistent identity in particular holds at $\bar{\theta}$,

$$Z_\infty(I) \subseteq H_{\bar{\theta}}. \tag{A.46}$$

If $N = 1$, the sum is empty and $H_{\bar{\theta}} = \{-F_0(\bar{\theta})\}$, a singleton of one-dimensional Lebesgue measure zero. Monotonicity then gives $\lambda_1(Z_\infty(I)) = 0$. This agrees with Lemma A.4, which says that the nonempty locus itself is a singleton.

Suppose $N \geq 2$. Let $\beta = (a_i)_{i \neq j_*} \in \mathbb{R}^{N-1}$ denote the original nonanchor coordinates. For every fixed β, the section of $H_{\bar{\theta}}$ in the original coordinate a_{j_*} is exactly the singleton

$$\left\{ -F_0(\bar{\theta}) - \sum_{i \neq j_*} \beta_i F_i(\bar{\theta}) \right\}. \tag{A.47}$$

After the measure-preserving permutation that places a_{j_*} last, Tonelli's theorem applied to the nonnegative Borel indicator of $H_{\bar{\theta}}$ yields

$$\begin{aligned} \lambda_N(H_{\bar{\theta}}) &= \int_{\mathbb{R}^{N-1}} \lambda_1(\{t \in \mathbb{R} : (\beta, t) \in H_{\bar{\theta}}\})\, d\lambda_{N-1}(\beta) \\ &= \int_{\mathbb{R}^{N-1}} 0 \, d\lambda_{N-1}(\beta) = 0. \end{aligned} \tag{A.48}$$

Here (β, t) means insertion into the original coordinate slots before the harmless permutation. The graph may be unbounded, but Tonelli is applied to a nonnegative indicator on the full product space, and the fiber-measure integrand is identically zero. Therefore

$$0 \leq \lambda_N(Z_\infty(I)) \leq \lambda_N(H_{\bar{\theta}}) = 0. \tag{A.49}$$

No regularity in θ, root multiplicity statement, or analyticity is used in this geometric nullity calculation. This proves the lemma. □

Proposition A.3 (Nullity under arbitrary full joint laws)**.** *Under Assumption 4 and Lemma A.5, for every $\mu \in \mathcal{D}_{N,R,\kappa}$ and every interval $I \subseteq \Theta$ with $|I| > 0$,*

$$\Pr_{\alpha \sim \mu}[F_0 + \langle \alpha, F \rangle \equiv 0 \text{ on } I] = \Pr_{\alpha \sim \mu}[\alpha \in Z_\infty(I)] = 0. \tag{A.50}$$

The probability is computed from one full N-dimensional joint density. No independence, marginal-density bound, or conditional-density bound is required.

Proof. Fix an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$. Assumption 4 supplies a nonnegative joint density f_μ on the actual space $\mathbb{R}^N$, supported on the closed cube $[-R, R]^N$, and satisfying $f_\mu \leq \kappa$ almost everywhere. By Lemma A.5, $Z_\infty(I)$ is Borel and $\lambda_N(Z_\infty(I)) = 0$. Consequently,

$$
\begin{aligned}
\mu(Z_\infty(I)) &= \int_{\mathbb{R}^N} \mathbf{1}_{Z_\infty(I)}(a) f_\mu(a)\, d\lambda_N(a) \\
&= \int_{Z_\infty(I)\cap[-R,R]^N} f_\mu(a)\, d\lambda_N(a) \\
&\le \kappa\, \lambda_N\big(Z_\infty(I)\cap[-R,R]^N\big) \\
&\le \kappa\, \lambda_N(Z_\infty(I)) = 0.
\end{aligned} \tag{A.51}
$$

This is a single joint-density calculation. It never factors f_μ, integrates a marginal-density bound, or conditions on one coordinate, so every correlation allowed by the law class is retained. If the affine locus is unbounded, only its support-cube intersection is integrated, and that intersection remains Borel and Lebesgue-null. The closed-cube boundary causes no separate term. Empty loci, nonempty proper loci, $F_0 \equiv 0$, dependent or constant features, and the $N = 1$ singleton branch all enter the same zero calculation. This proves the proposition. □

Synthesis of persistent-root nullity. Fix an interval $I \subseteq \Theta$ of positive length and $\mu \in \mathcal{D}_{N,R,\kappa}$. By Lemma A.4, either $Z_\infty(I)$ is empty or, for any $a^0 \in Z_\infty(I)$,

$$
Z_\infty(I) = a^0 + K_I, \qquad e_{j_*} \notin K_I. \tag{A.52}
$$

Thus it is a proper affine subspace; in particular, $a^0 + e_{j_*}$ is not in the locus. Lemma A.5 proves that this possibly unbounded locus is Borel and Lebesgue-null, and Proposition A.3 integrates its indicator against the single full joint density to obtain

$$
\Pr_{\alpha\sim\mu}\left[F_0 + \langle \alpha, F\rangle \equiv 0 \text{ on } I\right] = 0. \tag{A.53}
$$

This argument retains arbitrary correlation and does not exclude a nonempty persistent-root locus by assumption. □

A.3 Measurable Affine Pivot Sweep

Theorem A.1 (Equal-dimensional Euclidean area formula)**.** *Let $D \subseteq \mathbb{R}^N$ be Lebesgue measurable and let $G : D \to \mathbb{R}^N$ be Lipschitz, with a C^1 formula on an open neighborhood of D. Then, with the derivative of that ambient formula at almost every density point of D,*

$$
\int_D |\det DG(x)|\, d\lambda_N(x) = \int_{\mathbb{R}^N} N(G,D,y)\, d\lambda_N(y), \qquad N(G,D,y) := \#\{x \in D : G(x) = y\}. \tag{A.54}
$$

In particular,

$$
\lambda_N(G(D)) \le \int_D |\det DG(x)|\, d\lambda_N(x), \tag{A.55}
$$

without injectivity, noncriticality, simple-root, or finite-fiber hypotheses (Federer, 1969).

In Proposition A.4, $D = D_{j,n} \subseteq \mathbb{R} \times \mathbb{R}^{N-1}$ and $G = \Psi_j$; source and target both have dimension N. Lemma A.6 supplies completed measurability, and Lemma A.7 supplies the Lipschitz restriction, ambient C^1 formula, and exact determinant. Thus every hypothesis of Theorem A.1 is checked in the same coefficient coordinates used by the probability law.

Lemma A.6 (Measurable incidence and finite pivot exhaustion)**.** *Under Assumptions 1, 2, 3, and 4, and the Anchor nonvanishing Lemma A.1, fix an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$, an interval $I \subseteq \Theta$ with $|I| > 0$ and its literal endpoint convention, and a Lebesgue-measurable legal partition*

$$
I = \bigsqcup_{j=1}^{N} E_j, \qquad F_j \neq 0 \text{ on } E_j. \tag{A.56}
$$

Define

$$\mathcal{R}_I := \{a \in [-R,R]^N : \exists \theta \in I,\ F_0(\theta) + \langle a, F(\theta)\rangle = 0\}, \tag{A.57}$$

$$E_{j,n} := \{\theta \in E_j : |F_j(\theta)| \geq 1/n\}, \tag{A.58}$$

and

$$D_{j,n} := \{(\theta,\beta) : \theta \in E_{j,n},\ \beta \in [-R,R]^{N-1},\ |T_j(\theta,\beta)| \leq R\}. \tag{A.59}$$

Then $\mathcal{R}_I$ is Borel; $E_{j,n}$ and $D_{j,n}$ are completed-Lebesgue measurable; $D_{j,n} \subseteq D_{j,n+1}$; and

$$E_{j,n} \uparrow E_j. \tag{A.60}$$

Moreover T_j, $\partial_\theta T_j$, and

$$\mathbf{1}\{|T_j| \leq R\}|\partial_\theta T_j| \tag{A.61}$$

are measurable on the relevant pivot domains. For $N = 1$, the beta domain is $[-R,R]^0 = \{()\}$, its 0-dimensional Lebesgue mass is one, and all nonpivot sums are empty.

Proof. The primitive common-chain presentation makes each F_i continuously differentiable on U. Hence

$$h(\theta,a) := F_0(\theta) + \langle a, F(\theta)\rangle \tag{A.62}$$

is continuous on $U \times \mathbb{R}^N$.

Because I is a bounded interval, it admits compact sets $K_r \subseteq I$ with $K_r \uparrow I$, for every choice of open, closed, or half-open endpoints. Concretely, trim every excluded endpoint by $1/r$ and retain every included endpoint once the trimmed interval is nonempty. For each r,

$$C_r := \{(\theta,a) \in K_r \times [-R,R]^N : h(\theta,a) = 0\} \tag{A.63}$$

is closed in a compact product, hence compact. Its coefficient projection is compact. The exact identity

$$\mathcal{R}_I = \bigcup_{r=1}^{\infty} \mathrm{proj}_a(C_r) \tag{A.64}$$

therefore makes $\mathcal{R}_I$ an F_σ subset of $\mathbb{R}^N$. This proves Borel measurability without deleting an included parameter endpoint or a coefficient-cube face.

For fixed j, n, continuity of F_j makes $\{\theta \in \Theta : |F_j(\theta)| \geq 1/n\}$ closed. Its intersection with the Lebesgue-measurable cell E_j is $E_{j,n}$, hence is Lebesgue measurable. The product $E_{j,n} \times [-R,R]^{N-1}$ is measurable in the Euclidean product completion.

On the open nonzero-pivot set

$$\Omega_j := \{(\theta,\beta) \in U \times \mathbb{R}^{N-1} : F_j(\theta) \neq 0\}, \tag{A.65}$$

write

$$G_j(\theta,\beta) := F_0(\theta) + \sum_{i \neq j} \beta_i F_i(\theta). \tag{A.66}$$

Then

$$T_j = -\frac{G_j}{F_j}, \qquad \partial_\theta T_j = -\frac{(\partial_\theta G_j)F_j - G_j F_j'}{F_j^2}. \tag{A.67}$$

Both functions are continuous on Ω_j. Thus $D_{j,n}$ is the intersection of the measurable product domain with the measurable weak-threshold set $\{|T_j| \leq R\}$; it is completed measurable. This nonnegative integrand is measurable there as a product of measurable functions.

The threshold $1/n$ decreases with n, so $E_{j,n} \subseteq E_{j,n+1}$ and $D_{j,n} \subseteq D_{j,n+1}$. If $\theta \in E_j$, legality gives $|F_j(\theta)| > 0$. Choosing any integer

$$n \geq \frac{1}{|F_j(\theta)|} \tag{A.68}$$

puts θ in $E_{j,n}$. Hence $\bigcup_n E_{j,n} = E_j$, with no uniform lower bound on the pivot. For $N = 1$, the same proof has a one-point beta space and empty sums. □

Lemma A.7 (Original-coordinate finite charts and exact Jacobian)**.** *Under Assumptions 1, 2, and 3, and Lemma A.6, for every $j \in \{1, \dots, N\}$ and $n \geq 1$, define*

$$K_{j,n} := \{\theta \in \Theta : |F_j(\theta)| \geq 1/n\}. \tag{A.69}$$

The restrictions of T_j and Ψ_j to $K_{j,n} \times [-R, R]^{N-1}$, and hence to $D_{j,n}$, are Lipschitz even when the restriction is disconnected. The map inserts the solved root in the original j-th coefficient coordinate:

$$(\Psi_j(\theta, \beta))_j = T_j(\theta, \beta), \qquad (\Psi_j(\theta, \beta))_i = \beta_i \quad (i \neq j), \tag{A.70}$$

and it satisfies

$$F_0(\theta) + \langle \Psi_j(\theta, \beta), F(\theta) \rangle = 0. \tag{A.71}$$

On the ambient open pivot set, and therefore almost everywhere on every $D_{j,n}$,

$$|\det D\Psi_j(\theta, \beta)| = |\partial_\theta T_j(\theta, \beta)|. \tag{A.72}$$

For $N = 1$, this is the determinant of the 1×1 derivative matrix.

Proof. Compactness of Θ and primitive C^1 regularity give finite proof-local quantities

$$M_i := \sup_{\theta \in \Theta} |F_i(\theta)|, \qquad L_i := \sup_{\theta \in \Theta} |F_i'(\theta)|, \qquad 0 \leq i \leq N. \tag{A.73}$$

If $K_{j,n}$ is empty, every Lipschitz assertion is vacuous. Otherwise take $\theta, \vartheta \in K_{j,n}$. For every $i \in \{0, \dots, N\} \setminus \{j\}$, direct quotient algebra, the two endpoint denominator bounds, and the mean-value bound on the full interval Θ give

$$\begin{aligned} \left| \frac{F_i(\theta)}{F_j(\theta)} - \frac{F_i(\vartheta)}{F_j(\vartheta)} \right| &\leq n|F_i(\theta) - F_i(\vartheta)| + n^2 M_i |F_j(\theta) - F_j(\vartheta)| \\ &\leq (nL_i + n^2 M_i L_j)|\theta - \vartheta|. \end{aligned} \tag{A.74}$$

Also

$$\left| \frac{F_i(\theta)}{F_j(\theta)} \right| \leq n M_i. \tag{A.75}$$

No segment is required to stay in $K_{j,n}$; only its two endpoints use the pivot margin, while the mean-value estimate is taken on the ambient interval. The calculation therefore remains valid across different connected components and if F_j has opposite signs on those components.

For $\beta, \gamma \in [-R, R]^{N-1}$, split the difference first in θ and then in beta. These bounds and Cauchy–Schwarz yield

$$|T_j(\theta, \beta) - T_j(\vartheta, \gamma)| \leq C_{\theta,j,n}|\theta - \vartheta| + C_{\beta,j,n}\|\beta - \gamma\|_2, \tag{A.76}$$

where

$$C_{\theta,j,n} := nL_0 + n^2 M_0 L_j + R \sum_{\substack{1 \leq i \leq N \\ i \neq j}} (nL_i + n^2 M_i L_j), \tag{A.77}$$

$$C_{\beta,j,n} := n \left(\sum_{\substack{1 \le i \le N \\ i \ne j}} M_i^2 \right)^{1/2}. \tag{A.78}$$

These are finite explicit proof-level constants, not theorem-facing margins. The resulting bound proves that T_j is Lipschitz for the Euclidean product metric. Since Ψ_j inserts T_j and copies every beta coordinate, Ψ_j is Lipschitz on the same set.

The root identity is exact:

$$\begin{aligned} F_0(\theta) + \langle \Psi_j(\theta,\beta), F(\theta)\rangle &= F_0(\theta) + T_j(\theta,\beta)F_j(\theta) + \sum_{i\ne j} \beta_i F_i(\theta) \\ &= 0. \end{aligned} \tag{A.79}$$

Thus Ψ_j outputs the original coefficient vector, not a transformed or augmented coordinate.

Order the nonpivot indices as $i_1, \dots, i_{N-1}$, use input coordinates $(\theta, \beta_{i_1}, \dots, \beta_{i_{N-1}})$, and temporarily permute output rows into the order $(j, i_1, \dots, i_{N-1})$. The derivative matrix is

$$\begin{pmatrix} \partial_\theta T_j & \partial_{\beta_{i_1}} T_j & \cdots & \partial_{\beta_{i_{N-1}}} T_j \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}. \tag{A.80}$$

Its determinant is $\partial_\theta T_j$. Undoing the row permutation changes only the sign, so the absolute determinant is exactly $|\partial_\theta T_j|$. The formula is C^1 on Ω_j; derivative locality therefore supplies this ambient derivative almost everywhere on the measurable restriction $D_{j,n}$. When $N = 1$, there is no row permutation or beta coordinate, and the matrix is $(\partial_\theta T_1)$. $\square$

Proposition A.4 (Multiplicity-safe finite-level area and joint-density bound)**.** *Under Assumption 4, Lemmas A.6 and A.7, and the restated equal-dimensional Euclidean area formula, define*

$$A_{j,n} := \Psi_j(D_{j,n}), \qquad A_n := \bigcup_{j=1}^N A_{j,n}. \tag{A.81}$$

For every $\mu \in \mathcal{D}_{N,R,\kappa}$, every interval $I \subseteq \Theta$ with $|I| > 0$, and every measurable legal pivot partition fixed as in Lemma A.6, every $A_{j,n}$ and A_n is completed-Lebesgue measurable,

$$A_n \subseteq [-R,R]^N, \qquad A_n \subseteq A_{n+1}, \tag{A.82}$$

and

$$\mu(A_n) \le \kappa \sum_{j=1}^N \int_{E_{j,n}} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j(\theta,\beta)| \le R\} |\partial_\theta T_j(\theta,\beta)| \, d\beta \, d\theta. \tag{A.83}$$

The conclusion retains zero-Jacobian charts, tangent and multiple roots, finite and infinite fibers, included interval endpoints, every closed coefficient-cube face, and $N = 1$.

Proof. Fix j, n. The domain $D_{j,n}$ is completed measurable by Lemma A.6, while Lemma A.7 gives a Lipschitz restriction of the ambient C^1 chart and its exact determinant. Applying Theorem A.1 gives

$$\begin{aligned} \lambda_N(A_{j,n}) &\le \int_{D_{j,n}} |\det D\Psi_j(\theta,\beta)| \, d\theta \, d\beta \\ &= \int_{E_{j,n}} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j(\theta,\beta)| \le R\} |\partial_\theta T_j(\theta,\beta)| \, d\beta \, d\theta. \end{aligned} \tag{A.84}$$

Tonelli applies because the integrand is nonnegative and measurable, including if the integral is infinite.

For completeness, completed image measurability is not being assumed. Choose a Borel representative of $D_{j,n}$ modulo a null set and intersect that representative with

$$K_{j,n} \times [-R, R]^{N-1} \subseteq \Omega_j. \tag{A.85}$$

This does not change it modulo a null set because $D_{j,n}$ is already contained in that finite-pivot product. The map is continuous and Lipschitz there, so the image of the Borel part is analytic and Lebesgue measurable. The finite-level Lipschitz map sends the null symmetric difference to a null set by the equal-dimensional area inequality. Hence $A_{j,n}$, and then the finite union A_n, is measurable in the completed λ_N-space and in the completion of μ.

The exact multiplicity identity is

$$\int_{D_{j,n}} |\partial_\theta T_j| = \int_{\mathbb{R}^N} N(\Psi_j, D_{j,n}, a)\, da. \tag{A.86}$$

Every $a \in A_{j,n}$ has multiplicity at least one. Several roots or several parameter-beta preimages are all counted. If the fiber is infinite, the multiplicity is $+\infty$; the pointwise inequality

$$\mathbf{1}_{A_{j,n}}(a) \leq N(\Psi_j, D_{j,n}, a) \tag{A.87}$$

still has the same direction. Thus no injectivity, finite-root count, or simple-root condition enters the volume estimate.

Differentiate the exact chart identity from Lemma A.7 at fixed beta:

$$F_0'(\theta) + \langle \Psi_j(\theta, \beta), F'(\theta)\rangle + F_j(\theta)\partial_\theta T_j(\theta, \beta) = 0. \tag{A.88}$$

On $D_{j,n}$, $F_j \neq 0$. Therefore a tangent root, or any multiple root whose fixed-coefficient first derivative vanishes, is a critical preimage with $\partial_\theta T_j = 0$. For the measurable critical set

$$C_{j,n} := \{(\theta, \beta) \in D_{j,n} : \partial_\theta T_j(\theta, \beta) = 0\}, \tag{A.89}$$

the same area formula gives

$$\lambda_N(\Psi_j(C_{j,n})) \leq \int_{C_{j,n}} 0 = 0. \tag{A.90}$$

Critical roots remain in the image; they are controlled by the area formula rather than excluded by a transversality assumption. An identically zero chart Jacobian is the same zero-integral case.

If an endpoint θ_0 belongs to I, it belongs literally to its partition cell and, when active, to the corresponding finite-level domain. The slice $\{\theta_0\} \times [-R, R]^{N-1}$ has N-dimensional measure zero, so its Lipschitz image is null by the same area formula. Thus endpoint roots are included and controlled; an excluded endpoint never belongs to the event. The beta cube and the constraint $|T_j| \leq R$ use weak inequalities, so every cube face and corner is included.

Every output has nonpivot coordinates in $[-R, R]$ and pivot coordinate in $[-R, R]$, hence $A_n \subseteq [-R, R]^N$. Domain nesting gives image nesting. Since $f_\mu \leq \kappa$ almost everywhere under the one actual N-dimensional joint law,

$$\begin{aligned} \mu(A_n) &= \int_{A_n} f_\mu(a)\, da \\ &\leq \kappa \lambda_N(A_n) \\ &\leq \kappa \sum_{j=1}^{N} \lambda_N(A_{j,n}). \end{aligned} \tag{A.91}$$

Substituting the chart area bounds proves the asserted inequality. Nothing factors f_μ, conditions on a coordinate, or replaces it by a marginal density. For $N = 1$, the beta integral is integration over the one-point 0-dimensional cube with mass

$$\lambda_0([-R, R]^0) = 1 = (2R)^0. \tag{A.92}$$

□

Lemma A.8 (Complete root coverage and persistent-root removal)**.** *Under the Anchor nonvanishing Lemma A.1, the Borel and Lebesgue nullity Lemma A.5, the Nullity under arbitrary full joint laws Proposition A.3, and Lemma A.6, for every admissible law, positive-length interval, and measurable legal pivot partition,*

$$\mathcal{R}_I \subseteq Z_\infty(I) \cup \bigcup_{n=1}^{\infty} A_n, \qquad \mu(Z_\infty(I)) = 0. \tag{A.93}$$

Every nonpersistent root coefficient enters an actual finite chart in the original coefficient cube. The statement includes endpoint and cube-boundary roots, tangent and multiple roots, nonpersistent coefficients with finite or infinite root sets, and persistent or identically-zero affine combinations.

Proof. Lemma A.5 gives Borel measurability of the exact persistent locus, and the probability-zero statement is exactly Proposition A.3, both instantiated with the same F_0, F, I, μ and original coefficient space. These two results are the only ones used to remove persistent or identically-zero affine combinations; they are not assumed absent and no independent continuation or root theorem is invoked.

Take $a \in \mathcal{R}_I \setminus Z_\infty(I)$. By the definition of $\mathcal{R}_I$, choose one root $\theta \in I$. The exact partition has a unique j with $\theta \in E_j$, and legality gives $F_j(\theta) \neq 0$. Let $\beta = a_{-j}$ be the list of the original nonpivot coordinates. Since $a \in [-R, R]^N$,

$$\beta \in [-R, R]^{N-1}. \tag{A.94}$$

Solving the actual affine equation in the actual j-th coefficient gives

$$\begin{aligned} a_j &= -\frac{F_0(\theta)}{F_j(\theta)} - \sum_{i \neq j} a_i \frac{F_i(\theta)}{F_j(\theta)} \\ &= T_j(\theta, \beta). \end{aligned} \tag{A.95}$$

Thus $|T_j(\theta, \beta)| = |a_j| \leq R$. Choose an integer

$$n \geq \frac{1}{|F_j(\theta)|}. \tag{A.96}$$

Then $\theta \in E_{j,n}$, $(\theta, \beta) \in D_{j,n}$, and the exact original-coordinate insertion gives

$$\Psi_j(\theta, \beta) = a. \tag{A.97}$$

Therefore $a \in A_{j,n} \subseteq A_n$, proving the inclusion.

This argument selects only one root and never differentiates it. It applies unchanged to tangent or multiple roots and to a nonpersistent coefficient with infinitely many roots. Weak cube inequalities retain faces and corners. An included endpoint is a literal member of I and of its partition cell; an excluded endpoint is not part of the event. A pivot may be arbitrarily small, because the chosen root needs only one finite coefficient-dependent level. For $N = 1$, beta is the empty tuple and the same equation reads $a_1 = T_1(\theta)$. Persistent and identically-zero cases remain present in $Z_\infty(I)$ and are removed only by Proposition A.3. □

Proposition A.5 (Exhausted affine pivot-sweep inequality)**.** *Under Assumptions 1, 2, 3, and 4, the Anchor nonvanishing Lemma A.1, the Borel and Lebesgue nullity Lemma A.5, the Nullity under arbitrary full joint laws Proposition A.3, Proposition A.4, and Lemma A.8, for every $\mu \in \mathcal{D}_{N,R,\kappa}$, every interval $I \subseteq \Theta$ with $|I| > 0$, and every Lebesgue-measurable legal partition*

$$I = \bigsqcup_{j=1}^{N} E_j, \qquad F_j \neq 0 \text{ on } E_j, \tag{A.98}$$

one has, in the extended nonnegative reals,

$$\begin{aligned}\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] &\leq \kappa \sum_{j=1}^{N} \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j(\theta,\beta)| \leq R\} |\partial_\theta T_j(\theta,\beta)| \, d\beta \, d\theta \\ &\leq \kappa \sum_{j=1}^{N} \int_{E_j} \int_{[-R,R]^{N-1}} |\partial_\theta T_j(\theta,\beta)| \, d\beta \, d\theta.\end{aligned} \tag{A.99}$$

The constants are literal. The conclusion includes $N = 1$, empty cells, zero Jacobians, pivots without a uniform margin, every endpoint convention, all root multiplicities and fiber cardinalities, and identically-zero affine combinations.

Proof. Lemma A.8, completed-measure subadditivity, and the persistent-root nullity give

$$\begin{aligned}\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] &= \mu(\mathcal{R}_I) \\ &\leq \mu(Z_\infty(I)) + \mu\left(\bigcup_{n=1}^{\infty} A_n\right) \\ &= \mu\left(\bigcup_{n=1}^{\infty} A_n\right).\end{aligned} \tag{A.100}$$

The first equality uses only the primitive support statement $\mu([-R,R]^N) = 1$, since $\mathcal{R}_I$ is the full root event restricted to that support cube. The measurable sets A_n increase by Proposition A.4. Continuity from below and the finite-level bound yield

$$\begin{aligned}\mu\left(\bigcup_{n=1}^{\infty} A_n\right) &= \lim_{n\to\infty} \mu(A_n) \\ &\leq \kappa \lim_{n\to\infty} \sum_{j=1}^{N} \int_{E_{j,n}} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j| \leq R\} |\partial_\theta T_j| \, d\beta \, d\theta.\end{aligned} \tag{A.101}$$

For fixed j, extend the relevant nonnegative chart integrand by zero outside E_j. By Lemma A.6, $\mathbf{1}_{E_{j,n}} \uparrow \mathbf{1}_{E_j}$. Monotone convergence, without any integrability assumption, gives

$$\begin{aligned}&\lim_{n\to\infty} \int_{E_{j,n}} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j| \leq R\} |\partial_\theta T_j| \, d\beta \, d\theta \\ &\quad = \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j| \leq R\} |\partial_\theta T_j| \, d\beta \, d\theta.\end{aligned} \tag{A.102}$$

There are exactly $N < \infty$ chart terms in the proposition, so direct finite addition commutes with the limit and introduces no extra factor. This proves the indicator inequality, even if one or both sides are $+\infty$.

Finally, pointwise on every legal chart domain,

$$0 \le \mathbf{1}\{|T_j(\theta,\beta)| \le R\}|\partial_\theta T_j(\theta,\beta)| \le |\partial_\theta T_j(\theta,\beta)|. \tag{A.103}$$

Tonelli and finite addition give the indicator-dropped inequality with coefficient one. No term other than the indicator is dropped.

For $N = 1$, the inner measure is $\lambda_0([-R,R]^0) = 1$, so this is the exact one-dimensional area inequality. In every dimension,

$$\lambda_{N-1}([-R,R]^{N-1}) = (2R)^{N-1}; \tag{A.104}$$

the chart theorem leaves that beta integral intact, so this literal factor is available whenever a later application bounds an integrand uniformly, without this step asserting any later specialization. Empty cells give zero integrals, while divergent mass near a vanishing pivot is allowed by the extended-real statement.

□

Synthesis of the measurable pivot sweep. Fix $\mu \in \mathcal{D}_{N,R,\kappa}$, a positive-length interval $I \subseteq \Theta$ with its chosen endpoint convention, and a measurable legal pivot partition. Lemma A.6 makes the root event and finite chart domains measurable and gives $E_{j,n} \uparrow E_j$. Lemma A.7 supplies the original-coordinate root identity and

$$|\det D\Psi_j| = |\partial_\theta T_j| \tag{A.105}$$

on each finite domain. Proposition A.4 applies the equal-dimensional area formula with full multiplicity and the single joint-density bound $f_\mu \le \kappa$. Lemma A.8 then covers every nonpersistent root and removes persistent coefficients by the zero-probability result of Proposition A.3.

Continuity from below for the coefficient images and monotone convergence for the nonnegative Jacobian masses, as carried out in Proposition A.5, yield

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] \le \kappa \sum_{j=1}^{N} \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j| \le R\}|\partial_\theta T_j|\, d\beta\, d\theta. \tag{A.106}$$

The pointwise inequality

$$\mathbf{1}\{|T_j| \le R\}|\partial_\theta T_j| \le |\partial_\theta T_j| \tag{A.107}$$

gives the indicator-dropped form with coefficient one. The same argument covers tangent and multiple roots, infinite fibers, zero Jacobians, endpoints, empty cells, and $N = 1$, without a pivot margin or an independence assumption. □

A.4 Coordinate-Free Section Parametrization

Theorem A.2 (Injective affine Euclidean area formula)**.** *Let $n \ge k \ge 0$ be integers, let $L : \mathbb{R}^k \to \mathbb{R}^n$ be an injective linear map, let $b \in \mathbb{R}^n$, and define $\Xi(x) = b + Lx$. For every Lebesgue-measurable $D \subseteq \mathbb{R}^k$ and every nonnegative Borel function h on $\Xi(D)$,*

$$\int_{\Xi(D)} h(y)\, d\mathcal{H}^k(y) = \sqrt{\det(L^\top L)} \int_D h(\Xi(x))\, d\lambda_k(x). \tag{A.108}$$

Both sides are interpreted in $[0,\infty]$. When $k = 0$, the empty Gram determinant is one, $\mathbb{R}^0 = \{()\}$, $\lambda_0(\{()\}) = 1$, and $\mathcal{H}^0$ is counting measure. This is the injective affine specialization of the Euclidean area formula (Federer, 1969, Section 3.2.3).

Lemma A.9 (Exact fixed-section parametrization and graph Jacobian)**.** *Under Assumption 3, fix $\theta \in \Theta$ and $j \in \{1, \dots, N\}$ such that $F_j(\theta) \neq 0$. Define the exact fixed-section beta domain*

$$\mathcal{B}_{j,\theta} := \{\beta \in [-R, R]^{N-1} : |T_j(\theta, \beta)| \leq R\}. \tag{A.109}$$

Then $\mathcal{B}_{j,\theta}$ is Borel and

$$\Psi_j(\theta, \cdot) : \mathcal{B}_{j,\theta} \longrightarrow H_\theta \cap [-R, R]^N \tag{A.110}$$

is a bijection in the original coefficient coordinates. Its Euclidean $(N-1)$-dimensional Jacobian is constant in beta and equals

$$J_{N-1}\Psi_j(\theta, \cdot) = \frac{\|F(\theta)\|_2}{|F_j(\theta)|}. \tag{A.111}$$

Consequently, for every nonnegative Borel function h on the section,

$$\int_{H_\theta \cap [-R,R]^N} h(a)\, d\mathcal{H}^{N-1}(a) = \frac{\|F(\theta)\|_2}{|F_j(\theta)|} \int_{\mathcal{B}_{j,\theta}} h(\Psi_j(\theta, \beta))\, d\beta. \tag{A.112}$$

The assertions remain literal for an empty section and for $N = 1$, where $\mathcal{H}^0$ counts the unique point of a nonempty section.

Proof. At fixed θ, $T_j(\theta, \cdot)$ is affine and hence continuous. Therefore $\mathcal{B}_{j,\theta}$ is the intersection of the closed beta cube with the inverse image of the closed interval $[-R, R]$, so it is closed and Borel.

For every $\beta \in \mathcal{B}_{j,\theta}$, the setting-defined chart uses the original coefficient coordinates:

$$(\Psi_j(\theta, \beta))_j = T_j(\theta, \beta), \qquad (\Psi_j(\theta, \beta))_i = \beta_i \quad (i \neq j). \tag{A.113}$$

All nonpivot coordinates lie in $[-R, R]$, and the defining weak inequality for $\mathcal{B}_{j,\theta}$ puts the pivot coordinate in the same closed interval. Direct substitution into the affine equation gives

$$\begin{aligned} F_0(\theta) + \langle \Psi_j(\theta, \beta), F(\theta) \rangle &= F_0(\theta) + F_j(\theta) T_j(\theta, \beta) + \sum_{i \neq j} \beta_i F_i(\theta) \\ &= 0. \end{aligned} \tag{A.114}$$

Thus the chart image is contained in $H_\theta \cap [-R, R]^N$.

Conversely, take $a \in H_\theta \cap [-R, R]^N$ and set $\beta_i = a_i$ for $i \neq j$. Since the pivot is nonzero, the section equation has exactly one solution in the pivot coordinate:

$$a_j = -\frac{F_0(\theta)}{F_j(\theta)} - \sum_{i \neq j} a_i \frac{F_i(\theta)}{F_j(\theta)} = T_j(\theta, \beta). \tag{A.115}$$

The cube membership of a implies $\beta \in \mathcal{B}_{j,\theta}$, and then $a = \Psi_j(\theta, \beta)$. This proves surjectivity. Because the map copies all nonpivot coordinates, two equal images have equal beta tuples, proving injectivity. No coefficient transformation or extra random coordinate has been introduced; F_0 remains the same deterministic offset.

Assume first $N \geq 2$, and order the nonpivot indices as $i_1, \dots, i_{N-1}$. Differentiating only with respect to beta gives the columns

$$c_i := \partial_{\beta_i} \Psi_j(\theta, \beta) = e_i - \frac{F_i(\theta)}{F_j(\theta)} e_j, \qquad i \neq j. \tag{A.116}$$

Let $v \in \mathbb{R}^{N-1}$ have coordinates $v_i = F_i(\theta)/F_j(\theta)$ for $i \neq j$. The Gram matrix of the derivative columns is

$$G = (\langle c_i, c_k \rangle)_{i,k \neq j} = I_{N-1} + vv^\mathsf{T}. \tag{A.117}$$

If $v = 0$, $G = I_{N-1}$. If $v \neq 0$, G acts as the identity on $v^\perp$ and sends v to $(1 + \|v\|_2^2)v$. Hence in both cases

$$\det G = 1 + \|v\|_2^2 = 1 + \sum_{i \neq j} \frac{F_i(\theta)^2}{F_j(\theta)^2} = \frac{\|F(\theta)\|_2^2}{F_j(\theta)^2}. \tag{A.118}$$

The Euclidean graph Jacobian is the nonnegative square root of the Gram determinant, so

$$J_{N-1}\Psi_j(\theta, \cdot) = \sqrt{\det G} = \frac{\|F(\theta)\|_2}{|F_j(\theta)|}. \tag{A.119}$$

This computation explicitly retains either sign of the pivot. Apply Theorem A.2 with

$$k = N-1, \qquad n = N, \qquad D = \mathcal{B}_{j,\theta}, \qquad \Xi = \Psi_j(\theta, \cdot).$$

The domain D is Borel, hence Lebesgue measurable, by the first paragraph of this proof. The linear part of Ξ has the columns c_i in (A.116); it is injective because the chart copies all nonpivot coordinates. In fact, (A.110) and the preceding two-sided argument prove that Ξ is a bijection onto the exact affine section $H_\theta \cap [-R, R]^N$. Its Gram matrix is G, and (A.118)–(A.119) give

$$\det(L^\mathsf{T} L) = \frac{\|F(\theta)\|_2^2}{F_j(\theta)^2}, \qquad \sqrt{\det(L^\mathsf{T} L)} = \frac{\|F(\theta)\|_2}{|F_j(\theta)|}.$$

Thus, for every nonnegative Borel integrand h on the section, the theorem applies in the extended nonnegative reals and yields (A.112). The square-root Gram Jacobian introduces no orientation sign, injectivity introduces no multiplicity factor, and no boundary deletion occurs: the measurable domain retains all cube faces and corners because both its cube and pivot constraints are weak. When $N = 1$, equivalently $k = 0$, the theorem's empty-Gram and counting-measure conventions give the same identity. Because the parametrization is affine, its graph Jacobian is constant in beta.

If $N = 1$, then $j = 1$, the beta space is $\mathbb{R}^0 = \{()\}$, and the empty Gram determinant is one. Moreover

$$\frac{\|F(\theta)\|_2}{|F_1(\theta)|} = 1. \tag{A.120}$$

The hyperplane $H_\theta \subseteq \mathbb{R}$ is the singleton $\{-F_0(\theta)/F_1(\theta)\}$. It meets the cube exactly when $|T_1(\theta, ())| \leq R$. In the nonempty case both λ_0 on the beta point and $\mathcal{H}^0$ on the unique section point have mass one; in the empty case both integrals are zero. This also proves the asserted dimension-zero formula. Cube faces and corners are retained throughout because every cube inequality is weak. □

Lemma A.10 (Exact normal-density identity and pivot cancellation)**.** *Under Assumption 3 and Lemma A.9, fix $\theta \in \Theta$ and a legal pivot $F_j(\theta) \neq 0$. For every $\beta \in \mathbb{R}^{N-1}$, fixed-beta differentiation of the same chart root identity gives*

$$\partial_\theta T_j(\theta, \beta) = -\frac{F_0'(\theta) + \langle \Psi_j(\theta, \beta), F'(\theta) \rangle}{F_j(\theta)}. \tag{A.121}$$

On $\mathcal{B}_{j,\theta}$, the following is an equality of nonnegative measures under the bijection $a = \Psi_j(\theta, \beta)$:

$$|\partial_\theta T_j(\theta, \beta)|\, d\beta = \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a). \tag{A.122}$$

Precisely, for every nonnegative Borel function h on the section,

$$\begin{aligned}\int_{\mathcal{B}_{j,\theta}} & h(\Psi_j(\theta,\beta))|\partial_\theta T_j(\theta,\beta)|\, d\beta \\ &= \int_{H_\theta \cap [-R,R]^N} h(a) \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a).\end{aligned} \tag{A.123}$$

The identities include either pivot sign, a zero numerator, zero chart velocity, an empty section, cube faces and corners, and $N = 1$.

Proof. The setting-defined chart identity, already verified directly in Lemma A.9, is

$$F_0(\theta) + F_j(\theta) T_j(\theta, \beta) + \sum_{i \neq j} \beta_i F_i(\theta) = 0. \tag{A.124}$$

Hold beta fixed and differentiate with respect to θ on the open set where $F_j \neq 0$. The setting-defined feature functions and the quotient chart are continuously differentiable there. The product rule gives

$$F_0'(\theta) + F_j'(\theta) T_j(\theta, \beta) + F_j(\theta) \partial_\theta T_j(\theta, \beta) + \sum_{i \neq j} \beta_i F_i'(\theta) = 0. \tag{A.125}$$

Because Ψ_j inserts T_j in the original j-th coordinate and beta in every other original coordinate,

$$F_j'(\theta) T_j(\theta, \beta) + \sum_{i \neq j} \beta_i F_i'(\theta) = \langle \Psi_j(\theta, \beta), F'(\theta)\rangle. \tag{A.126}$$

Dividing by the legal nonzero pivot yields exactly

$$\partial_\theta T_j(\theta, \beta) = -\frac{F_0'(\theta) + \langle \Psi_j(\theta,\beta), F'(\theta)\rangle}{F_j(\theta)}. \tag{A.127}$$

This is the derivative of the same algebraic chart root identity, not a root theorem and not an implicit-root argument. No root is selected as a function of theta, and no simple-root or transversality condition appears.

Taking absolute values retains both pivot signs:

$$|\partial_\theta T_j(\theta, \beta)| = \frac{|F_0'(\theta) + \langle \Psi_j(\theta,\beta), F'(\theta)\rangle|}{|F_j(\theta)|}. \tag{A.128}$$

Lemma A.9 gives, under the same bijection,

$$d\mathcal{H}^{N-1}(a) = \frac{\|F(\theta)\|_2}{|F_j(\theta)|}\, d\beta, \qquad d\beta = \frac{|F_j(\theta)|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a). \tag{A.129}$$

Multiplying these two literal identities cancels the same absolute pivot factor exactly and gives

$$|\partial_\theta T_j|\, d\beta = \frac{|F_0' + \langle a, F'\rangle|}{\|F\|_2}\, d\mathcal{H}^{N-1}(a). \tag{A.130}$$

Applying the affine change-of-variables formula from Lemma A.9 with the nonnegative test function

$$h(a)\frac{|F_0'(\theta)+\langle a,F'(\theta)\rangle|}{\|F(\theta)\|_2} \tag{A.131}$$

makes this differential notation a rigorous integral equality. In particular, taking $h \equiv 1$ gives the exact fixed-section identity used in Lemma A.11.

If the numerator is zero at a point, the signed derivative identity makes $\partial_\theta T_j = 0$ at its unique beta preimage, so both densities vanish. If the whole chart velocity vanishes, both measures are zero. If the section is empty, both measures are supported on the empty set. Touching cube faces or corners does not change the affine area formula. For $N = 1$, the Hausdorff Jacobian is one, $\|F\|_2 = |F_1|$, and $\mathcal{H}^0$ evaluates the right-hand density at the unique point of a nonempty section; hence the same formulas remain literal. $\square$

Lemma A.11 (Measurable section mass and exact partition summation)**.** *Under Assumption 3 and Lemma A.10, fix an interval $I \subseteq \Theta$ with $|I| > 0$, retaining its literal endpoint convention, and a Lebesgue-measurable legal partition*

$$I = \bigsqcup_{j=1}^{N} E_j. \tag{A.132}$$

For $\theta \in E_j$, define the chart mass

$$\mathcal{W}_j(\theta) := \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j(\theta,\beta)| \le R\}|\partial_\theta T_j(\theta,\beta)|\,d\beta, \tag{A.133}$$

and define the coordinate-free section mass

$$\mathcal{V}(\theta) := \int_{H_\theta \cap [-R,R]^N} \frac{|F_0'(\theta)+\langle a,F'(\theta)\rangle|}{\|F(\theta)\|_2}\,d\mathcal{H}^{N-1}(a). \tag{A.134}$$

Then the cellwise functions are Lebesgue measurable, $\mathcal{V}$ is Lebesgue measurable on I, and

$$\mathcal{W}_j(\theta) = \mathcal{V}(\theta) \qquad (\theta \in E_j). \tag{A.135}$$

Consequently, in $[0,\infty]$,

$$\sum_{j=1}^{N} \int_{E_j} \mathcal{W}_j(\theta)\,d\theta = \int_I \mathcal{V}(\theta)\,d\theta. \tag{A.136}$$

This equality holds for every measurable legal partition, including disconnected or empty cells, and its right-hand side is independent of the legal pivot assignment.

Proof. For each j, let

$$\Omega_j := \{\theta \in \Theta : F_j(\theta) \ne 0\}. \tag{A.137}$$

The setting-defined feature functions are continuous, so Ω_j is relatively open. On $\Omega_j \times \mathbb{R}^{N-1}$, the quotient formula for T_j and its fixed-beta theta derivative are continuous. Define on the entire product $\Theta \times [-R,R]^{N-1}$

$$g_j(\theta,\beta) := \begin{cases} \mathbf{1}\{|T_j(\theta,\beta)| \le R\}|\partial_\theta T_j(\theta,\beta)|, & \theta \in \Omega_j, \\ 0, & \theta \notin \Omega_j. \end{cases} \tag{A.138}$$

This piecewise function is Borel and nonnegative: it is a Borel function on the Borel open set $\Omega_j \times [-R, R]^{N-1}$ and is zero on its Borel complement. Tonelli's measurability conclusion gives the Borel function

$$\widehat{\mathcal{W}}_j(\theta) := \int_{[-R,R]^{N-1}} g_j(\theta, \beta)\, d\beta \quad (\theta \in \Theta), \tag{A.139}$$

with values in $[0, \infty]$. Since legality gives $E_j \subseteq \Omega_j$, the restriction of $\widehat{\mathcal{W}}_j$ to E_j is exactly $\mathcal{W}_j$. It is therefore Lebesgue measurable even when E_j is not Borel.

For each $\theta \in E_j$, the beta points selected by the indicator are exactly $\mathcal{B}_{j,\theta}$. Taking $h \equiv 1$ in Lemma A.10 gives the pointwise identity

$$\mathcal{W}_j(\theta) = \mathcal{V}(\theta). \tag{A.140}$$

Thus, pointwise on I,

$$\mathcal{V}(\theta) = \sum_{j=1}^{N} \mathbf{1}_{E_j}(\theta) \widehat{\mathcal{W}}_j(\theta). \tag{A.141}$$

The right-hand side is a finite sum of nonnegative Lebesgue-measurable functions, so this identity proves measurability of the moving-section mass $\mathcal{V}$ without invoking a separate moving-hyperplane, coarea, or root theorem.

Finite additivity for the exact disjoint partition now yields

$$\begin{aligned} \sum_{j=1}^{N} \int_{E_j} \mathcal{W}_j(\theta)\, d\theta &= \sum_{j=1}^{N} \int_{E_j} \mathcal{V}(\theta)\, d\theta \\ &= \int_I \mathcal{V}(\theta)\, d\theta. \end{aligned} \tag{A.142}$$

Every term is nonnegative, so the calculation is valid if the common value is $+\infty$; no expression of the form $\infty - \infty$ occurs. The cells are disjoint, so different legal charts never double charge a single theta in the sum. Although two pivot coordinates may both be legal at the same theta, the exact fixed-section equality shows that either chosen chart represents the same $\mathcal{V}(\theta)$. Empty cells contribute zero. Included interval endpoints remain literal members of their cells and excluded endpoints do not enter; no closure of I is substituted.

For $N = 1$, the sole beta integral is integration over $\mathbb{R}^0$ with mass one, the partition has the single legal cell $E_1 = I$, and the pointwise equality is exactly the $\mathcal{H}^0$ evaluation established in Lemma A.10. □

Proposition A.6 (Coordinate-free first affine swept-area inequality)**.** *Under Assumption 3, the Exhausted affine pivot-sweep Proposition A.5, and Lemma A.11, for every $\mu \in \mathcal{D}_{N,R,\kappa}$ and every interval $I \subseteq \Theta$ with $|I| > 0$, retaining the interval's literal endpoint convention,*

$$\Pr_{\alpha \sim \mu}[\exists \theta \in I : \phi_\alpha(\theta) = 0] \le \kappa \int_I \int_{H_\theta \cap [-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta) \rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta. \tag{A.143}$$

The inequality is in $[0, \infty]$, keeps the original coefficient coordinates and deterministic offset, and has the literal factor κ. It applies to every arbitrary correlated full joint law in $\mathcal{D}_{N,R,\kappa}$, and includes all endpoint, root-multiplicity, cube-boundary, empty-section, zero-velocity, and $N = 1$ cases covered by its stated inputs.

Proof. Fix the deterministic setting instance. Then fix an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$ and, after the law, an arbitrary interval $I \subseteq \Theta$ with $|I| > 0$ and its literal endpoint convention.

At least one measurable legal partition exists directly from the primitive anchor: take

$$E_{j_*} = I, \qquad E_j = \varnothing \quad (j \neq j_*), \tag{A.144}$$

because $F_{j_*} \equiv 1$. More generally, fix any measurable legal partition. The indicator form of the Proposition A.5 gives, without any new event or root argument,

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] \leq \kappa \sum_{j=1}^{N} \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j(\theta,\beta)| \leq R\} |\partial_\theta T_j(\theta,\beta)|\, d\beta\, d\theta. \tag{A.145}$$

Lemma A.11 identifies the entire chart sum, exactly and in the extended nonnegative reals, with

$$\int_I \int_{H_\theta\cap[-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta. \tag{A.146}$$

Substitution proves the claimed inequality with no remainder, pivot factor, chart-count factor, or change to κ. Proposition A.5 already treats the original full joint density, so the deterministic fixed-section conversion neither conditions on nor marginalizes a coordinate and preserves arbitrary correlation. It also deletes no coefficient, parameter endpoint, root, cube face, or corner. Tangent, multiple, and persistent roots remain covered exactly as in that proposition; this proof has invoked no second root theorem. Because μ and then I were arbitrary, the quantifier order is preserved. □

Section-formula synthesis. Fix the deterministic instance, then a law, a positive-length interval, and a measurable legal pivot partition. For every $\theta \in E_j$, Lemma A.9 gives a bijection from

$$\{\beta \in [-R,R]^{N-1} : |T_j(\theta,\beta)| \leq R\} \tag{A.147}$$

onto $H_\theta \cap [-R,R]^N$ with graph Jacobian $\|F(\theta)\|_2/|F_j(\theta)|$. On the same chart, Lemma A.10 gives

$$\partial_\theta T_j(\theta,\beta) = -\frac{F_0'(\theta) + \langle \Psi_j(\theta,\beta), F'(\theta)\rangle}{F_j(\theta)}. \tag{A.148}$$

The pivot denominator therefore cancels exactly:

$$|\partial_\theta T_j|\, d\beta = \frac{|F_0' + \langle a, F'\rangle|}{\|F\|_2}\, d\mathcal{H}^{N-1}(a). \tag{A.149}$$

Lemma A.11 proves measurability and sums this equality over the finite partition in the extended nonnegative reals. Substitution into the indicator form of Proposition A.5 is precisely Proposition A.6. It preserves the original coefficient coordinates, the single correlated joint law, literal κ, all endpoint conventions, empty cells, either pivot sign, zero velocity, cube boundaries, and the $N = 1$ convention. □

A.5 Translated Cube Sections

Theorem A.3 (Brunn–Minkowski inequality)**.** *Let $d \geq 1$, let X, Y be nonempty compact subsets of a d-dimensional Euclidean affine space, and let $0 \leq \lambda \leq 1$. Then*

$$\operatorname{vol}_d((1-\lambda)X + \lambda Y)^{1/d} \geq (1-\lambda)\operatorname{vol}_d(X)^{1/d} + \lambda \operatorname{vol}_d(Y)^{1/d}. \tag{A.150}$$

The statement permits either input to have zero d-volume (Gardner, 2002).

Theorem A.4 (Ball's central cube-section theorem)**.** *For $n \geq 2$, let $Q_n = [-1/2, 1/2]^n$. For every Euclidean hyperplane L through the origin,*

$$\mathcal{H}^{n-1}(Q_n \cap L) \leq \sqrt{2}. \tag{A.151}$$

This is the main cube-slicing theorem of Ball (1986).

Lemma A.12 (Central maximality of parallel cube sections)**.** *Under Assumption 1, if $N \geq 2$, $u \in \mathbb{R}^N$ satisfies $\|u\|_2 = 1$, and*

$$v(t) := \mathcal{H}^{N-1}\big([-R, R]^N \cap (u^\perp + tu)\big), \qquad t \in \mathbb{R}, \tag{A.152}$$

then the function $t \mapsto v(t)^{1/(N-1)}$ is concave on the closed interval on which the sections are nonempty, $v(-t) = v(t)$, and

$$v(t) \leq v(0) \qquad \textit{for every } t \in \mathbb{R}. \tag{A.153}$$

Proof. Put $K = [-R, R]^N$, set $d = N - 1$, and identify each affine slice isometrically with a subset of $u^\perp$ by defining

$$A_t := \{x \in u^\perp : x + tu \in K\}. \tag{A.154}$$

The translation $x \mapsto x + tu$ is a Euclidean isometry from A_t onto $K \cap (u^\perp + tu)$. Therefore

$$v(t) = \mathrm{vol}_d(A_t). \tag{A.155}$$

The projection of K onto the line spanned by u is

$$\big\{\langle x, u\rangle : x \in K\big\} = \left[-R\sum_{i=1}^{N} |u_i|,\ R\sum_{i=1}^{N} |u_i|\right]. \tag{A.156}$$

Indeed, the two extrema follow by choosing each cube coordinate with the sign opposite to, or equal to, the sign of u_i, and convexity fills the interval between them. Thus this is exactly the closed interval of nonempty sections; in particular, its endpoints are retained.

Fix s, t in this interval and $0 \leq \lambda \leq 1$. If $x_s \in A_s$ and $x_t \in A_t$, convexity of K gives

$$(1-\lambda)(x_s + su) + \lambda(x_t + tu) = \big((1-\lambda)x_s + \lambda x_t\big) + \big((1-\lambda)s + \lambda t\big)u \in K. \tag{A.157}$$

Because $(1-\lambda)x_s + \lambda x_t \in u^\perp$, this proves

$$(1-\lambda)A_s + \lambda A_t \subseteq A_{(1-\lambda)s+\lambda t}. \tag{A.158}$$

Theorem A.3 and monotonicity of Euclidean volume now yield

$$v((1-\lambda)s + \lambda t)^{1/d} \geq (1-\lambda)v(s)^{1/d} + \lambda v(t)^{1/d}. \tag{A.159}$$

This argument remains valid when an endpoint slice has zero d-volume; Brunn–Minkowski does not require positive input volume. Hence $v^{1/d}$ is concave on the full closed interval of nonempty sections, including tangent slices supported on faces, lower-dimensional faces, edges, or corners.

Central symmetry $K = -K$ gives $A_{-t} = -A_t$, so $v(-t) = v(t)$. Applying concavity to the midpoint $0 = (t + (-t))/2$ gives, for every t in the support interval,

$$v(0)^{1/d} \geq \frac{1}{2}v(t)^{1/d} + \frac{1}{2}v(-t)^{1/d} = v(t)^{1/d}. \tag{A.160}$$

Since $d \geq 1$, raising both sides to the power d gives $v(t) \leq v(0)$. For t outside the support interval the translated section is empty, so $v(t) = 0 \leq v(0)$. This proves the assertion for all real offsets, without extending the concave function by zero outside an endpoint where a facet section may have positive measure. $\square$

Lemma A.13 (Euclidean scaling of Ball's cube-slicing bound)**.** *Under Assumption 1, if $N \geq 2$ and $u \in \mathbb{R}^N$ satisfies $\|u\|_2 = 1$, then*

$$\mathcal{H}^{N-1}\big([-R,R]^N \cap u^\perp\big) \leq \sqrt{2}(2R)^{N-1}. \tag{A.161}$$

Proof. Let $Q_N = [-1/2, 1/2]^N$, which has N-dimensional volume one. Theorem A.4, in the same Euclidean metric, gives

$$\mathcal{H}^{N-1}(Q_N \cap u^\perp) \leq \sqrt{2}. \tag{A.162}$$

Assumption 1 gives $R > 0$, and scalar dilation by $2R$ satisfies the literal set identity

$$(2R)(Q_N \cap u^\perp) = ([-R,R]^N) \cap u^\perp. \tag{A.163}$$

For every subset E of a Euclidean $(N-1)$-plane and every $c > 0$, Hausdorff measure obeys the exact metric scaling identity

$$\mathcal{H}^{N-1}(cE) = c^{N-1}\mathcal{H}^{N-1}(E). \tag{A.164}$$

Taking $c = 2R$ proves

$$\mathcal{H}^{N-1}\big([-R,R]^N \cap u^\perp\big) = (2R)^{N-1}\mathcal{H}^{N-1}(Q_N \cap u^\perp) \leq \sqrt{2}(2R)^{N-1}. \tag{A.165}$$

No normalization factor is hidden: both the source and target use Euclidean Hausdorff measure, and a scalar dilation multiplies every tangent length by exactly $2R$. $\square$

Proposition A.7 (Uniform translated cube-section certificate)**.** *Under Assumption 1 and, when $N \geq 2$, Lemmas A.12 and A.13, for every Euclidean unit vector $u \in \mathbb{R}^N$ and every $t \in \mathbb{R}$,*

$$\mathcal{H}^{N-1}\big([-R,R]^N \cap (u^\perp + tu)\big) \leq \mathcal{H}^{N-1}\big([-R,R]^N \cap u^\perp\big) \leq \sqrt{2}(2R)^{N-1}. \tag{A.166}$$

Moreover, if a setting-defined affine root section has $F(\theta) \neq 0$, then with

$$u_\theta := \frac{F(\theta)}{\|F(\theta)\|_2}, \qquad t_\theta := -\frac{F_0(\theta)}{\|F(\theta)\|_2}, \tag{A.167}$$

one has the exact same-normal identity

$$H_\theta = u_\theta^\perp + t_\theta u_\theta \tag{A.168}$$

and therefore

$$\mathcal{H}^{N-1}\big(H_\theta \cap [-R,R]^N\big) \leq \mathcal{H}^{N-1}\big(F(\theta)^\perp \cap [-R,R]^N\big) \leq \sqrt{2}(2R)^{N-1}. \tag{A.169}$$

Proof. First suppose $N \geq 2$. Lemma A.12 bounds every translated section by its parallel central section, including support endpoints and empty sections. Lemma A.13 bounds that central section uniformly over the Euclidean unit normal u. Their composition proves the asserted two inequalities.

Now suppose $N = 1$. A Euclidean unit normal is $u = 1$ or $u = -1$, and $u^\perp = \{0\}$. Thus $u^\perp + tu = \{tu\}$. Its intersection with $[-R,R]$ is either one point or empty, while the central section is the singleton $\{0\}$. Since $\mathcal{H}^0$ counts a singleton as one and the empty set as zero,

$$\mathcal{H}^0([-R,R]\cap\{tu\}) \le 1 = \mathcal{H}^0([-R,R]\cap\{0\}) \le \sqrt{2}(2R)^0. \tag{A.170}$$

This proves both inequalities literally in dimension one.

It remains to verify that the actual affine section is the same object, rather than a substituted central section. For $F(\theta)\neq 0$, division of

$$F_0(\theta) + \langle a, F(\theta)\rangle = 0 \tag{A.171}$$

by $\|F(\theta)\|_2$ shows that $a \in H_\theta$ exactly when

$$\langle a, u_\theta\rangle = t_\theta. \tag{A.172}$$

Every $a \in \mathbb{R}^N$ has the orthogonal decomposition

$$a = \big(a - \langle a, u_\theta\rangle u_\theta\big) + \langle a, u_\theta\rangle u_\theta, \tag{A.173}$$

so this level set is precisely $u_\theta^\perp + t_\theta u_\theta$. Its parallel central section is precisely $u_\theta^\perp = F(\theta)^\perp$. In the actual branch, the setting anchor gives $F_{j_*}(\theta) = 1$, hence $F(\theta)\neq 0$ at every θ. In particular, when $N=1$, $F(\theta)$ is a nonzero scalar, $H_\theta = \{-F_0(\theta)/F(\theta)\}$, and its intersection with $[-R,R]$ is a singleton or empty, as proved in the preceding dimension-one calculation.

The proof uses no regularity of t_θ as θ varies. A central offset, a noncentral offset, tangency to a cube face, intersection through a face, edge, or corner, a coordinate-aligned normal, a diagonal normal, a support endpoint, and a zero-measure or empty section are all already covered by the quantified geometric argument. □

Synthesis of the translated-section bound. For $N \ge 2$, Lemma A.12, through the stated Brunn–Minkowski theorem, bounds every translated section by the parallel central section. Lemma A.13 applies Ball's central cube-section theorem with the exact Euclidean Hausdorff scaling from the unit-volume cube to $[-R,R]^N$. Proposition A.7 also proves the $N=1$ case and identifies the actual affine section using

$$u_\theta = \frac{F(\theta)}{\|F(\theta)\|_2}, \qquad t_\theta = -\frac{F_0(\theta)}{\|F(\theta)\|_2}. \tag{A.174}$$

Consequently, for every $N \ge 1$, every $R > 0$, and every actual section,

$$\mathcal{H}^{N-1}(H_\theta \cap [-R,R]^N) \le \mathcal{H}^{N-1}(F(\theta)^\perp \cap [-R,R]^N) \le \sqrt{2}(2R)^{N-1}. \tag{A.175}$$

The conclusion includes every orientation and offset, empty sections, support endpoints, cube faces and corners, and the stated zero-dimensional convention. □

A.6 Affine Root-Section Velocity

Lemma A.14 (Root-section Euclidean coupling)**.** *Under Assumptions 1 and 3, if $\theta \in \Theta$ and $a \in H_\theta \cap [-R,R]^N$, then*

$$|F_0(\theta)| = |\langle a, F(\theta)\rangle| \le R\sqrt{N}\,\|F(\theta)\|_2, \tag{A.176}$$

$$\|\widetilde{F}(\theta)\|_2 \le \sqrt{1+NR^2}\,\|F(\theta)\|_2, \tag{A.177}$$

and

$$\|(1,a)\|_2 \le \sqrt{1+NR^2}. \tag{A.178}$$

Proof. Membership in the actual affine root section means, by the setting definition of H_θ,

$$F_0(\theta) + \langle a, F(\theta)\rangle = 0. \tag{A.179}$$

Thus, pointwise for this identical a and θ,

$$F_0(\theta) = -\langle a, F(\theta)\rangle, \qquad |F_0(\theta)| = |\langle a, F(\theta)\rangle|. \tag{A.180}$$

No bound on F_0 away from the root section has been introduced. Cube membership gives, coordinate by coordinate, $|a_i| \le R$, and hence

$$\|a\|_2^2 = \sum_{i=1}^{N} a_i^2 \le NR^2, \qquad \|a\|_2 \le R\sqrt{N}. \tag{A.181}$$

Applying Euclidean Cauchy–Schwarz to the right-hand side of (A.180) and then using (A.181) proves (A.176):

$$|F_0(\theta)| = |\langle a, F(\theta)\rangle| \le \|a\|_2 \|F(\theta)\|_2 \le R\sqrt{N}\, \|F(\theta)\|_2. \tag{A.182}$$

Squaring this derived bound, and using the setting's actual augmented feature vector rather than a surrogate, gives

$$\begin{aligned} \|\widetilde{F}(\theta)\|_2^2 &= |F_0(\theta)|^2 + \|F(\theta)\|_2^2 \\ &\le NR^2 \|F(\theta)\|_2^2 + \|F(\theta)\|_2^2 \\ &= (1 + NR^2)\|F(\theta)\|_2^2. \end{aligned} \tag{A.183}$$

Both sides are nonnegative, so taking square roots proves (A.177). Finally, the same cube calculation gives

$$\|(1, a)\|_2^2 = 1 + \|a\|_2^2 \le 1 + NR^2, \tag{A.184}$$

and taking the nonnegative square root proves (A.178). Every constant in (A.176)–(A.178) has been derived from the actual root equation and cube membership. □

Proposition A.8 (Exact affine root-section velocity)**.** *Under Assumptions 1 and 3, Lemma A.14, and Lemma A.2, if $\theta \in \Theta$ and $a \in H_\theta \cap [-R, R]^N$, then*

$$F_0'(\theta) + \langle a, F'(\theta)\rangle = \langle (1, a), B(\theta)\widetilde{F}(\theta)\rangle, \tag{A.185}$$

and

$$\frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} \le (1 + NR^2)\widehat{\Lambda}_{B,T}. \tag{A.186}$$

More precisely, the coefficient and feature square-root factors in (A.186) multiply exactly as

$$\sqrt{1 + NR^2}\,\sqrt{1 + NR^2} = 1 + NR^2. \tag{A.187}$$

Proof. The vector a is held fixed in this pointwise velocity evaluation; no θ-dependent section parametrization and no derivative of a is introduced. The exact primitive closure identity is

$$\widetilde{F}'(\theta) = \big(F_0'(\theta), F'(\theta)\big) = B(\theta)\widetilde{F}(\theta). \tag{A.188}$$

Taking its Euclidean inner product with the identical augmented coefficient vector $(1, a)$ gives

$$
\begin{aligned}
F_0'(\theta) + \langle a, F'(\theta)\rangle &= \langle (1,a), \big(F_0'(\theta), F'(\theta)\big)\rangle \\
&= \langle (1,a), \widetilde{F}'(\theta)\rangle \\
&= \langle (1,a), B(\theta)\widetilde{F}(\theta)\rangle.
\end{aligned}
\tag{A.189}
$$

This proves the identity (A.185) on the actual objects. Euclidean Cauchy–Schwarz and the definition of the induced operator norm then give

$$
\begin{aligned}
|F_0'(\theta) + \langle a, F'(\theta)\rangle| &= |\langle (1,a), B(\theta)\widetilde{F}(\theta)\rangle| \\
&\le \|(1,a)\|_2 \, \|B(\theta)\widetilde{F}(\theta)\|_2 \\
&\le \|(1,a)\|_2 \, \|B(\theta)\|_{\mathrm{op}} \, \|\widetilde{F}(\theta)\|_2.
\end{aligned}
\tag{A.190}
$$

Assumption 3 supplies the legal normalization

$$
\|F(\theta)\|_2 \ge |F_{j_*}(\theta)| = 1. \tag{A.191}
$$

Lemma A.2 yields

$$
\|B(\theta)\|_{\mathrm{op}} \le \sup_{\vartheta\in\Theta} \|B(\vartheta)\|_{\mathrm{op}} \le \widehat{\Lambda}_{B,T}. \tag{A.192}
$$

Divide (A.190) by the positive quantity in (A.191), and substitute (A.177), (A.178), and (A.192) without dropping or absorbing any term. The two separate square-root factors remain visible:

$$
\begin{aligned}
\frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} &\le \frac{\underbrace{\sqrt{1+NR^2}}_{\text{from } \|(1,a)\|_2} \, \widehat{\Lambda}_{B,T} \, \underbrace{\sqrt{1+NR^2}\|F(\theta)\|_2}_{\text{from } \|\widetilde{F}(\theta)\|_2}}{\|F(\theta)\|_2} \\
&= \big(\sqrt{1+NR^2}\big)\big(\sqrt{1+NR^2}\big)\widehat{\Lambda}_{B,T} \\
&= (1+NR^2)\widehat{\Lambda}_{B,T}.
\end{aligned}
\tag{A.193}
$$

The cancellation in the first equality of (A.193) is exact because the anchor proves the denominator is nonzero. Equation (A.193) proves (A.186) and (A.187), with no residual or hidden factor. ☐

Proposition A.9 (Boundary and zero-certificate consistency)**.** *Under Assumptions 1 and 3, use Lemma A.14, Proposition A.8, and Lemma A.2. Their conclusions remain valid when* $N = 1$*,* $F_0 \equiv 0$*,* $a = 0$*, the root section is empty,* θ *is an endpoint of* Θ*,* $R > 0$*, or* $\widehat{\Lambda}_{B,T} = 0$*. In the zero-certificate case, for every* $\theta \in \Theta$*,*

$$
B(\theta) = 0, \qquad \widetilde{F}'(\theta) = 0, \tag{A.194}
$$

and for every $a \in H_\theta \cap [-R,R]^N$*,*

$$
\frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} = 0. \tag{A.195}
$$

Proof. Each requested case is a direct specialization of the same pointwise identities.

1. If $N = 1$, cube membership gives $\|a\|_2 = |a| \le R$, so the two factors are $\sqrt{1+R^2}$, and their product is exactly $1 + R^2$. The anchor is the sole feature coordinate and still gives $\|F(\theta)\|_2 \ge 1$. No positive-dimensional section geometry is used.

2. If $F_0 \equiv 0$, then at every root-section point $\langle a, F(\theta)\rangle = 0$ and $\|\widetilde{F}(\theta)\|_2 = \|F(\theta)\|_2$ exactly. Also $F_0' = 0$, so (A.185) specializes to

$$\langle a, F'(\theta)\rangle = \langle (1,a), B(\theta)\widetilde{F}(\theta)\rangle. \tag{A.196}$$

The general velocity bound remains valid. If only $F_0(\theta) = 0$ at the tested point, the root and augmented-norm conclusions remain exact there; no derivative conclusion about F_0' is inferred.

3. If $a = 0$ belongs to the tested section, the actual root equation gives $F_0(\theta) = 0$, while $\|(1,a)\|_2 = 1$ and $\|\widetilde{F}(\theta)\|_2 = \|F(\theta)\|_2$. Membership at one point does not imply $F_0'(\theta) = 0$; the derivative numerator is controlled by the exact closure calculation from (A.185) through (A.193).

4. If $H_\theta \cap [-R,R]^N$ is empty, the assertion quantified over its elements is vacuous. No normalization, section measure, or event is evaluated on a nonexistent coefficient point.

5. If θ is an endpoint of the compact interval Θ, the exact closure identity remains valid because it is assumed on the open interval $U \supseteq \Theta$. Thus all derivatives and every pointwise equality in the proposition is the restriction of an identity valid on an open neighborhood of the endpoint.

6. The primitive regime requires $R > 0$. Therefore $R\sqrt{N}$, $1 + NR^2$, and $\sqrt{1+NR^2}$ are finite, with the square-root factor strictly positive. The proof never divides by R and needs no additional lower bound or limiting convention for it.

7. If $\widehat{\Lambda}_{B,T} = 0$, the certificate and nonnegativity of the operator norm give, for every $\theta \in \Theta$,

$$0 \le \|B(\theta)\|_{\mathrm{op}} \le \sup_{\vartheta\in\Theta} \|B(\vartheta)\|_{\mathrm{op}} \le 0. \tag{A.197}$$

Hence $\|B(\theta)\|_{\mathrm{op}} = 0$, so $B(\theta) = 0$. The exact primitive closure, rather than a law-level argument, now gives

$$\widetilde{F}'(\theta) = B(\theta)\widetilde{F}(\theta) = 0. \tag{A.198}$$

The primitive anchor still gives $\|F(\theta)\|_2 \ge 1$, so the normalization in (A.195) is legal, while its numerator is exactly zero. This proves (A.195). No root-event probability conclusion is asserted or needed in this local velocity step.

These checks use the identical pointwise objects and introduce no section-size, density, transversality, root-simplicity, pivot, or probabilistic claim. □

Synthesis of the affine normal-velocity bound. Fix $\theta \in \Theta$ and $a \in H_\theta \cap [-R,R]^N$. Lemma A.14 gives

$$|F_0(\theta)| = |\langle a, F(\theta)\rangle| \le R\sqrt{N}\,\|F(\theta)\|_2, \qquad \|\widetilde{F}(\theta)\|_2 \le \sqrt{1+NR^2}\,\|F(\theta)\|_2. \tag{A.199}$$

Proposition A.8 applies the closure identity to the identical objects and obtains

$$F_0'(\theta) + \langle a, F'(\theta)\rangle = \langle (1,a), B(\theta)\widetilde{F}(\theta)\rangle. \tag{A.200}$$

Together with $\|(1,a)\|_2 \le \sqrt{1+NR^2}$ and Lemma A.2, this yields

$$\frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} \le (1+NR^2)\widehat{\Lambda}_{B,T}. \tag{A.201}$$

Proposition A.9 verifies endpoints, empty sections, $a = 0$, vanishing F_0, and zero certificate. Since the pair (θ, a) was arbitrary, this is the required bound on every actual affine root section, without a section-size, density, or probability assertion. □

A.7 General Affine Rate

Lemma A.15 (Fixed-law affine probability rate with literal constants)**.** *Under Assumption 4, the setting parameter definitions, and Propositions A.6, A.7, and A.8, if $\mu \in \mathcal{D}_{N,R,\kappa}$ and $I \subseteq \Theta$ is an interval with $|I| > 0$, then*

$$\begin{aligned}\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] &\le \kappa \int_I \int_{H_\theta\cap[-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta \\ &\le \kappa\sqrt{2}(2R)^{N-1}(1 + NR^2)\widehat{\Lambda}_{B,T}|I| \\ &= \frac{A(1 + NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}|I|.\end{aligned} \tag{A.202}$$

This holds for every arbitrarily correlated admissible law, with the endpoint and zero-dimensional-section conventions.

Proof. Fix the deterministic presentation data. Next fix an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$, and after the law fix an arbitrary interval $I \subseteq \Theta$ with $|I| > 0$. Proposition A.6 gives the first line of (A.202) directly:

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] \le \kappa \int_I \int_{H_\theta\cap[-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta. \tag{A.203}$$

The factor κ in (A.203) is already the full-joint-density domination. It is neither repeated nor converted into a marginal bound.

For every fixed $\theta \in I$ and every $a \in H_\theta \cap [-R, R]^N$, Proposition A.8 applies to this identical section point and gives

$$0 \le \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} \le (1 + NR^2)\widehat{\Lambda}_{B,T}. \tag{A.204}$$

Integrating (A.204) on the actual section, including the empty-section case, yields

$$\begin{aligned}&\int_{H_\theta\cap[-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a) \\ &\qquad \le (1 + NR^2)\widehat{\Lambda}_{B,T}\, \mathcal{H}^{N-1}(H_\theta \cap [-R, R]^N).\end{aligned} \tag{A.205}$$

Proposition A.7 applies to the same H_θ, coefficient cube, Euclidean metric, and Hausdorff convention, so

$$\mathcal{H}^{N-1}(H_\theta \cap [-R, R]^N) \le \sqrt{2}(2R)^{N-1}. \tag{A.206}$$

Substituting (A.206) into (A.205), then integrating the resulting constant over the literal interval I, gives

$$\begin{aligned}&\kappa \int_I \int_{H_\theta\cap[-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta \\ &\qquad \le \kappa \int_I \sqrt{2}(2R)^{N-1}(1 + NR^2)\widehat{\Lambda}_{B,T}\, d\theta \\ &\qquad = \kappa\sqrt{2}(2R)^{N-1}(1 + NR^2)\widehat{\Lambda}_{B,T}|I|.\end{aligned} \tag{A.207}$$

There is no chart sum in (A.205)–(A.207), hence no chart-count factor. There is no residual, surrogate section, second law domination, or hidden absorption. The factor in (A.207) is the direct product of the actual translated-section measure and the actual pointwise normal velocity.

It remains only to verify the final equality in (A.202). Assumption 1 gives $R > 0$, and the setting defines $A = (2R)^N\kappa$. Therefore

$$\begin{aligned}\kappa\sqrt{2}(2R)^{N-1} &= \frac{\sqrt{2}}{2R}\,\kappa(2R)^N \\ &= \frac{\sqrt{2}}{2R}\,A \\ &= \frac{A}{\sqrt{2}R},\end{aligned} \tag{A.208}$$

where $\sqrt{2}/(2R) = 1/(\sqrt{2}R)$ follows from $2 = (\sqrt{2})^2$ and $R > 0$. Multiplying (A.208) by the unchanged nonnegative factors $(1+NR^2)\widehat{\Lambda}_{B,T}|I|$ proves the last line of (A.202).

For $N = 1$, (A.203) uses the $\mathcal{H}^0$ counting convention and zero-dimensional beta mass one; (A.206) reads 0 or $1 \le \sqrt{2}(2R)^0$; and (A.204) has the literal factor $(1+R^2)\widehat{\Lambda}_{B,T}$. Empty sections contribute zero to (A.205). Included interval endpoints are inherited from (A.203), while changing endpoint conventions does not alter (A.207). No lower bound on $|I|$ beyond strict positivity appears, so the argument applies unchanged to arbitrarily short positive-length intervals. Because the law and interval were arbitrary, (A.202) holds separately for every such pair, without independence or a law/interval union bound. □

Lemma A.16 (Exact interval-then-law capacity conversion)**.** *Under Assumption 4, the setting parameter definitions, Propositions A.6, A.7, and A.8, and Lemma A.15,*

$$C^{\mathrm{aff}}_{\mathcal{D}}(F_0, F; \Theta) \le \frac{A(1+NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}. \tag{A.209}$$

The conversion first takes the setting-defined inner supremum over positive-length intervals for a fixed law and then the outer supremum over admissible laws.

Proof. Fix an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$. For every interval $I \subseteq \Theta$ with $|I| > 0$, Lemma A.15 and positivity of $|I|$ give

$$\frac{\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0]}{|I|} \le \frac{A(1+NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}. \tag{A.210}$$

The right-hand side is deterministic and independent of I. Taking exactly the inner interval supremum in the setting definition yields, for this fixed μ,

$$\sup_{\substack{I\subseteq\Theta \text{ interval} \\ |I|>0}} \frac{\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0]}{|I|} \le \frac{A(1+NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}. \tag{A.211}$$

Now take the outer supremum over $\mu \in \mathcal{D}$. By the setting definition,

$$\begin{aligned}C^{\mathrm{aff}}_{\mathcal{D}}(F_0, F; \Theta) &= \sup_{\mu\in\mathcal{D}} \sup_{\substack{I\subseteq\Theta \text{ interval} \\ |I|>0}} \frac{\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0]}{|I|} \\ &\le \frac{A(1+NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}.\end{aligned} \tag{A.212}$$

Equations (A.210)–(A.212) divide only by the positive scalar $|I|$. They do not form a simultaneous random event, exchange random and deterministic suprema, or apply any law/interval union bound. Ordinary probability for each fixed law is preserved. □

Proposition A.10 (General affine coefficient-sweep rate and zero certificate)**.** *Under Assumption 4, the setting parameter definitions, Propositions A.6, A.7, and A.8, and Lemmas A.15 and A.16, every $\mu \in \mathcal{D}_{N,R,\kappa}$ and every interval $I \subseteq \Theta$ with $|I| > 0$ satisfy*

$$
\begin{aligned}
\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] &\le \kappa \int_I \int_{H_\theta\cap[-R,R]^N} \frac{|F_0'(\theta) + \langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta \\
&\le \kappa\sqrt{2}(2R)^{N-1}(1 + NR^2)\widehat{\Lambda}_{B,T}|I| \\
&= \frac{A(1 + NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}|I|,
\end{aligned}
\tag{A.213}
$$

and

$$
C_{\mathcal{D}}^{\mathrm{aff}}(F_0, F; \Theta) \le \frac{A(1 + NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}. \tag{A.214}
$$

If $\widehat{\Lambda}_{B,T} = 0$, then for every such law and interval,

$$
\Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] = 0. \tag{A.215}
$$

These conclusions retain arbitrary coefficient correlation, ordinary probability, uniformity over every positive-length interval, the $N = 1$ zero-dimensional section convention, empty root sections, literal interval endpoints, and the exact dependence on $A, N, R, \kappa, \widehat{\Lambda}_{B,T}$, and $|I|$. There are no hidden constants. Once the supplied certificate is fixed, additional dependence on q, M, Δ is exactly degree zero.

Proof. Equation (A.213) is Lemma A.15, and (A.214) is Lemma A.16. Their proofs use the identical section, coefficient, feature tuple, matrix certificate, Euclidean norm, and Hausdorff measure convention used by the three propositions, so no change of objects or constants occurs.

For the zero-certificate case, substitute $\widehat{\Lambda}_{B,T} = 0$ into (A.213). Since $A = (2R)^N\kappa$ is finite, $R > 0$, and probabilities are nonnegative,

$$
0 \le \Pr_{\alpha\sim\mu}[\exists\theta \in I : \phi_\alpha(\theta) = 0] \le \frac{A(1 + NR^2)\cdot 0}{\sqrt{2}R}|I| = 0. \tag{A.216}
$$

Thus (A.215) is an exact probability conclusion for each law and interval. It is not inferred from a vacuous empty-section statement: even if an actual root section is nonempty, Proposition A.8 makes the nonnegative integrand in (A.203) equal to zero when $\widehat{\Lambda}_{B,T} = 0$, and Proposition A.6 then bounds the actual root event by that zero integral. No separate persistent-root or affine-hyperplane nullity result is imported.

The $N = 1$, empty-section, endpoint, and arbitrarily short positive-length interval cases were checked in Lemma A.15; positivity of interval length is the only condition used in the capacity division. This completes every clause of the proposition. □

Synthesis of the general affine rate. Proposition A.6 supplies the normal-velocity integral with coefficient κ. On the same sections, Propositions A.8 and A.7 supply, respectively,

$$
\frac{|F_0' + \langle a, F'\rangle|}{\|F\|_2} \le (1 + NR^2)\widehat{\Lambda}_{B,T}, \qquad \mathcal{H}^{N-1}(H_\theta \cap [-R, R]^N) \le \sqrt{2}(2R)^{N-1}. \tag{A.217}
$$

Their literal product in Lemma A.15 is

$$\kappa\,[\sqrt{2}(2R)^{N-1}][(1+NR^2)\widehat{\Lambda}_{B,T}]\,|I|, \tag{A.218}$$

and the only algebraic conversion is

$$\kappa\sqrt{2}(2R)^{N-1} = \frac{\sqrt{2}}{2R}\kappa(2R)^N = \frac{A}{\sqrt{2}R}. \tag{A.219}$$

Lemma A.16 divides only by positive $|I|$, takes the interval supremum for a fixed law, and only then takes the law supremum. Proposition A.10 combines these conclusions and proves the zero-certificate branch. Thus no chart-count factor, second density domination, probability conversion, confidence parameter, independence condition, pivot margin, transversality, or hidden constant enters the rate. □

A.8 Homogeneous Projective Rate

Lemma A.17 (Exact homogeneous radial cancellation on the actual section)**.** *Under Assumptions 1, 2, and 3, Lemma A.1, and the exact specialization $F_0 \equiv 0$, define*

$$r(\theta) := \|F(\theta)\|_2, \qquad \gamma_F(\theta) := \frac{F(\theta)}{r(\theta)}. \tag{A.220}$$

Then, for every $\theta \in \Theta$, $r(\theta) \geq 1$, both definitions and all differentiations are legal, and the actual setting-defined coefficient section is

$$H_\theta = \{a \in \mathbb{R}^N : \langle a, F(\theta)\rangle = 0\} = F(\theta)^\perp = \gamma_F(\theta)^\perp. \tag{A.221}$$

For every $a \in H_\theta$,

$$\frac{|\langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} = |\langle a, \gamma_F'(\theta)\rangle|. \tag{A.222}$$

This is an equality on the same actual coefficient section, with no bound on r beyond anchor legality and no bound on r'.

Proof. Lemma A.1 gives

$$F_{j_*}(\theta) = 1, \qquad r(\theta) = \|F(\theta)\|_2 \geq 1 \tag{A.223}$$

for every $\theta \in \Theta$, and gives differentiability of γ_F. Thus normalization and division by $r(\theta)$ are legal, including at interval endpoints by restriction from U.

Because $F_0 \equiv 0$, the setting definition of the actual section gives

$$\begin{aligned} H_\theta &= \{a : F_0(\theta) + \langle a, F(\theta)\rangle = 0\} \\ &= \{a : \langle a, F(\theta)\rangle = 0\} \\ &= \{a : r(\theta)\langle a, \gamma_F(\theta)\rangle = 0\} \\ &= \gamma_F(\theta)^\perp. \end{aligned} \tag{A.224}$$

This also proves $H_\theta = F(\theta)^\perp$. In particular, for the same a appearing in the actual section integral,

$$\langle a, \gamma_F(\theta)\rangle = 0. \tag{A.225}$$

Differentiate the object identity $F = r\gamma_F$:

$$F' = r'\gamma_F + r\gamma_F'. \tag{A.226}$$

Taking the inner product with that same $a \in H_\theta$, dividing by the positive $r = \|F\|_2$, and using actual-section orthogonality gives

$$\frac{\langle a, F' \rangle}{\|F\|_2} = \frac{r'}{r}\langle a, \gamma_F \rangle + \langle a, \gamma_F' \rangle = \langle a, \gamma_F' \rangle. \tag{A.227}$$

Taking absolute values proves the asserted equality. The term involving r' vanishes algebraically; it is not bounded, integrated, absorbed, or replaced by an amplitude condition. $\square$

Lemma A.18 (Literal homogeneous coefficient algebra)**.** *Under Assumption 1 and the setting definition* $A = (2R)^N \kappa$*,*

$$\kappa R\sqrt{N}\,\sqrt{2}(2R)^{N-1} = A\sqrt{\frac{N}{2}}. \tag{A.228}$$

Proof. Assumption 1 gives $R > 0$, $N \geq 1$, and $\kappa > 0$. Hence every power and square root in the proposition is legal, and

$$R(2R)^{N-1} = \frac{(2R)^N}{2}. \tag{A.229}$$

Therefore

$$\begin{aligned} \kappa R\sqrt{N}\,\sqrt{2}(2R)^{N-1} &= \kappa(2R)^N \frac{\sqrt{N}\,\sqrt{2}}{2} \\ &= \kappa(2R)^N \sqrt{\frac{N}{2}} \\ &= A\sqrt{\frac{N}{2}}. \end{aligned} \tag{A.230}$$

Every equality is literal; no term is dominated or hidden. $\square$

Proposition A.11 (Stationary projective branch is one fixed law-null central hyperplane)**.** *Under Assumptions 1, 4, and 3, Lemma A.1, and the exact specialization* $F_0 \equiv 0$*, suppose*

$$\Gamma_{\mathrm{proj}}(F) = 0. \tag{A.231}$$

Then there is a fixed unit vector $\gamma_0 \in \mathbb{R}^N$ *such that* $\gamma_F(\theta) = \gamma_0$ *for every* $\theta \in \Theta$*. For every nonempty interval* $I \subseteq \Theta$*,*

$$\big\{a \in \mathbb{R}^N : \exists\theta \in I,\ \langle a, F(\theta) \rangle = 0\big\} = \gamma_0^\perp. \tag{A.232}$$

Thus the union root event is one fixed proper nonempty central hyperplane, not an empty section, and every $\mu \in \mathcal{D}_{N,R,\kappa}$ *satisfies*

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta) \rangle = 0] = 0. \tag{A.233}$$

For $N = 1$*, the same statement reads* $\gamma_0^\perp = \{0\}$*, and the conclusion remains ordinary probability zero under every admissible one-dimensional density.*

Proof. By definition,

$$\Gamma_{\text{proj}}(F) = \sup_{\theta \in \Theta} \|\gamma_F'(\theta)\|_2. \tag{A.234}$$

All terms under the supremum are nonnegative. Hence $\Gamma_{\text{proj}}(F) = 0$ implies $\gamma_F'(\theta) = 0$ for every $\theta \in \Theta$. The set Θ is a connected interval, and γ_F is differentiable by Lemma A.1. The coordinatewise zero-derivative criterion therefore gives a fixed vector γ_0 with $\gamma_F(\theta) = \gamma_0$ on all of Θ. Since every $\gamma_F(\theta)$ is normalized, $\|\gamma_0\|_2 = 1$.

Lemma A.1 also gives $r(\theta) = \|F(\theta)\|_2 \geq 1$. Thus, for every θ,

$$F(\theta) = r(\theta)\gamma_0, \qquad \langle a, F(\theta)\rangle = 0 \iff \langle a, \gamma_0\rangle = 0. \tag{A.235}$$

Every positive-length interval is nonempty, so taking the union over $\theta \in I$ gives exactly $\gamma_0^{\perp}$. Because $\gamma_0 \neq 0$, this is a proper hyperplane. Because $0 \in \gamma_0^{\perp}$, it is central and nonempty. The conclusion depends only on the projective direction, so even an arbitrary positive differentiable radial factor would leave this root set unchanged.

A proper hyperplane is N-dimensional Lebesgue-null. Assumption 4 supplies one full joint density f_μ, so

$$\mu(\gamma_0^{\perp}) = \int_{\gamma_0^{\perp}} f_\mu(a)\, d\,\mathrm{Leb}^N(a) = 0. \tag{A.236}$$

This conclusion uses neither independence nor marginal densities. When $N = 1$, a unit γ_0 has either normal orientation and its orthogonal hyperplane is the singleton $\{0\}$. Its intersection with $[-R, R]$ is nonempty and has $\mathcal{H}^0$-mass one, but it has one-dimensional Lebesgue measure zero, so every admissible density still assigns it probability zero.

□

Proposition A.12 (Sharp pairwise homogeneous interval rate)**.** *Under Assumptions 1, 2, 4, and 3, the current Propositions A.1, A.6, and A.7, Lemma A.1, Lemmas A.17 and A.18, and Proposition A.11, if $F_0 \equiv 0$, then for every $\mu \in \mathcal{D}_{N,R,\kappa}$ and every interval $I \subseteq \Theta$ with $|I| > 0$,*

$$\begin{aligned} \Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta)\rangle = 0] &\leq A\sqrt{\frac{N}{2}}\,\Gamma_{\text{proj}}(F)|I| \\ &\leq A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}|I|. \end{aligned} \tag{A.237}$$

This is ordinary probability for the arbitrary, possibly correlated, full-joint-density law. The coefficients and both inequalities are literal, with no clipping at one.

Proof. Fix the complete deterministic instance, including $F_0 \equiv 0$, before selecting a law or interval. Now fix an arbitrary $\mu \in \mathcal{D}_{N,R,\kappa}$, and only afterward fix an arbitrary interval $I \subseteq \Theta$ with $|I| > 0$, retaining its literal endpoint convention.

Because $F_0 \equiv 0$, one has $F_0' \equiv 0$ and $\phi_\alpha(\theta) = \langle \alpha, F(\theta)\rangle$. Apply only the coordinate-free first swept-area inequality, Proposition A.6:

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta)\rangle = 0] \leq \kappa \int_I \int_{H_\theta \cap [-R,R]^N} \frac{|\langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2}\, d\mathcal{H}^{N-1}(a)\, d\theta. \tag{A.238}$$

Lemma A.17 identifies the same actual section as $H_\theta = \gamma_F(\theta)^{\perp}$ and replaces the integrand by equality, not by an estimate:

$$\Pr_{\alpha\sim\mu}[\exists\theta\in I:\langle\alpha,F(\theta)\rangle=0]\le\kappa\int_I\int_{\gamma_F(\theta)^\perp\cap[-R,R]^N}|\langle a,\gamma_F'(\theta)\rangle|\,d\mathcal{H}^{N-1}(a)\,d\theta. \tag{A.239}$$

For every $a\in[-R,R]^N$,

$$\|a\|_2^2=\sum_{i=1}^N a_i^2\le NR^2,\qquad \|a\|_2\le R\sqrt{N}. \tag{A.240}$$

Euclidean Cauchy–Schwarz therefore gives, on the actual central section,

$$|\langle a,\gamma_F'(\theta)\rangle|\le R\sqrt{N}\,\|\gamma_F'(\theta)\|_2. \tag{A.241}$$

The integrands are nonnegative. Hence Proposition A.7, applied with the actual Euclidean unit normal $u=\gamma_F(\theta)$ and $t=0$, yields

$$\begin{aligned}\int_{\gamma_F(\theta)^\perp\cap[-R,R]^N}&|\langle a,\gamma_F'(\theta)\rangle|\,d\mathcal{H}^{N-1}(a)\\ &\le R\sqrt{N}\,\|\gamma_F'(\theta)\|_2\,\mathcal{H}^{N-1}\big(\gamma_F(\theta)^\perp\cap[-R,R]^N\big)\\ &\le R\sqrt{N}\,\sqrt{2}(2R)^{N-1}\|\gamma_F'(\theta)\|_2.\end{aligned} \tag{A.242}$$

This calculation is literal also for $N=1$: the actual section is the singleton $\{0\}$, $\mathcal{H}^0$ counts it once, and the integrand at $a=0$ is zero. Thus no positive point mass is inferred from the counting measure.

Integrating the pointwise inequality and using the definition of projective speed gives

$$\begin{aligned}\Pr_{\alpha\sim\mu}[\exists\theta\in I:\langle\alpha,F(\theta)\rangle=0]&\le\kappa R\sqrt{N}\,\sqrt{2}(2R)^{N-1}\int_I\|\gamma_F'(\theta)\|_2\,d\theta\\ &\le\kappa R\sqrt{N}\,\sqrt{2}(2R)^{N-1}\Gamma_{\mathrm{proj}}(F)|I|.\end{aligned} \tag{A.243}$$

No interval-length lower bound other than $|I|>0$ is used. If $\Gamma_{\mathrm{proj}}(F)=0$, Proposition A.11 independently identifies the event as one fixed proper central hyperplane and proves its probability is zero; the nonnegative-integral derivation in (A.243) also remains valid without dividing by $\Gamma_{\mathrm{proj}}(F)$.

Apply the literal identity in Lemma A.18:

$$\Pr_{\alpha\sim\mu}[\exists\theta\in I:\langle\alpha,F(\theta)\rangle=0]\le A\sqrt{\frac{N}{2}}\,\Gamma_{\mathrm{proj}}(F)|I|. \tag{A.244}$$

Finally, apply the exact named projective certificate, Proposition A.1,

$$\Gamma_{\mathrm{proj}}(F)\le\widehat{\Lambda}_{B,T}, \tag{A.245}$$

to obtain

$$\Pr_{\alpha\sim\mu}[\exists\theta\in I:\langle\alpha,F(\theta)\rangle=0]\le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}|I|. \tag{A.246}$$

The full-joint-density factor κ entered only once, inside the first swept-area inequality. There is no conditioning, marginalization, independence reduction, union bound, second probability conversion, chart-count factor, or auxiliary tolerance. If either stated right-hand side exceeds one, it remains a valid upper bound and is not clipped. □

Proposition A.13 (Defining-supremum closure for the homogeneous capacity)**.** *Under Assumptions 1, 2, 4, and 3, Proposition A.12, and the exact specialization $F_0 \equiv 0$,*

$$C_{\mathcal{D}}^{\mathrm{Pf}}(F;\Theta) \le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}. \tag{A.247}$$

The interval supremum is taken for each fixed law before the outer law supremum, exactly as in the setting definition.

Proof. Fix $\mu \in \mathcal{D}$. For every interval $I \subseteq \Theta$ with $|I| > 0$, Proposition A.12 gives

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta)\rangle = 0] \le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}|I|. \tag{A.248}$$

Only now divide by the strictly positive $|I|$:

$$\frac{\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta)\rangle = 0]}{|I|} \le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}. \tag{A.249}$$

The right-hand side is deterministic and independent of I. Therefore, still for this fixed law,

$$\sup_{\substack{I\subseteq\Theta \text{ interval}\\ |I|>0}} \frac{\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta)\rangle = 0]}{|I|} \le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}. \tag{A.250}$$

The same deterministic inequality holds for every $\mu \in \mathcal{D}$. Taking the outer law supremum only after the interval supremum gives

$$\begin{aligned} C_{\mathcal{D}}^{\mathrm{Pf}}(F;\Theta) &= \sup_{\mu\in\mathcal{D}} \sup_{\substack{I\subseteq\Theta \text{ interval}\\ |I|>0}} \frac{\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta)\rangle = 0]}{|I|} \\ &\le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}. \end{aligned} \tag{A.251}$$

No zero-length interval is divided by, and arbitrarily short positive-length intervals remain within the supremum. ☐

Synthesis of the homogeneous projective rate. Fix the homogeneous deterministic presentation before selecting a law and interval. Lemma A.1 makes $r = \|F\|_2 \ge 1$, and Lemma A.17 gives on the actual section

$$H_\theta = F(\theta)^\perp = \gamma_F(\theta)^\perp, \qquad \frac{|\langle a, F'(\theta)\rangle|}{\|F(\theta)\|_2} = |\langle a, \gamma_F'(\theta)\rangle|. \tag{A.252}$$

Proposition A.11 handles $\Gamma_{\mathrm{proj}}(F) = 0$ as one fixed proper law-null central hyperplane. Apply Cauchy–Schwarz and $\|a\|_2 \le R\sqrt{N}$ in Proposition A.6. The section estimate is Proposition A.7. Together they give the coefficient $\kappa R\sqrt{N}\sqrt{2}(2R)^{N-1}$, which Lemma A.18 identifies exactly with $A\sqrt{N/2}$. Proposition A.12 therefore gives

$$\Pr_{\alpha\sim\mu}[\exists\theta \in I : \langle \alpha, F(\theta)\rangle = 0] \le A\sqrt{\frac{N}{2}}\Gamma_{\mathrm{proj}}(F)|I| \le A\sqrt{\frac{N}{2}}\widehat{\Lambda}_{B,T}|I|, \tag{A.253}$$

where the last inequality is Proposition A.1. Finally, Proposition A.13 preserves the interval-then-law supremum order and proves

$$C_{\mathcal{D}}^{\mathrm{Pf}}(F;\Theta) \le A\sqrt{\frac{N}{2}}\widehat{\Lambda}_{B,T}. \tag{A.254}$$

☐

A.9 Exact Monic Presentation

Proposition A.14 (Augmented Monomial Presentation)**.** *Under Assumptions 1, 2, and the explicit monic specialization of Assumption 3, let $d \geq 1$, let $J \subset \mathbb{R}$ be a bounded interval, and let $\alpha = (\alpha_0, \ldots, \alpha_{d-1}) \in [-R, R]^d$. If*

$$F_0(\theta) = \theta^d, \qquad F_{k+1}(\theta) = \theta^k \quad (0 \leq k \leq d-1), \tag{A.255}$$

then, with the ordered augmented tuple

$$\widetilde{F} = (F_0, F_1, \ldots, F_d) = (\theta^d, 1, \theta, \ldots, \theta^{d-1}), \tag{A.256}$$

one has pointwise on $\mathbb{R}$

$$F_0(\theta) + \langle \alpha, F(\theta) \rangle = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k = p_\alpha(\theta). \tag{A.257}$$

The Balcan descriptor values are exactly $q = 0$, $M = 0$, $\Delta = d$, $N = d$, and $A = (2R)^d \kappa$, and the literal anchor is $F_1 = 1$. The coefficient vector contains exactly the d lower coefficients; the leading coefficient one is deterministic and is not a coordinate of α.

Proof. Take the empty common chain. By the setting's explicit convention for an empty chain,

$$q = 0, \qquad M = 0. \tag{A.258}$$

The output polynomials in the one scalar parameter are

$$Q_0(\theta) = \theta^d, \qquad Q_{k+1}(\theta) = \theta^k \quad (0 \leq k \leq d-1). \tag{A.259}$$

Their coordinate index map is

$$k \in \{0, \ldots, d-1\} \longmapsto i = k+1 \in \{1, \ldots, d\}. \tag{A.260}$$

Thus there are d random-feature coordinates, so $N = d$, and

$$\Delta = \max \left\{ \deg Q_0, \max_{0 \leq k \leq d-1} \deg Q_{k+1} \right\} = \max\{d, d-1\} = d. \tag{A.261}$$

This calculation also covers $d = 1$, when the second maximum is the single value 0. By the setting definition of A,

$$A = (2R)^N \kappa = (2R)^d \kappa. \tag{A.262}$$

The first random feature corresponds to $k = 0$, hence

$$F_1(\theta) = \theta^0 = 1 \quad \text{for every } \theta, \tag{A.263}$$

which verifies the literal anchor required by Assumption 3.

Finally, using the index map from Proposition A.14 coefficient by coefficient,

$$\begin{aligned} F_0(\theta) + \langle \alpha, F(\theta) \rangle &= F_0(\theta) + \sum_{i=1}^{d} \alpha_{i-1} F_i(\theta) \\ &= \theta^d + \sum_{i=1}^{d} \alpha_{i-1} \theta^{i-1} \\ &= \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k = p_\alpha(\theta). \end{aligned} \tag{A.264}$$

This is a pointwise identity for every real θ. The domain $[-R,R]^d$ is the actual lower-coefficient cube. There is no coordinate α_d, and the deterministic term $F_0=\theta^d$ is not part of the random feature inner product. $\square$

Lemma A.19 (Exact Derivative-Shift Certificate)**.** *Under Assumptions 1, 2, and 3, let $d\geq 1$ and let $\widetilde{F}$ be the ordered tuple in Proposition A.14. Define a constant matrix $B\in\mathbb{R}^{(d+1)\times(d+1)}$, with rows and columns indexed by $0,1,\ldots,d$, by declaring its only nonzero entries to be*

$$B_{0,d}=d,\qquad B_{k+1,k}=k\quad(1\leq k\leq d-1).\tag{A.265}$$

Then $\widetilde{F}'=B\widetilde{F}$ coefficient by coefficient, $m=0$, and

$$\widehat{\Lambda}_{B,T}=\left(\sum_{k=1}^{d}k^2\right)^{1/2}.\tag{A.266}$$

For $d=1$, B is the 2-by-2 matrix with the sole nonzero entry $B_{0,1}=1$.

Proof. The row actions of B follow the ordered coordinate map exactly.

Row 0 has only column d nonzero. Since $F_d(\theta)=\theta^{d-1}$,

$$(B\widetilde{F})_0=dF_d=d\theta^{d-1}=F_0'(\theta).\tag{A.267}$$

Row 1 is zero because it is not among the declared nonzero rows. Hence

$$(B\widetilde{F})_1=0=(1)'=F_1'(\theta).\tag{A.268}$$

Every remaining row is uniquely $r=k+1$ for an integer $1\leq k\leq d-1$. Its only nonzero column is $s=k$, and the feature index map gives $F_k(\theta)=\theta^{k-1}$. Therefore

$$(B\widetilde{F})_{k+1}=kF_k=k\theta^{k-1}=(\theta^k)'=F_{k+1}'(\theta).\tag{A.269}$$

Rows 0, 1, and $k+1$ for $1\leq k\leq d-1$ are exactly all rows $0,1,\ldots,d$. Thus every coordinate agrees and

$$\widetilde{F}'(\theta)=B\widetilde{F}(\theta)\quad\text{for every }\theta\in\mathbb{R}.\tag{A.270}$$

This verifies the closure identity of Assumption 3; it was not imported as an unverified property of the monic tuple.

Because every entry of B is constant, its setting representation is

$$B_{rs}(\theta)=b_{rs,0},\qquad b_{rs,0}=B_{rs},\tag{A.271}$$

with no coefficient of positive degree. Hence the polynomial matrix degree is exactly $m=0$. Since $T_*^0=1$, the setting-defined coefficient height satisfies

$$\begin{aligned}\widehat{\Lambda}_{B,T}^2&=\sum_{r=0}^{d}\sum_{s=0}^{d}\left(\sum_{\ell=0}^{0}|b_{rs,\ell}|T_*^\ell\right)^2\\&=\sum_{r=0}^{d}\sum_{s=0}^{d}|B_{rs}|^2\\&=\|B\|_{\mathrm{F}}^2\\&=d^2+\sum_{k=1}^{d-1}k^2\\&=\sum_{k=1}^{d}k^2.\end{aligned}\tag{A.272}$$

Taking the nonnegative square root proves the asserted certificate. In particular, neither T nor the location of J appears.

When $d = 1$, the range $1 \le k \le d-1$ is empty and

$$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{A.273}$$

Its row actions are $(B\widetilde{F})_0 = F_1 = 1 = (\theta)'$ and $(B\widetilde{F})_1 = 0 = (1)'$, while the height calculation gives $\widehat{\Lambda}^2_{B,T} = 1 = \sum_{k=1}^{1} k^2$. $\square$

Proposition A.15 (Two-Pivot Coefficient Charts)**.** *Under Assumptions 1, 2, and 3, let $d \ge 2$, let $J \subset \mathbb{R}$ be a bounded interval, and use the tuple and coefficient vector of Proposition A.14. Then*

$$E_1 = J \cap \{|\theta| \le 1\}, \qquad E_d = J \cap \{|\theta| > 1\} \tag{A.274}$$

is a measurable disjoint partition of J, with $\theta = \pm 1$ assigned to E_1. The pivot α_0 is legal on E_1, the pivot α_{d-1} is legal on E_d, and the exact pivot-coordinate maps are

$$T_1(\theta, \beta) = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k, \qquad \beta = (\beta_1, \ldots, \beta_{d-1}) \in [-R, R]^{d-1}, \tag{A.275}$$

and

$$T_d(\theta, \beta) = -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1}, \qquad \beta = (\beta_0, \ldots, \beta_{d-2}) \in [-R, R]^{d-1}. \tag{A.276}$$

Inserting the pivot values into their respective coefficient coordinates gives a pointwise zero of the same polynomial p_α.

Proof. Every interval is Borel. The set $\{|\theta| \le 1\}$ is closed and the set $\{|\theta| > 1\}$ is open, so E_1 and E_d are measurable. Their defining inequalities are complementary and disjoint, hence

$$E_1 \cap E_d = \varnothing, \qquad E_1 \cup E_d = J. \tag{A.277}$$

Both boundary points 1 and -1 satisfy the weak inner inequality and therefore belong to E_1 whenever they belong to J.

The coefficient-to-feature index map of Proposition A.14 sends α_0 to $F_1 = 1$. Thus the inner pivot denominator is exactly one on all of E_1. Solving

$$0 = \theta^d + \alpha_0 + \sum_{k=1}^{d-1} \alpha_k \theta^k \tag{A.278}$$

for α_0, and writing the remaining coefficient α_k as β_k, gives

$$T_1(\theta, \beta) = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k. \tag{A.279}$$

The full coefficient insertion map is

$$\Psi_1(\theta, \beta) = \big(T_1(\theta, \beta), \beta_1, \ldots, \beta_{d-1}\big) \in \mathbb{R}^d, \tag{A.280}$$

and direct substitution verifies

$$p_{\Psi_1(\theta,\beta)}(\theta) = \theta^d + T_1(\theta, \beta) + \sum_{k=1}^{d-1} \beta_k \theta^k = 0. \tag{A.281}$$

On E_d, the outer pivot α_{d-1} multiplies

$$F_d(\theta) = \theta^{d-1}. \tag{A.282}$$

Because $d \geq 2$ and $|\theta| > 1$, $|F_d(\theta)| = |\theta|^{d-1} > 1$, so this pivot is nonzero. Solving

$$0 = \theta^d + \sum_{k=0}^{d-2} \alpha_k \theta^k + \alpha_{d-1}\theta^{d-1} \tag{A.283}$$

for α_{d-1}, with $\beta_k = \alpha_k$ for $0 \leq k \leq d-2$, gives

$$\begin{aligned} T_d(\theta,\beta) &= -\frac{\theta^d}{\theta^{d-1}} - \sum_{k=0}^{d-2} \beta_k \frac{\theta^k}{\theta^{d-1}} \\ &= -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1}. \end{aligned} \tag{A.284}$$

All divisions are legal because $\theta \neq 0$ on E_d. The full insertion map is

$$\Psi_d(\theta,\beta) = \big(\beta_0, \dots, \beta_{d-2}, T_d(\theta,\beta)\big) \in \mathbb{R}^d. \tag{A.285}$$

Multiplying the pivot formula by θ^{d-1} yields

$$T_d(\theta,\beta)\theta^{d-1} = -\theta^d - \sum_{k=0}^{d-2} \beta_k \theta^k, \tag{A.286}$$

and consequently

$$p_{\Psi_d(\theta,\beta)}(\theta) = \theta^d + \sum_{k=0}^{d-2} \beta_k \theta^k + T_d(\theta,\beta)\theta^{d-1} = 0. \tag{A.287}$$

The nonpivot input is the actual cube $[-R,R]^{d-1}$. The inserted pivot value may or may not lie in $[-R,R]$; the downstream chart indicator, not this deterministic specialization, handles that separate condition. $\square$

Lemma A.20 (Inner-Region Chart Velocity)**.** *Under Assumptions 1, 2, and 3, and Proposition A.15, let $d \geq 2$, $\theta \in E_1$, and $\beta = (\beta_1, \dots, \beta_{d-1}) \in [-R,R]^{d-1}$. Then*

$$|\partial_\theta T_1(\theta,\beta)| \leq d + R\sum_{k=1}^{d-1} k = d + \frac{Rd(d-1)}{2}. \tag{A.288}$$

Proof. Term-by-term differentiation of the finite inner chart gives

$$\partial_\theta T_1(\theta,\beta) = -d\theta^{d-1} - \sum_{k=1}^{d-1} k\beta_k\theta^{k-1}. \tag{A.289}$$

Here every exponent $d-1$ and $k-1$ is nonnegative. Because $\theta \in E_1$ implies $|\theta| \leq 1$, and because $|\beta_k| \leq R$, the triangle inequality gives every intermediate term explicitly:

$$\begin{aligned} |\partial_\theta T_1(\theta,\beta)| &\leq d|\theta|^{d-1} + \sum_{k=1}^{d-1} k|\beta_k||\theta|^{k-1} \\ &\leq d + R\sum_{k=1}^{d-1} k. \end{aligned} \tag{A.290}$$

At $\theta = 0$, the $k = 1$ factor is $|\theta|^0 = 1$, while every term with a positive exponent vanishes, so the same calculation remains literal.

For completeness, the finite arithmetic sum follows by pairing the list $1, \dots, d-1$ with its reverse. Since $k \mapsto d-k$ permutes $\{1, \dots, d-1\}$,

$$\begin{aligned} 2\sum_{k=1}^{d-1} k &= \sum_{k=1}^{d-1} k + \sum_{k=1}^{d-1}(d-k) \\ &= \sum_{k=1}^{d-1}\big(k + d - k\big) \\ &= \sum_{k=1}^{d-1} d \\ &= d(d-1). \end{aligned} \tag{A.291}$$

Therefore $\sum_{k=1}^{d-1} k = d(d-1)/2$, which proves

$$|\partial_\theta T_1| \leq d + \frac{Rd(d-1)}{2}. \tag{A.292}$$

□

Lemma A.21 (Outer-Region Chart Velocity)**.** *Under Assumptions 1, 2, and 3, and Proposition A.15, let $d \geq 2$, $\theta \in E_d$, and $\beta = (\beta_0, \dots, \beta_{d-2}) \in [-R, R]^{d-1}$. Then*

$$|\partial_\theta T_d(\theta, \beta)| \leq 1 + R\sum_{m=1}^{d-1} \frac{m}{|\theta|^{m+1}} \leq 1 + \frac{Rd(d-1)}{2} \leq d + \frac{Rd(d-1)}{2}. \tag{A.293}$$

Proof. Every power in the outer chart is defined because $|\theta| > 1$. Direct differentiation gives

$$\begin{aligned} \partial_\theta T_d(\theta, \beta) &= -1 - \sum_{k=0}^{d-2} \beta_k (k - d + 1)\theta^{k-d} \\ &= -1 + \sum_{k=0}^{d-2}(d - 1 - k)\beta_k \theta^{k-d}. \end{aligned} \tag{A.294}$$

Using $|\beta_k| \leq R$ and the triangle inequality,

$$|\partial_\theta T_d(\theta, \beta)| \leq 1 + R\sum_{k=0}^{d-2}(d - 1 - k)|\theta|^{k-d}. \tag{A.295}$$

Now set

$$m = d - 1 - k. \tag{A.296}$$

As k increases from 0 to $d-2$, m decreases from $d-1$ to 1, and

$$k = d - 1 - m, \qquad k - d = -m - 1. \tag{A.297}$$

Reversing this finite order gives the exact reindexing

$$\begin{aligned} \sum_{k=0}^{d-2}(d - 1 - k)|\theta|^{k-d} &= \sum_{m=1}^{d-1} m|\theta|^{-m-1} \\ &= \sum_{m=1}^{d-1} \frac{m}{|\theta|^{m+1}}. \end{aligned} \tag{A.298}$$

This is the only use of negative powers. Since $|\theta| > 1$, every $m \geq 1$ satisfies

$$0 < \frac{1}{|\theta|^{m+1}} < 1. \tag{A.299}$$

Therefore

$$|\partial_\theta T_d(\theta, \beta)| \leq 1 + R \sum_{m=1}^{d-1} m. \tag{A.300}$$

Again displaying the entire finite-sum calculation, the reverse-list pairing gives

$$2 \sum_{m=1}^{d-1} m = \sum_{m=1}^{d-1} \big(m + d - m\big) = \sum_{m=1}^{d-1} d = d(d-1), \tag{A.301}$$

so

$$|\partial_\theta T_d(\theta, \beta)| \leq 1 + \frac{Rd(d-1)}{2}. \tag{A.302}$$

Finally, $d \geq 2$ implies $1 \leq d$, and hence

$$1 + \frac{Rd(d-1)}{2} \leq d + \frac{Rd(d-1)}{2}. \tag{A.303}$$

For negative outer θ, integer powers may change sign inside the derivative, but $|\theta^{k-d}| = |\theta|^{k-d}$; this absolute-value calculation is unchanged. No upper bound on $|\theta|$ was used. □

Proposition A.16 (Global Linear-Case Chart)**.** *Under Assumptions 1, 2, and 3, and Proposition A.14 specialized to $d = 1$, let $J \subset \mathbb{R}$ be bounded and let $\alpha_0 \in [-R, R]$. Then, with the explicit zero-dimensional convention*

$$[-R, R]^0 = \{()\}, \tag{A.304}$$

$$\widetilde{F}(\theta) = (\theta, 1), \qquad p_{\alpha_0}(\theta) = \theta + \alpha_0, \tag{A.305}$$

the sole coefficient α_0 is a legal pivot on every $\theta \in J$, the zero-dimensional-beta chart is

$$T_1(\theta) = -\theta, \tag{A.306}$$

and

$$|\partial_\theta T_1(\theta)| = 1 = d + \frac{Rd(d-1)}{2}. \tag{A.307}$$

Proof. For $d = 1$, Proposition A.14 gives the augmented tuple $(F_0, F_1) = (\theta, 1)$, the actual coefficient vector $(\alpha_0) \in [-R, R]^1$, and

$$p_{\alpha_0}(\theta) = \theta + \alpha_0. \tag{A.308}$$

The only random feature is the constant $F_1 = 1$, so it is nonzero on all of J. Solving the root equation gives $\alpha_0 = -\theta$.

The nonpivot coordinate set has dimension $d - 1 = 0$. Under the standard product convention,

$$[-R, R]^0 = \{()\}, \tag{A.309}$$

so there is one empty beta tuple and no beta sum. Thus the full coefficient chart is simply

$$\Psi_1(\theta) = \big(T_1(\theta)\big) = (-\theta), \tag{A.310}$$

and pointwise

$$p_{\Psi_1(\theta)}(\theta) = \theta - \theta = 0. \tag{A.311}$$

Direct differentiation gives $\partial_\theta T_1 = -1$ and hence $|\partial_\theta T_1| = 1$. Since $d(d-1)/2 = 0$ at $d = 1$, this is exactly the common cap $d + Rd(d-1)/2 = 1$. Lemma A.19 separately verifies that the matrix is 2-by-2, has only $B_{0,1} = 1$, and has height one. □

Lemma A.22 (Location-Free Boundary and Index Closure)**.** *Under Assumptions 1, 2, and 3, let $d \geq 1$, let $J \subset \mathbb{R}$ be any bounded interval, and use Proposition A.14, Lemma A.19, Proposition A.15 when $d \geq 2$, Lemmas A.20 and A.21 when $d \geq 2$, and Proposition A.16 when $d = 1$. Then all chart, certificate, and object conclusions remain valid at $\theta = 0$, $\theta = 1$, $\theta = -1$, negative outer θ, empty partition pieces, $d = 1$, and $d = 2$. They hold for every location of J, use exactly d lower-coordinate inputs and a $(d+1)$-square augmented matrix, and contain neither a singular/randomized leading coordinate nor a bound depending on $\sup_{\theta \in J} |\theta|$.*

Proof. At $\theta = 0$, the point is in E_1 for $d \geq 2$, and the pivot is the literal constant $F_1 = 1$. The inner formula contains only nonnegative powers, and Lemma A.20 explicitly treats the exponent-zero term. At $\theta = 1$ and $\theta = -1$, the weak inequality $|\theta| \leq 1$ again assigns the point to E_1, so no negative power is evaluated at the transition.

For negative θ with $|\theta| > 1$, the outer pivot θ^{d-1} is nonzero. Proposition A.15 derives the chart algebraically at that same signed θ, while Lemma A.21 takes absolute values only after differentiation and therefore retains all signs correctly.

Either E_1 or E_d may be empty. The set identities

$$E_1 \cap E_d = \varnothing, \qquad E_1 \cup E_d = J \tag{A.312}$$

remain valid, and a chart assertion on an empty cell is vacuous. If J itself is empty, both cells are empty and every pointwise assertion is vacuous. If J lies wholly in one region, only that region's chart is active.

The case $d = 1$ is closed by Proposition A.16. For the first genuinely two-pivot case $d = 2$, every index can be inspected explicitly:

$$\widetilde{F} = (\theta^2, 1, \theta), \qquad B = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \widehat{\Lambda}_{B,T} = \sqrt{2^2 + 1^2} = \sqrt{5}. \tag{A.313}$$

The two charts and their derivatives are

$$T_1(\theta, \beta_1) = -\theta^2 - \beta_1 \theta, \qquad \partial_\theta T_1 = -2\theta - \beta_1, \tag{A.314}$$

and

$$T_2(\theta, \beta_0) = -\theta - \beta_0 \theta^{-1}, \qquad \partial_\theta T_2 = -1 + \beta_0 \theta^{-2}. \tag{A.315}$$

Consequently, on their respective regions,

$$|\partial_\theta T_1| \leq 2 + R, \qquad |\partial_\theta T_2| \leq 1 + \frac{R}{|\theta|^2} \leq 1 + R \leq 2 + R. \tag{A.316}$$

Thus the index conventions and bounds agree at $d = 2$ without a missing row, feature, or coefficient.

For any bounded interval J, Assumption 1 can be instantiated on a compact $\Theta \supseteq J$ inside some $[-T, T]$. Lemma A.19 has $m = 0$, so $T_*^0 = 1$ eliminates T from the certificate. The inner velocity uses only $|\theta| \leq 1$, and the outer velocity uses reciprocal powers bounded by one on $|\theta| > 1$. Hence the bounds do not use $\sup_{\theta \in J} |\theta|$, even when the location of J varies without a common bound across problem instances.

Finally, Proposition A.14 maps coefficient indices $k = 0, \ldots, d-1$ bijectively to feature indices $1, \ldots, d$, and Lemma A.19 indexes the augmented matrix by $0, \ldots, d$. Thus the coefficient cube is exactly $[-R, R]^d$, each beta cube has dimension $d-1$, and the matrix has size $(d+1)$-by-$(d+1)$. The leading coefficient one belongs to deterministic $F_0 = \theta^d$; no extra random coordinate, augmented density, or singular law is introduced.

□

Synthesis of the exact monic presentation. For every $d \geq 1$, Proposition A.14 gives

$$\widetilde{F} = (\theta^d, 1, \theta, \ldots, \theta^{d-1}), \qquad F_0 + \langle \alpha, F \rangle = p_\alpha, \tag{A.317}$$

with $q = M = 0$, $\Delta = N = d$, $A = (2R)^d\kappa$, $F_1 = 1$, and exactly $\alpha = (\alpha_0, \ldots, \alpha_{d-1}) \in [-R, R]^d$ random. Lemma A.19 gives $m = 0$, proves $\widetilde{F}' = B\widetilde{F}$ coefficient by coefficient, and yields

$$B_{0,d} = d, \qquad B_{k+1,k} = k \quad (1 \leq k \leq d-1), \qquad \widehat{\Lambda}_{B,T} = \left(\sum_{k=1}^{d} k^2 \right)^{1/2}. \tag{A.318}$$

For $d \geq 2$, Proposition A.15 gives the measurable disjoint inner and outer cells, their legal pivots, and

$$T_1 = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k, \qquad T_d = -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1}. \tag{A.319}$$

Lemmas A.20 and A.21 give

$$|\partial_\theta T_1| \leq d + \frac{Rd(d-1)}{2} \tag{A.320}$$

and

$$|\partial_\theta T_d| \leq 1 + R \sum_{r=1}^{d-1} \frac{r}{|\theta|^{r+1}} \leq 1 + \frac{Rd(d-1)}{2} \leq d + \frac{Rd(d-1)}{2}. \tag{A.321}$$

Proposition A.16 gives the same cap for $d = 1$, and Lemma A.22 covers both signs, $|\theta| = 1$, empty cells, $d = 2$, arbitrary bounded interval location, and all index conventions. The leading coefficient one remains deterministic; no additional random coordinate or density is introduced. □

A.10 Affine-Monic Probability Recovery

Lemma A.23 (Exact monic chart measure accounting)**.** *Under Proposition A.15 when $d \geq 2$ and Proposition A.16 when $d = 1$, if $d \geq 1$, $R > 0$, and $J \subset \mathbb{R}$ is a bounded interval, then every chart's nonpivot cube has exact volume*

$$\lambda_{d-1}([-R, R]^{d-1}) = (2R)^{d-1}, \tag{A.322}$$

where

$$\lambda_0([-R, R]^0) = 1 = (2R)^0 \tag{A.323}$$

when $d = 1$. Moreover, the active cells are measurable and disjoint and have total Lebesgue length $|J|$. For $d \geq 2$,

$$|E_1| + |E_d| = |J|, \tag{A.324}$$

and the intermediate cells $E_j = \varnothing$, $2 \leq j \leq d-1$, have zero length; for $d = 1$, $E_1 = J$ and $|E_1| = |J|$.

Proof. For $d \geq 2$, either chart has exactly $d - 1$ nonpivot coordinates, each ranging over $[-R, R]$. Product measure gives

$$\lambda_{d-1}([-R, R]^{d-1}) = \prod_{r=1}^{d-1} \lambda_1([-R, R]) = \prod_{r=1}^{d-1} 2R = (2R)^{d-1}. \tag{A.325}$$

Proposition A.15 gives

$$E_1 = J \cap \{|\theta| \leq 1\}, \qquad E_d = J \cap \{|\theta| > 1\}. \tag{A.326}$$

The cells are measurable, disjoint, and have union J; $\theta = \pm 1$ belongs only to E_1. Adding the empty intermediate cells yields a legal partition indexed by every $j = 1, \ldots, d$, and finite additivity gives

$$\sum_{j=1}^{d} |E_j| = |E_1| + |E_d| = |J|. \tag{A.327}$$

This remains exact if J, E_1, or E_d is empty, or if J lies entirely in one region.

For $d = 1$, Proposition A.16 gives one empty beta tuple and the sole cell $E_1 = J$. By the zero-dimensional product convention,

$$[-R, R]^0 = \{()\}, \qquad \lambda_0(\{()\}) = 1 = (2R)^0, \tag{A.328}$$

and $|E_1| = |J|$. Thus no dimension introduces a multiplier for the number of active charts. $\square$

Proposition A.17 (Exact positive-length affine-monic sweep)**.** *Under Assumption 4, Proposition A.5, Proposition A.14, Lemma A.19, Proposition A.15, Lemmas A.20 and A.21, Proposition A.16, Lemma A.22, and Lemma A.23, let $d \geq 1$, let $J \subset \mathbb{R}$ be a bounded interval with $|J| > 0$, and let*

$$\alpha = (\alpha_0, \ldots, \alpha_{d-1}) \sim \mu, \tag{A.329}$$

where μ is any Borel full joint-density law supported on $[-R, R]^d$ with $f_\mu \leq \kappa$ almost everywhere. Then, in ordinary probability,

$$\Pr_{\alpha \sim \mu}[\exists \theta \in J : p_\alpha(\theta) = 0] \leq \kappa (2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right) |J|. \tag{A.330}$$

Proof. Proposition A.14 gives $N = d$, the original coefficient order $(\alpha_0, \ldots, \alpha_{d-1})$, and the exact pointwise identity

$$F_0(\theta) + \langle \alpha, F(\theta) \rangle = p_\alpha(\theta). \tag{A.331}$$

Thus the event in Proposition A.5 is exactly the event in this proposition. The leading coefficient remains the deterministic coefficient of $F_0 = \theta^d$.

Lemma A.22 permits a nondegenerate compact $\Theta \supseteq J$ inside some $[-T, T]$. Proposition A.14 and Lemma A.19 supply on that containing interval the exact presentation, anchor, and derivative closure required by Proposition A.5. Because the matrix is constant and $m = 0$, this instantiation introduces no T-dependent conclusion.

Assumption 4 supplies one full joint density $f_\mu \leq \kappa$ on the original cube. Proposition A.5 already performs direct joint-density domination. This specialization neither factors f_μ nor takes a marginal or conditional density.

First suppose $d \geq 2$. Use the cells E_1, E_d and define $E_j = \varnothing$ for $2 \leq j \leq d-1$. Proposition A.15 makes this a measurable legal partition for the d features. Applying both chart inequalities gives

$$\begin{aligned} \Pr_{\alpha \sim \mu}[\exists \theta \in J : p_\alpha(\theta) = 0] \leq{}& \kappa \int_{E_1} \int_{[-R,R]^{d-1}} \mathbf{1}\{|T_1(\theta, \beta)| \leq R\} |\partial_\theta T_1(\theta, \beta)| \, d\beta \, d\theta \\ &+ \kappa \int_{E_d} \int_{[-R,R]^{d-1}} \mathbf{1}\{|T_d(\theta, \beta)| \leq R\} |\partial_\theta T_d(\theta, \beta)| \, d\beta \, d\theta \\ \leq{}& \kappa \int_{E_1} \int_{[-R,R]^{d-1}} |\partial_\theta T_1(\theta, \beta)| \, d\beta \, d\theta \\ &+ \kappa \int_{E_d} \int_{[-R,R]^{d-1}} |\partial_\theta T_d(\theta, \beta)| \, d\beta \, d\theta. \end{aligned} \tag{A.332}$$

No other chart contributes because its cell is empty.

Set only as inherited shorthand

$$V_d := d + \frac{Rd(d-1)}{2}. \tag{A.333}$$

Lemmas A.20 and A.21 give, on their complete chart domains,

$$|\partial_\theta T_1| \le V_d, \qquad |\partial_\theta T_d| \le 1 + \frac{Rd(d-1)}{2} \le V_d. \tag{A.334}$$

The outer-chart bound in (A.334) is valid for both signs of θ. Lemma A.22 assigns $0, \pm 1$ to the legal inner branch and confirms that empty cells and arbitrary interval locations require no alteration. Lemma A.23 now yields

$$\begin{aligned}
\Pr_{\alpha\sim\mu}[\exists \theta \in J : p_\alpha(\theta) = 0] &\le \kappa V_d (2R)^{d-1}|E_1| + \kappa V_d (2R)^{d-1}|E_d| \\
&= \kappa(2R)^{d-1} V_d(|E_1| + |E_d|) \\
&= \kappa(2R)^{d-1} V_d |J|.
\end{aligned} \tag{A.335}$$

The two chart integrals charge disjoint parameter cells; they do not produce a factor of two.

Now suppose $d = 1$. Proposition A.16 supplies the sole legal cell $E_1 = J$, the chart $T_1 = -\theta$, and

$$|\partial_\theta T_1| = 1 = V_1. \tag{A.336}$$

The $N = 1$ case of Proposition A.5 and Lemma A.23 give

$$\begin{aligned}
\Pr_{\alpha_0\sim\mu}[\exists \theta \in J : p_{\alpha_0}(\theta) = 0] &\le \kappa \int_J \int_{[-R,R]^0} \mathbf{1}\{|T_1(\theta)| \le R\} |\partial_\theta T_1(\theta)| \, d\lambda_0 \, d\theta \\
&\le \kappa \int_J \int_{[-R,R]^0} 1 \, d\lambda_0 \, d\theta \\
&= \kappa |J| \\
&= \kappa (2R)^0 \left(1 + \frac{R \cdot 1 \cdot 0}{2}\right) |J|.
\end{aligned} \tag{A.337}$$

In both branches, Proposition A.15 or Proposition A.16 inserts the pivot into the same original coefficient vector and gives $p_{\Psi_j(\theta,\beta)}(\theta) = 0$. Thus the chart vector represents the identical polynomial root event. $\square$

Lemma A.24 (Degenerate bounded intervals are law-null)**.** *Under Assumption 4 and Proposition A.14, let $d \ge 1$, let $J \subset \mathbb{R}$ be a bounded interval with $|J| = 0$, and let $\alpha \sim \mu$ have any Borel full joint density supported on $[-R, R]^d$ with $f_\mu \le \kappa$ almost everywhere. Then*

$$\Pr_{\alpha\sim\mu}[\exists \theta \in J : p_\alpha(\theta) = 0] = 0. \tag{A.338}$$

Consequently,

$$\Pr_{\alpha\sim\mu}[\exists \theta \in J : p_\alpha(\theta) = 0] \le \kappa (2R)^{d-1} \left(d + \frac{Rd(d-1)}{2}\right) |J| = 0. \tag{A.339}$$

Proof. If $J = \varnothing$, the event is empty. Otherwise, a real interval of length zero is a singleton, say $J = \{\theta_0\}$. Proposition A.14 identifies the event with

$$\left\{\alpha \in [-R, R]^d : \alpha_0 = -\theta_0^d - \sum_{k=1}^{d-1} \alpha_k \theta_0^k \right\}. \tag{A.340}$$

This is a closed affine-hyperplane slice because the coefficient of α_0 is exactly one. For each fixed $(\alpha_1, \ldots, \alpha_{d-1})$, its section in the α_0-coordinate is empty or a singleton and therefore has one-dimensional Lebesgue measure zero. Tonelli gives zero d-dimensional Lebesgue measure. When $d = 1$, the nonpivot tuple is empty and the same set is the singleton $\{\alpha_0 = -\theta_0\} \cap [-R, R]$, so the argument remains literal.

Applying the full joint-density cap directly,

$$\Pr_{\alpha\sim\mu}[p_\alpha(\theta_0) = 0] = \int_{\{\alpha:p_\alpha(\theta_0)=0\}} f_\mu(\alpha)\, d\lambda_d(\alpha) \le \kappa\lambda_d(\{\alpha : p_\alpha(\theta_0) = 0\}) = 0. \tag{A.341}$$

No independence, marginal or conditional density, polynomial-root theorem, or randomized leading coordinate is used. Since $|J| = 0$, the target right-hand side is also zero. □

Proposition A.18 (Complete affine-monic presentation, certificate, and root-probability wrapper)**.** *Under Assumption 4 in dimension d, Proposition A.5, Proposition A.14, Lemma A.19, Proposition A.15, Lemmas A.20 and A.21, Proposition A.16, Lemma A.22, Lemma A.23, Proposition A.17, and Lemma A.24, let $d \ge 1$, $R > 0$, $0 < \kappa < \infty$, and let $J \subset \mathbb{R}$ be any bounded interval. Let*

$$\alpha = (\alpha_0, \ldots, \alpha_{d-1}) \sim \mu, \tag{A.342}$$

where μ is any Borel probability law on $\mathbb{R}^d$ with one full joint density f_μ, supported on $[-R, R]^d$, satisfying $f_\mu \le \kappa$ almost everywhere; its coordinates may be arbitrarily correlated. Then all conclusions below hold simultaneously.

1. ***Actual augmented presentation and random object.*** *The tuple and polynomial are*

$$(F_0, F_1, \ldots, F_d) = (\theta^d, 1, \theta, \ldots, \theta^{d-1}), \qquad F = (1, \theta, \ldots, \theta^{d-1}), \tag{A.343}$$

$$F_0(\theta) + \langle \alpha, F(\theta)\rangle = \theta^d + \sum_{k=0}^{d-1} \alpha_k \theta^k = p_\alpha(\theta) \quad \text{for every } \theta \in \mathbb{R}. \tag{A.344}$$

The vector $\alpha \in [-R, R]^d$ consists of exactly the d lower coefficients. The leading coefficient one is deterministic, belongs to $F_0 = \theta^d$, and is not a random or singular additional coordinate. The exact presentation parameters are

$$q = M = m = 0, \qquad \Delta = N = d, \qquad A = (2R)^d\kappa. \tag{A.345}$$

2. ***Exact constant derivative-shift certificate.*** *Index the rows and columns by $0, 1, \ldots, d$. Define the constant $(d + 1)$-square matrix by*

$$B_{rs} = \begin{cases} d, & (r, s) = (0, d), \\ k, & (r, s) = (k + 1, k) \text{ for some } 1 \le k \le d - 1, \\ 0, & \text{otherwise.} \end{cases} \tag{A.346}$$

Equivalently, its exactly nonzero entries are

$$B_{0,d} = d, \qquad B_{k+1,k} = k \quad (1 \le k \le d - 1), \tag{A.347}$$

and

$$\widetilde{F}'(\theta) = B\widetilde{F}(\theta) \quad \text{for every } \theta. \tag{A.348}$$

Its exact setting certificate is

$$\widehat{\Lambda}_{B,T} = \left(\sum_{k=1}^{d} k^2\right)^{1/2}. \tag{A.349}$$

Because $m = 0$, this equality contains no interval-location factor and no hidden T factor. At $d = 1$,

$$B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \widehat{\Lambda}_{B,T} = 1. \tag{A.350}$$

3. ***Legal charts and velocities for*** $d \geq 2$***.*** *The cells*

$$E_1 = J \cap \{|\theta| \leq 1\}, \qquad E_d = J \cap \{|\theta| > 1\}, \qquad E_j = \varnothing \quad (2 \leq j \leq d-1) \tag{A.351}$$

form a measurable disjoint legal partition of J, with

$$|E_1| + |E_d| = |J|. \tag{A.352}$$

On E_1, the legal pivot is the original coefficient α_0 multiplying $F_1 = 1$, and

$$T_1(\theta,\beta) = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k, \qquad \Psi_1(\theta,\beta) = (T_1(\theta,\beta), \beta_1, \dots, \beta_{d-1}). \tag{A.353}$$

On E_d, the legal pivot is the original coefficient α_{d-1} multiplying $F_d = \theta^{d-1} \neq 0$, and

$$T_d(\theta,\beta) = -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1}, \qquad \Psi_d(\theta,\beta) = (\beta_0, \dots, \beta_{d-2}, T_d(\theta,\beta)). \tag{A.354}$$

In both cases $\beta \in [-R,R]^{d-1}$, and

$$\lambda_{d-1}([-R,R]^{d-1}) = (2R)^{d-1}. \tag{A.355}$$

The insertion maps are in the original order $(\alpha_0, \dots, \alpha_{d-1})$, and

$$F_0(\theta) + \langle \Psi_1(\theta,\beta), F(\theta) \rangle = p_{\Psi_1(\theta,\beta)}(\theta) = 0, \tag{A.356}$$

$$F_0(\theta) + \langle \Psi_d(\theta,\beta), F(\theta) \rangle = p_{\Psi_d(\theta,\beta)}(\theta) = 0. \tag{A.357}$$

Thus either chart vector gives the identical target polynomial. The inserted vector belongs to the support cube exactly when its pivot also satisfies $|T_1| \leq R$ or $|T_d| \leq R$, respectively.

With

$$V_d := d + \frac{Rd(d-1)}{2}, \tag{A.358}$$

the exact derivative formulas and bounds are

$$\partial_\theta T_1 = -d\theta^{d-1} - \sum_{k=1}^{d-1} k\beta_k \theta^{k-1}, \qquad |\partial_\theta T_1| \leq V_d, \tag{A.359}$$

$$\partial_\theta T_d = -1 + \sum_{k=0}^{d-2} (d-1-k)\beta_k \theta^{k-d}, \tag{A.360}$$

$$|\partial_\theta T_d| \leq 1 + R \sum_{m=1}^{d-1} \frac{m}{|\theta|^{m+1}} \leq 1 + \frac{Rd(d-1)}{2} \leq V_d. \tag{A.361}$$

4. ***Separate linear chart.*** *At $d = 1$,*

$$(F_0, F_1) = (\theta, 1), \qquad p_{\alpha_0}(\theta) = \theta + \alpha_0, \qquad E_1 = J, \tag{A.362}$$

and the beta space is genuinely zero-dimensional:

$$[-R,R]^0=\{()\},\qquad \lambda_0([-R,R]^0)=1=(2R)^0. \tag{A.363}$$

The sole legal chart is

$$T_1(\theta)=-\theta,\qquad \Psi_1(\theta)=(-\theta),\qquad p_{\Psi_1(\theta)}(\theta)=0, \tag{A.364}$$

and

$$|\partial_\theta T_1|=1=V_1. \tag{A.365}$$

No beta coordinate is introduced.

5. ***Boundary, dimension, and location closure.*** *The inner chart handles $\theta=0,\theta=1,\theta=-1$; the outer chart handles both signs of θ whenever $|\theta|>1$. Empty J, empty active cells, and intervals lying in only one region require no change. At $d=2$, the generic formulas specialize to*

$$(F_0,F_1,F_2)=(\theta^2,1,\theta),\qquad B=\begin{pmatrix}0&0&2\\0&0&0\\0&1&0\end{pmatrix},\qquad \widehat{\Lambda}_{B,T}=\sqrt{5}, \tag{A.366}$$

$$T_1=-\theta^2-\beta_1\theta,\qquad T_2=-\theta-\beta_0\theta^{-1},\qquad |\partial_\theta T_1|\le 2+R,\qquad |\partial_\theta T_2|\le 1+R\le 2+R. \tag{A.367}$$

Every coefficient vector has dimension d, every beta cube has dimension $d-1$, and B has dimension $d+1$. None of the presentation, certificate, chart, velocity, or probability conclusions depends on the location of J.

6. ***Exact ordinary-probability bridge for every bounded interval.*** *For the same original law μ, with no independence assumption and for every bounded interval J, including empty and singleton intervals,*

$$\boxed{\Pr_{\alpha\sim\mu}[\exists\theta\in J: p_\alpha(\theta)=0]\le\kappa(2R)^{d-1}\left(d+\frac{Rd(d-1)}{2}\right)|J|.} \tag{A.368}$$

When $|J|=0$, the event is empty or a proper affine-hyperplane slice and has probability zero under the same full joint density. When $d=1$, this bound is exactly $\kappa|J|$.

Proof. Proposition A.14 gives clause 1 exactly. Lemma A.19 gives clause 2 exactly, including the $d=1$ matrix and the absence of a T factor. Proposition A.15, Lemmas A.20 and A.21, and Lemma A.23 give clause 3. Proposition A.16 and Lemma A.23 give clause 4. Lemma A.22 gives clause 5, including the explicit $d=2$ specialization and arbitrary interval location.

For $|J|>0$, Proposition A.17 applies Proposition A.5 to these exact original-coordinate charts, uses the same full joint-density cap, and proves clause 6. For $|J|=0$, Lemma A.24 proves clause 6 by affine-hyperplane nullity. These cases exhaust every bounded interval.

Every item in the statement is therefore either a conclusion of the named results or the integration and nullity argument proved here. This completes the proposition. □

Synthesis of affine-monic probability recovery. Proposition A.18 combines the exact d-coordinate monic presentation, constant derivative-shift matrix, certificate, legal chart cells, insertion maps, regional velocity bounds, zero-dimensional $d=1$ chart, and all sign, boundary, dimension, empty-cell, and interval-location cases. For $|J|>0$ its probability clause follows from Proposition A.17, which applies Proposition A.5 to the original coefficient coordinates and uses the exact beta-cube volume from Lemma A.23. For $|J| = 0$, Lemma A.24 gives probability zero under the same full joint density. Hence, for every bounded interval and every possibly correlated admissible law,

$$\Pr_{\alpha\sim\mu}[\exists\theta \in J : p_\alpha(\theta) = 0] \le \kappa(2R)^{d-1}\left(d + \frac{Rd(d-1)}{2}\right)|J|. \tag{A.369}$$

This is one complete presentation, certificate, and probability result; it uses neither an independent polynomial-root theorem nor a random leading coordinate. $\square$

A.11 Counter-example 1 Scale Calculation

Lemma A.25 (Exact one-entry derivative certificate)**.** *Under Assumptions 1, 2, and 3, suppose* $0 < \delta \le 1$, $\Theta = [-1, 1]$, *and, in the exact coordinate ordering* $0, 1, 2$,

$$\widetilde{F}(\theta) = \begin{pmatrix} F_0(\theta) \\ F_1(\theta) \\ F_2(\theta) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ \theta/\delta \end{pmatrix}. \tag{A.370}$$

Then

$$q = 0, \qquad M = 0, \qquad \Delta = 1, \qquad N = 2, \qquad m = 0, \tag{A.371}$$

and the constant matrix

$$B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1/\delta & 0 \end{pmatrix} \tag{A.372}$$

has the sole nonzero entry $B_{2,1} = 1/\delta$, *satisfies*

$$\widetilde{F}'(\theta) = B\widetilde{F}(\theta), \tag{A.373}$$

and has exact static certificate

$$\widehat{\Lambda}_{B,T} = \frac{1}{\delta}. \tag{A.374}$$

This value is independent of the legal containing parameter T *satisfying* $[-1, 1] \subseteq [-T, T]$.

Proof. The chain is empty, so $q = 0$ and the setting convention gives $M = 0$. The three output polynomials are $Q_0 = 0$, $Q_1 = 1$, and $Q_2(\theta) = \theta/\delta$; their maximum total degree is one, so $\Delta = 1$. There are two random-coordinate features, hence $N = 2$. This matrix is constant, so $m = 0$, and direct multiplication in the stated row-column ordering gives

$$B\widetilde{F}(\theta) = \begin{pmatrix} 0 \\ 0 \\ (1/\delta)F_1(\theta) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1/\delta \end{pmatrix} = \widetilde{F}'(\theta). \tag{A.375}$$

Thus this is the supplied anchored closure matrix, with $F_1 \equiv 1$ as the anchor. Its coefficient list has only $b_{2,1,0} = 1/\delta$. Since $m = 0$, $T_*^0 = 1$, and therefore the setting definition gives

$$\begin{aligned} \widehat{\Lambda}_{B,T} &= \left(\sum_{r=0}^{2}\sum_{s=0}^{2}\left(\sum_{\ell=0}^{0}|b_{rs,\ell}|T_*^\ell\right)^2\right)^{1/2} \\ &= \left(\left|\frac{1}{\delta}\right|^2\right)^{1/2} = \frac{1}{\delta}, \end{aligned} \tag{A.376}$$

where the last equality uses $\delta > 0$. This direct calculation agrees with the one-entry-shear specialization in Proposition A.2. Since the only power is $T_*^0 = 1$, changing the legal containing T changes neither the coefficient list nor the certificate. The certificate is therefore a raw Euclidean coefficient-height calculation, not a value inferred from a later probability claim.

□

Lemma A.26 (Euclidean projective speed)**.** *Under Assumptions 1, 2, and 3, and Lemma A.25, let $F(\theta) = (1, \theta/\delta)$ on $\Theta = [-1, 1]$, where $0 < \delta \le 1$, and set $x = \theta/\delta$. Then*

$$\gamma_F(\theta) = \frac{(1, x)}{\sqrt{1 + x^2}}, \qquad \gamma_F'(\theta) = \frac{(-x, 1)}{\delta(1 + x^2)^{3/2}}, \tag{A.377}$$

so

$$\|\gamma_F'(\theta)\|_2 = \frac{1}{\delta(1 + x^2)}, \qquad \Gamma_{\text{proj}}(F) = \frac{1}{\delta}. \tag{A.378}$$

At $\theta = 0$,

$$\gamma_F(0) = (1, 0), \qquad \gamma_F'(0) = (0, 1/\delta), \qquad \|\gamma_F'(0)\|_2 = 1/\delta. \tag{A.379}$$

Proof. The Euclidean norm is

$$\|F(\theta)\|_2 = \sqrt{1 + x^2} > 0, \tag{A.380}$$

so normalization is valid everywhere, including the endpoints. Since $x' = 1/\delta$, direct differentiation gives

$$\begin{aligned} \gamma_F'(\theta) &= \frac{1}{\delta} \frac{d}{dx} \Big((1, x)(1 + x^2)^{-1/2} \Big) \\ &= \frac{1}{\delta} \left(-\frac{x}{(1 + x^2)^{3/2}}, \frac{1}{(1 + x^2)^{3/2}} \right). \end{aligned} \tag{A.381}$$

Taking the Euclidean norm, without changing the parameter coordinate, yields

$$\begin{aligned} \|\gamma_F'(\theta)\|_2 &= \frac{1}{\delta} \frac{\sqrt{x^2 + 1}}{(1 + x^2)^{3/2}} \\ &= \frac{1}{\delta(1 + x^2)}. \end{aligned} \tag{A.382}$$

Because $1 + x^2 \ge 1$, this is at most $1/\delta$. The point $\theta = 0 \in [-1, 1]$ has $x = 0$ and attains equality, so the ordinary supremum is exactly $1/\delta$. Substitution at $x = 0$ gives the stated active vector value. The projective certificate from Proposition A.1 therefore holds with equality for this instance. □

Proposition A.19 (Exact homogeneous upper coefficient)**.** *Under Assumptions 1, 2, 4, and 3, the Propositions A.12 and A.13, and Lemmas A.25 and A.26, specialize to the following statement. For every possibly correlated law $\mu \in \mathcal{D}_{2,1,1/4}$ and every interval $I \subseteq [-1, 1]$ with $|I| > 0$,*

$$\Pr_{\alpha \sim \mu}[\exists \theta \in I : \alpha_1 + \alpha_2 \theta/\delta = 0] \le \frac{1}{\delta}|I|, \tag{A.383}$$

and

$$C_{\mathcal{D}}^{\text{Pf}}(F; [-1, 1]) \le \frac{1}{\delta}. \tag{A.384}$$

Proof. For $N=2$, $R=1$, and $\kappa=1/4$, the setting quantity is

$$A=(2R)^N\kappa=2^2\cdot\frac{1}{4}=1,\qquad A\sqrt{\frac{N}{2}}=1\cdot\sqrt{\frac{2}{2}}=1. \tag{A.385}$$

Proposition A.12 applies to every law in the original class, without an independence hypothesis, and gives

$$\begin{aligned}\Pr_{\alpha\sim\mu}[\exists\theta\in I:\alpha_1+\alpha_2\theta/\delta=0]&\le A\sqrt{\frac{N}{2}}\,\Gamma_{\text{proj}}(F)|I|\\&=\frac{1}{\delta}|I|.\end{aligned} \tag{A.386}$$

Here Lemma A.26 supplies the exact equality $\Gamma_{\text{proj}}(F)=1/\delta$. The certificate version has the identical coefficient because Lemma A.25 gives $\widehat{\Lambda}_{B,T}=1/\delta$. Proposition A.13, with the same substitution, gives the capacity bound. Neither bound is clipped at one. No feature of the particular independent uniform law is used in this upper bound. ☐

Lemma A.27 (Exact root-wedge equivalence)**.** *Under Assumptions 1 and 3, Lemma A.25, and $0<\epsilon\le\delta\le1$, define*

$$t:=\frac{\epsilon}{\delta}\in(0,1], \tag{A.387}$$

and the two closed subsets of $[-1,1]^2$

$$\begin{aligned}W_+(t)&:=\{(\alpha_1,\alpha_2):0\le\alpha_2\le1,-t\alpha_2\le\alpha_1\le0\},\\W_-(t)&:=\{(\alpha_1,\alpha_2):-1\le\alpha_2\le0,0\le\alpha_1\le-t\alpha_2\}.\end{aligned} \tag{A.388}$$

Then the full coefficient event is exactly

$$\{\alpha\in[-1,1]^2:\exists\theta\in[0,\epsilon],\ \alpha_1+\alpha_2\theta/\delta=0\}=W_+(t)\cup W_-(t). \tag{A.389}$$

Off the coefficient axes this is equivalently the union of both opposite-sign wedges

$$\alpha_1\alpha_2<0,\qquad|\alpha_1|\le t|\alpha_2|. \tag{A.390}$$

The boundary $\alpha_1=0$ gives the endpoint root $\theta=0$, the equality boundary $|\alpha_1|=t|\alpha_2|$ gives the endpoint root $\theta=\epsilon$, and the only root with $\alpha_2=0$ is the origin, where the function is identically zero.

Proof. Write $u=\theta/\delta$. Since $\delta>0$, the range $\theta\in[0,\epsilon]$ is exactly $u\in[0,t]$, and the root equation becomes

$$\alpha_1+\alpha_2u=0. \tag{A.391}$$

If $\alpha_2>0$, the unique candidate is $u=-\alpha_1/\alpha_2$, and

$$0\le-\frac{\alpha_1}{\alpha_2}\le t\quad\Longleftrightarrow\quad-t\alpha_2\le\alpha_1\le0. \tag{A.392}$$

This is exactly the positive-α_2 part of $W_+(t)$. If $\alpha_2<0$, division reverses the inequalities and gives

$$0\le-\frac{\alpha_1}{\alpha_2}\le t\quad\Longleftrightarrow\quad0\le\alpha_1\le-t\alpha_2, \tag{A.393}$$

which is exactly the negative-α_2 part of $W_-(t)$. If $\alpha_2 = 0$, the equation has a root if and only if $\alpha_1 = 0$; this origin belongs to both closed wedges. These three exhaustive cases prove the event equality.

Because $t \le 1$, the bounds $|\alpha_1| \le t|\alpha_2| \le 1$ keep both wedges inside the coefficient square, including at $t = 1$. Away from the axes, the inequalities force strict opposite signs. The candidate root is $u = 0$ when $\alpha_1 = 0$ and $u = t$ on the sloping equality boundary, proving the two endpoint assertions rather than discarding their coefficient sets. $\square$

Proposition A.20 (Exact wedge probability for the uniform source law)**.** *Under Assumption 4, let α_1, α_2 be independent uniform random variables on $[-1, 1]$, solely for the required source example. Under the event identity of Lemma A.27, for every $0 < \epsilon \le \delta \le 1$,*

$$\lambda_2(W_+(\epsilon/\delta) \cup W_-(\epsilon/\delta)) = \frac{\epsilon}{\delta}, \tag{A.394}$$

and

$$\Pr[\exists\theta \in [0, \epsilon] : \alpha_1 + \alpha_2\theta/\delta = 0] = \frac{\epsilon}{4\delta}. \tag{A.395}$$

Proof. Retain $t = \epsilon/\delta \in (0, 1]$. At a fixed $\alpha_2 = s \in [0, 1]$, the $W_+(t)$ slice in the α_1-coordinate is $[-ts, 0]$, of length ts. Hence

$$\lambda_2(W_+(t)) = \int_0^1 ts\, ds = \frac{t}{2}. \tag{A.396}$$

For $W_-(t)$, substitute $s = -\alpha_2 \in [0, 1]$; its slice length is again ts, so

$$\lambda_2(W_-(t)) = \int_0^1 ts\, ds = \frac{t}{2}. \tag{A.397}$$

The two wedges intersect only at the origin. Therefore

$$\lambda_2(W_+(t) \cup W_-(t)) = 2\int_0^1 ts\, ds = t = \frac{\epsilon}{\delta}. \tag{A.398}$$

This is the combined area of both sign quadrants, not the area of one wedge. The coefficient axes, the two sloping equality edges, and the two outer square-edge segments are finite unions of line segments and hence have planar Lebesgue measure zero. All closed boundaries were included in the exact event in Lemma A.27; their nullity is used only when evaluating area, not to remove either endpoint root or any square-boundary coefficient from the event.

The selected product law has joint density exactly $1/4$ on the square. Integrating that density over the exact event gives

$$\Pr[\exists\theta \in [0, \epsilon] : \alpha_1 + \alpha_2\theta/\delta = 0] = \frac{1}{4}\lambda_2(W_+(t) \cup W_-(t)) = \frac{\epsilon}{4\delta}. \tag{A.399}$$

At $\epsilon = \delta$, $t = 1$, the two triangles have total area one and the probability is $1/4$. At $\delta = 1$, the same calculation holds for every $0 < \epsilon \le 1$. The $\theta = 0$ endpoint roots and the $\theta = \epsilon$ endpoint roots are already included exactly. $\square$

Proposition A.21 (Exact normalized upper/lower scale comparison)**.** *Under Lemma A.25 and Propositions A.19 and A.20, for the selected uniform source law and every $0 < \epsilon \le \delta \le 1$,*

$$\frac{\Pr[\exists\theta \in [0, \epsilon] : \alpha_1 + \alpha_2\theta/\delta = 0]}{|[0, \epsilon]|} = \frac{1}{4\delta}. \tag{A.400}$$

The three quantities are therefore:

$$\begin{aligned}\textit{selected-law lower ratio} &= \frac{1}{4\delta},\\ \textit{general upper coefficient} &= \frac{1}{\delta},\\ \textit{raw one-entry certificate} &= \frac{1}{\delta}.\end{aligned} \tag{A.401}$$

The first quantity is smaller by the literal factor four; it is not asserted to equal either of the latter two, and no optimal upper constant is claimed. At $\epsilon = \delta$ and at $\delta = 1$ these values remain literal. For fixed $\delta > 0$, the notation $\epsilon \downarrow 0$ refers only to the limit through positive interval lengths, for which the ratio remains $1/(4\delta)$; no ratio is asserted at $\epsilon = 0$.

Proof. Since $\epsilon > 0$, the interval has positive length $|[0, \epsilon]| = \epsilon$, so Proposition A.20 may be divided by that length:

$$\frac{\epsilon/(4\delta)}{\epsilon} = \frac{1}{4\delta}. \tag{A.402}$$

Proposition A.19 supplies $1/\delta$ from the general theorem, not from the direct wedge calculation. Separately, Lemma A.25 supplies the raw Euclidean coefficient height $1/\delta$ directly from the single matrix entry. Thus the lower test ratio is $1/4$ times each comparison scale, while the upper theorem and certificate retain logically distinct mathematical roles. If $\epsilon \downarrow 0$ with $\epsilon > 0$, the probability $\epsilon/(4\delta)$ tends to zero while the normalized ratio remains constant for fixed $\delta > 0$; at $\epsilon = 0$, division is not defined and is not used. □

Synthesis of the Counter-example 1 scales. For $0 < \delta \leq 1$, Lemma A.25 gives

$$(q, M, \Delta, N, m) = (0, 0, 1, 2, 0), \qquad B_{2,1} = \frac{1}{\delta}, \qquad \widehat{\Lambda}_{B,T} = \frac{1}{\delta}, \tag{A.403}$$

with no other nonzero closure entry. Lemma A.26 independently gives

$$\|\gamma_F'(\theta)\|_2 = \frac{1}{\delta(1 + (\theta/\delta)^2)}, \qquad \Gamma_{\text{proj}}(F) = \frac{1}{\delta}. \tag{A.404}$$

With $A = 1$ and $N = 2$, Proposition A.19 therefore gives the coefficient $1/\delta$ for every admissible, possibly correlated law. For the selected uniform law only, Lemma A.27 identifies both closed opposite-sign wedges, and Proposition A.20 computes their combined area and probability as

$$\lambda_2(W_+(\epsilon/\delta) \cup W_-(\epsilon/\delta)) = \frac{\epsilon}{\delta}, \qquad \Pr[\exists\theta \in [0, \epsilon] : \alpha_1 + \alpha_2\theta/\delta = 0] = \frac{\epsilon}{4\delta}. \tag{A.405}$$

Proposition A.21 divides only by positive ϵ and gives the selected-law ratio $1/(4\delta)$, distinct from the all-law upper coefficient and raw certificate, both $1/\delta$. The deterministic-leading-coefficient monic baseline is unaffected. □

A.12 Anchored Coefficient-Sweep Theorem

Proposition A.22 (Anchored derivative-closure coefficient sweep)**.** *Under Assumptions 1, 2, 3, and 4, with each conclusion using only the assumption subsets stated in the listed results, use the following named theorem-style conclusions with exactly their stated scopes:*

Lemma A.1, Lemma A.2, Lemma A.3,
Proposition A.1, Proposition A.2;
Lemma A.6, Lemma A.7, Proposition A.4,
Lemma A.8, Proposition A.5;
Proposition A.10;
Lemma A.17, Proposition A.11, (A.406)
Lemma A.18, Proposition A.12, Proposition A.13;
Proposition A.18;
Lemma A.25, Lemma A.26, Proposition A.19,
Lemma A.27, Proposition A.20, Proposition A.21.

Then all of the following conclusions hold simultaneously.
1. ***Static certificate and homogeneous closure.*** *For every $\theta \in \Theta$,*

$$F_{j_*}(\theta) = 1, \qquad \|F(\theta)\|_2 \geq 1, \qquad F(\theta) \neq 0, \tag{A.407}$$

and

$$\|B(\theta)\|_{\mathrm{op}} \leq \|B(\theta)\|_{\mathrm{F}} \leq \widehat{\Lambda}_{B,T}, \qquad \sup_{\vartheta \in \Theta} \|B(\vartheta)\|_{\mathrm{op}} \leq \widehat{\Lambda}_{B,T}. \tag{A.408}$$

If $F_0 \equiv 0$, then

$$F' = B_F F, \qquad \gamma_F' = (I_N - \gamma_F \gamma_F^{\mathsf{T}}) B_F \gamma_F, \qquad \Gamma_{\mathrm{proj}}(F) \leq \widehat{\Lambda}_{B,T}. \tag{A.409}$$

These conclusions remain literal for $q = 0$, $m = 0$, constant B, constant $\widetilde{F}$, $N = 1$, both endpoints of Θ, zero certificate height, and stationary γ_F. The two fixed-presentation certificate reductions are

$$\widehat{\Lambda}_{B,T} = \left(\sum_{k=1}^{d} k^2 \right)^{1/2} \quad \text{for the augmented monic shift}, \qquad \widehat{\Lambda}_{B,T} = \frac{1}{\delta} \quad \text{for the one-entry shear.} \tag{A.410}$$

2. ***Equivalent chart form and general affine theorem.*** *For every $\mu \in \mathcal{D}_{N,R,\kappa}$, every interval $I \subseteq \Theta$ with $|I| > 0$, and every Lebesgue-measurable legal partition $I = \bigsqcup_{j=1}^{N} E_j$ with $F_j \neq 0$ on E_j,*

$$\begin{aligned} \Pr_{\alpha \sim \mu}[\exists \theta \in I : \phi_\alpha(\theta) = 0] &\leq \kappa \sum_{j=1}^{N} \int_{E_j} \int_{[-R,R]^{N-1}} \mathbf{1}\{|T_j(\theta, \beta)| \leq R\} |\partial_\theta T_j(\theta, \beta)| \, d\beta \, d\theta \\ &\leq \kappa \sum_{j=1}^{N} \int_{E_j} \int_{[-R,R]^{N-1}} |\partial_\theta T_j(\theta, \beta)| \, d\beta \, d\theta. \end{aligned} \tag{A.411}$$

Independently of the choice of legal partition, every such law and positive-length interval satisfies the coordinate-free chain

$$\begin{aligned}\Pr_{\alpha\sim\mu}[\exists\theta\in I:\phi_\alpha(\theta)=0]&\le\kappa\int_I\int_{H_\theta\cap[-R,R]^N}\frac{|F_0'(\theta)+\langle a,F'(\theta)\rangle|}{\|F(\theta)\|_2}\,d\mathcal{H}^{N-1}(a)\,d\theta\\&\le\kappa\sqrt{2}(2R)^{N-1}(1+NR^2)\widehat{\Lambda}_{B,T}|I|\\&=\frac{A(1+NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}|I|,\end{aligned}\tag{A.412}$$

and

$$C^{\mathrm{aff}}_{\mathcal{D}}(F_0,F;\Theta)\le\frac{A(1+NR^2)\widehat{\Lambda}_{B,T}}{\sqrt{2}R}.\tag{A.413}$$

If $\widehat{\Lambda}_{B,T}=0$, the event probability is exactly zero for every such law and interval. The chart conclusion includes completed-measurable validation, finite localized charts, exact area with multiplicity, persistent-root probability zero, arbitrary correlation, empty cells, zero Jacobians, tangent and multiple roots, every endpoint convention, pivots with no uniform margin, and $N=1$, without adding the finite exhaustion variables to the conclusion.

*3. **Sharper homogeneous theorem.** If $F_0\equiv0$, then on the actual central section*

$$H_\theta=F(\theta)^\perp=\gamma_F(\theta)^\perp,\qquad\frac{|\langle a,F'(\theta)\rangle|}{\|F(\theta)\|_2}=|\langle a,\gamma_F'(\theta)\rangle|\quad(a\in H_\theta).\tag{A.414}$$

For every admissible law and every positive-length interval,

$$\Pr_{\alpha\sim\mu}[\exists\theta\in I:\langle\alpha,F(\theta)\rangle=0]\le A\sqrt{\frac{N}{2}}\,\Gamma_{\mathrm{proj}}(F)|I|\le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}|I|,\tag{A.415}$$

and

$$C^{\mathrm{Pf}}_{\mathcal{D}}(F;\Theta)\le A\sqrt{\frac{N}{2}}\,\widehat{\Lambda}_{B,T}.\tag{A.416}$$

If $\Gamma_{\mathrm{proj}}(F)=0$, the normalized curve is constant, every nonempty-interval root event is one fixed proper central hyperplane, and every admissible full-density law assigns it probability zero. The coefficient identity

$$\kappa R\sqrt{N}\,\sqrt{2}(2R)^{N-1}=A\sqrt{\frac{N}{2}}\tag{A.417}$$

is literal. The statements retain $N=1$, stationary and zero-certificate branches, radial rescaling, $a=0$, literal endpoints, and arbitrarily short positive intervals.

*4. **Complete exact affine-monic baseline.** For every $d\ge1$, $R>0$, $0<\kappa<\infty$, every bounded interval $J\subset\mathbb{R}$, and every arbitrary possibly correlated d-dimensional law described in Proposition A.18, all six clauses of Proposition A.18 hold verbatim. In particular, the exact tuple, object identity, descriptor values, and certificate are*

$$(F_0,F_1,\dots,F_d)=(\theta^d,1,\theta,\dots,\theta^{d-1}),\quad p_\alpha(\theta)=F_0(\theta)+\langle\alpha,F(\theta)\rangle=\theta^d+\sum_{k=0}^{d-1}\alpha_k\theta^k,\tag{A.418}$$

$$q=M=m=0,\quad\Delta=N=d,\quad A=(2R)^d\kappa,\quad\widehat{\Lambda}_{B,T}=\left(\sum_{k=1}^{d}k^2\right)^{1/2},\tag{A.419}$$

with exactly the derivative-shift entries

$$B_{0,d} = d, \qquad B_{k+1,k} = k \quad (1 \le k \le d-1). \tag{A.420}$$

The random vector contains exactly the d lower coefficients; the leading coefficient one is deterministic. For $d \ge 2$, the exact legal cells, original-coordinate pivots, charts, insertion maps, beta volume, same-polynomial identities, and velocity formulas and bounds are

$$E_1 = J \cap \{|\theta| \le 1\}, \qquad E_d = J \cap \{|\theta| > 1\}, \qquad E_j = \varnothing \quad (2 \le j \le d-1), \qquad |E_1| + |E_d| = |J|, \tag{A.421}$$

$$T_1 = -\theta^d - \sum_{k=1}^{d-1} \beta_k \theta^k, \qquad \Psi_1 = (T_1, \beta_1, \dots, \beta_{d-1}), \tag{A.422}$$

$$T_d = -\theta - \sum_{k=0}^{d-2} \beta_k \theta^{k-d+1}, \qquad \Psi_d = (\beta_0, \dots, \beta_{d-2}, T_d), \tag{A.423}$$

$$\lambda_{d-1}([-R, R]^{d-1}) = (2R)^{d-1}, \qquad p_{\Psi_1(\theta,\beta)}(\theta) = p_{\Psi_d(\theta,\beta)}(\theta) = 0, \tag{A.424}$$

$$\partial_\theta T_1 = -d\theta^{d-1} - \sum_{k=1}^{d-1} k\beta_k \theta^{k-1}, \qquad |\partial_\theta T_1| \le d + \frac{Rd(d-1)}{2}, \tag{A.425}$$

$$\partial_\theta T_d = -1 + \sum_{k=0}^{d-2} (d-1-k)\beta_k \theta^{k-d}, \tag{A.426}$$

$$|\partial_\theta T_d| \le 1 + R \sum_{m=1}^{d-1} \frac{m}{|\theta|^{m+1}} \le 1 + \frac{Rd(d-1)}{2} \le d + \frac{Rd(d-1)}{2}. \tag{A.427}$$

At $d = 1$, $B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\widehat{\Lambda}_{B,T} = 1$, $E_1 = J$, the sole pivot is α_0, the beta mass is one, $T_1 = -\theta$, and $|\partial_\theta T_1| = 1$. Every boundary, sign, empty-cell, dimension, and interval-location branch stated by the wrapper is retained. Finally, for every such bounded J, including empty and singleton intervals,

$$\Pr_{\alpha \sim \mu}[\exists \theta \in J : p_\alpha(\theta) = 0] \le \kappa (2R)^{d-1} \left(d + \frac{Rd(d-1)}{2} \right) |J|. \tag{A.428}$$

*5. **Counter-example 1 scale calculation.** For $0 < \delta \le 1$, $\Theta = [-1, 1]$, $\widetilde{F} = (0, 1, \theta/\delta)$, $R = 1$, $\kappa = 1/4$, and $A = 1$,*

$$q = M = 0, \qquad \Delta = 1, \qquad N = 2, \qquad m = 0, \tag{A.429}$$

and the sole nonzero closure entry is $B_{2,1} = 1/\delta$, with

$$\widehat{\Lambda}_{B,T} = \Gamma_{\mathrm{proj}}(F) = \frac{1}{\delta}. \tag{A.430}$$

For every possibly correlated law in $\mathcal{D}_{2,1,1/4}$ and every positive-length interval $I \subseteq [-1, 1]$,

$$\Pr_{\alpha \sim \mu}[\exists \theta \in I : \alpha_1 + \alpha_2 \theta/\delta = 0] \le \frac{1}{\delta} |I|, \qquad C_{\mathcal{D}}^{\mathrm{Pf}}(F; [-1, 1]) \le \frac{1}{\delta}. \tag{A.431}$$

For the selected independent uniform law on $[-1,1]^2$ and every $0<\epsilon\le\delta$, the complete closed two-wedge event includes both sign branches, coefficient axes, square boundaries, and the roots at $\theta=0$ and $\theta=\epsilon$, and

$$\Pr[\exists\theta\in[0,\epsilon]:\alpha_1+\alpha_2\theta/\delta=0]=\frac{\epsilon}{4\delta}. \tag{A.432}$$

Consequently the selected-law positive-length lower ratio is exactly $1/(4\delta)$, separately from the all-law upper coefficient $1/\delta$ and raw certificate $1/\delta$. No optimality claim is made.

6. ***Dependence, mode, baseline, and nonoutput closure.*** *The deterministic instance $(\Theta,T,q,M,\Delta,N,R,\kappa,A,m,B)$ is fixed before a law and interval are selected. The law may have arbitrary coefficient correlation; probability is ordinary probability for each fixed law; every general interval statement ranges over all positive-length subintervals with its literal endpoint convention; and each capacity takes the interval supremum for a fixed law before the outer law supremum. All vector, operator, and Frobenius norms are Euclidean, and section measure is Euclidean Hausdorff measure, with the supplied (N=1) convention. Every stated constant is literal, there is no hidden constant or confidence parameter, and once $\widehat{\Lambda}_{B,T}$ is fixed its additional dependence on q,M,Δ is exactly degree zero.*

The exact monic deterministic-leading-coefficient theorem and the Counter-example $1/\delta$ upper versus $1/(4\delta)$ lower scale are preserved without a remainder, changed mode, or surrogate object. No result here asserts that every unrestricted raw Pfaffian presentation admits the supplied derivative-closure certificate or a polynomial presentation-format bound. No independence is claimed outside the selected uniform lower-law computation; no random leading coordinate, chart-count factor, interval-location factor, auxiliary tolerance, new source result, term absorption, new probability conversion, or proof-local general exhaustion object is introduced.

Proof. We verify each clause of the conjunction.

For conclusion 1, Lemma A.1, Lemma A.2, Lemma A.3, Proposition A.1, and Proposition A.2 give exactly the static, homogeneous, projective, endpoint, stationary, zero-height, and certificate-baseline statements. Nothing is inferred beyond those conclusions.

For the chart portion of conclusion 2, Lemma A.6 supplies measurable completed domains and exact exhaustion, Lemma A.7 supplies finite localized charts and the determinant, Proposition A.4 supplies exact area/multiplicity and joint density control, Lemma A.8 supplies persistent-root removal and full finite-chart coverage, and Proposition A.5 supplies both public chart inequalities. The finite domains, images, and exhaustion levels establish the chart bounds but do not appear in the present conclusion. For the coordinate-free rate, capacity, and zero-height portion, Proposition A.10 supplies every conclusion with the same objects and modes.

For conclusion 3, Lemma A.17 supplies the exact actual-section identity; Proposition A.11 supplies the fixed law-null stationary branch; Lemma A.18 supplies the literal coefficient equality; Proposition A.12 supplies the two pairwise rates; and Proposition A.13 supplies the capacity bound and its exact supremum order.

For conclusion 4, Proposition A.18 gives the result directly. Its six-clause statement already contains the tuple, coefficient dimension, deterministic leader, descriptor values, full shift matrix and (d=1) branch, certificate, cells, pivots, charts, insertion maps, beta volume, velocity bounds, all boundary/location branches, zero-length nullity, and exact probability inequality. No other monic result is needed.

For conclusion 5, Lemma A.25 supplies the instance and exact certificate, Lemma A.26 supplies the exact Euclidean projective speed, Proposition A.19 supplies the all-law upper statements, Lemma A.27 supplies the complete closed event geometry, Proposition A.20 supplies the selected-law exact probability, and Proposition A.21 supplies the distinct positive-length lower ratio and scale comparison.

Conclusion 6 repeats the exact dependence, probability, interval, norm, boundary, baseline, and nonoutput statements already contained in those named results. Conjunction changes no quantifier, object, dimension, constant, law class, interval class, probability mode, norm, or measure. Each result is applied to the identical setting object. There is no term absorption, new cited result, new source invocation, new probability conversion, hidden constant, or additional mathematical claim. This proves the proposition. □

Synthesis of the theorem clauses. The static and projective clauses are exactly the conclusions of Lemmas A.1, A.2, and A.3 and Propositions A.1 and A.2. The equivalent chart clause is exactly Proposition A.5, with its measurability, multiplicity, and persistent-root conclusions supplied by the other named results in that subsection. Proposition A.10 supplies the coordinate-free affine rate, capacity, and zero-certificate case. The five named homogeneous results in the projective-rate subsection supply the sharper rate and preserve the interval-then-law supremum order. Proposition A.18 gives the complete deterministic-leading-coefficient monic clause. Finally, the six named Counter-example results supply the exact certificate, projective speed, all-law upper coefficient, closed wedge geometry, selected-law probability, and normalized lower ratio.

All six groups use their original deterministic presentation, coefficient dimension, law class, interval class, probability mode, and Euclidean norm and measure conventions. Their conjunction changes no assumption, quantifier, object, constant, probability conversion, or supremum order. This gives Proposition A.22 with the exact deterministic-presentation-first order and both baseline invariance conclusions. □

A.13 Proof of the Main Theorem

Proof of Theorem 3.1. Under Assumptions 1–4, Proposition A.22 applies to the identical fixed presentation. Its first conclusion is the theorem's static derivative-closure and normalized projective certificate. Its second conclusion gives both original-coordinate chart forms, the coordinate-free affine rate and capacity, and the zero-certificate branch. Its third conclusion is the sharper homogeneous rate and capacity, including the stationary branch and the interval-before-law supremum order. Its fourth conclusion is the exact affine-monic presentation, certificate, chart, and ordinary-probability recovery with the deterministic leading coefficient outside the random vector. Its fifth conclusion is Counter-example 1 with the selected-law lower ratio $1/(4\delta)$, all-law upper coefficient $1/\delta$, and raw certificate and projective speed $1/\delta$. Its sixth conclusion supplies all dependence, mode, norm, measure, baseline-invariance, and nonoutput declarations, including the boundary excluding any claim for unrestricted raw Pfaffian presentations. These are exactly the clauses of Theorem 3.1, so the theorem follows. □